

CHAPTER 1

General Strategies

Shawn Doonan

1. Defining the Problem

The chapters that follow in this volume give detailed instructions on how to use the various methods that are available for purification of proteins. The question arises, however, of which of these methods to use and in which order to use them to achieve purification in any particular case. That is, the purification problem must be clearly defined. What follows outlines the sorts of questions that need to be asked as part of that definition and how the answers affect the approach that might be taken to developing a purification schedule. It should be noted here that the discussion does not touch on the special cases of purification of proteins at industrial scale or for therapeutic applications; these raise very specific problems that are outside the scope of this chapter (*see refs. 1 and 2, respectively, for a coverage of these topics*).

1.1. How Much Do I Need?

The answer to this question depends on the purpose for which the protein is required. For example, to carry out a full chemical and physical analysis of a protein may require several hundreds of milligrams of purified material while a kinetic analysis of the reaction catalyzed by an enzyme could perhaps be done with a few milligrams and <1 mg would be required to raise a polyclonal antibody. At the extreme end of the scale, if the objective is to obtain limited sequence information from the *N*-terminus of a protein as a preliminary to design of an oligonucleotide probe for clone screening, then using modern microsequencing techniques, a few micrograms will be sufficient. These different requirements for quantity

From *Methods in Molecular Biology, Vol. 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

may well dictate the source of the protein chosen (*see* Section 1.4.) and will certainly influence the approach to purification. Purification of large quantities of protein requires use of techniques, at least in the early stages, which have high capacity but low resolving power, such as fractional precipitation with salt or organic solvents (Chapter 13). Only when the volume and protein content of the extract has been reduced to manageable levels can methods of medium resolution and capacity, such as ion-exchange chromatography (Chapter 14) be used leading on, if necessary, to high-resolution but generally lower capacity techniques, such as affinity chromatography (Chapter 16) and isoelectric focusing (Chapter 23). On the other hand for isolation of small to medium amounts of proteins, it will usually be possible to move directly to the more refined methods of purification without the need for initial use of bulk methods. This is, of course, important because the fewer the steps that have to be used, the higher the final yield of the protein will be and the less time it will take to purify it.

1.2. Do I Want to Retain Biological Activity?

If the answer to this is positive then it restricts to some extent the range of techniques that can be employed and the conditions under which they can be performed. Most proteins retain activity when handled in neutral aqueous buffers at low temperature (although there are exceptions and these exceptions lend themselves to somewhat different approaches to purification). This consideration then rules out use of those techniques in which the conditions are likely to deviate substantially from the above. For example, immunoaffinity chromatography is a very powerful method but the conditions required to elute bound proteins are often rather severe, for example, the use of buffers of low pH, because of the tightness of binding between antibodies and antigens (*see* Chapters 16 and 19 for a discussion of this problem). Similarly, reversed-phase chromatography (Chapter 27) requires the use of organic solvents to elute proteins and rarely will be compatible with recovering an active species. Ion-exchange chromatography provides the most general method for isolation of proteins with retention of activity unless the protein has special characteristics that offer alternative strategies (*see* Section 2.4.). With labile molecules it is important to plan the purification schedule to contain as few steps as possible and with minimum requirement for changing buffers (Chapter 11), since this will reduce losses of activity.

In some cases, retention of biological activity is not required. This would be the case, for example, if the protein is needed for sequence analysis or perhaps for raising an antiserum. There is then no restriction on the methods that can be used and, indeed, the very powerful separation method of polyacrylamide gel electrophoresis in the presence of sodium dodecyl sulphate (SDS-PAGE) followed by blotting or elution from the gel can be used to isolate small amounts of pure protein either from partially purified extracts or even from crude extracts (Chapters 33–35). It is important in this context to differentiate between loss of biological activity arising from loss of three-dimensional structure, which will not be of concern in the applications outlined above, from loss of activity owing to modification of the chemical structure of the protein, which certainly would be a major concern. The most important route to chemical modification is proteolytic cleavage and ways in which this can be detected and avoided are discussed in Chapter 9.

1.3. Do I Need a Completely Pure Protein?

The concept of purity as applied to proteins is not entirely straightforward. It ought to mean that the protein sample contains, in addition to water and things like buffer ions that have been purposefully added, only one population of molecules all with identical covalent and three-dimensional structures. This is an unattainable goal and indeed an unnecessary one. What is required is a sample of protein that does not contain any species which will interfere with the experiments for which the protein is intended. This is not simply an academic point since it will usually become more and more difficult to remove residual contaminants from a protein sample as purification progresses. Extra purification steps will be required which take time (effectively an increase in cost of the product) and will inevitably lead to decreasing yields. What is required is an operational definition of purity for the particular project in hand because this will not only define the approach to the purification problem but may also govern its feasibility. It may not be possible to obtain a highly purified sample of a labile protein but it may be possible to obtain it in a sufficient state of purity for the purposes of a particular investigation.

The usual criterion of purity used for proteins is that a few micrograms of the sample produces a single band after electrophoresis on SDS-PAGE when stained with a reagent such as Coomassie blue or some similar non-specific stain (*see ref. 3 for practical details of this procedure and other*

chapters in the same volume for many other basic protein protocols). This simple criterion begs several questions. The most important of these is that SDS-PAGE separates proteins effectively on the basis of size and it may be that the sample contains two or more components that are sufficiently similar not to be resolved; the answer here is to subject the sample to an additional procedure, such as nondenaturing PAGE (4) since it is unlikely that two proteins will migrate identically in both systems. It must always be born in mind, however, that even if a single band is observed in two such systems, minor contaminants will inevitably become visible if the gel is more heavily loaded or if staining is carried out using a more sensitive procedure, such as silver staining (5).

The major question is: Does it matter if the protein is 50, 90, or 99% pure? The answer is that it depends on the purpose of the purification. For example, a 50% pure protein may be entirely acceptable for use in raising a monoclonal antibody but a 95% pure protein may be entirely unacceptable for raising a monospecific polyclonal antibody particularly if the contaminants are highly immunogenic. Similarly, a relatively impure preparation of an enzyme may be acceptable for kinetic studies provided that it does not contain any competing activities; an affinity chromatography method might provide a rapid way of obtaining such a preparation. As a final example, a 95% pure protein sample is perfectly adequate for amino acid sequence analysis and indeed a lower state of purity is acceptable if proper quantitation is carried out to ensure that a particular sequence does not arise from a contaminant.

The message here is that preparation of a sample of protein approaching homogeneity is difficult and may not always be necessary so long as one knows what else is there. By taking account of the purpose for which the protein is required, it may be possible to decide on an acceptable level of contaminants and consideration of the nature of acceptable contaminants may suggest a purification strategy to be adopted.

1.4. What Source Should I Use?

The answer to this question may be partly or entirely dictated by the problem in hand. Clearly if the objective is to study the enzyme ribulose biphosphate carboxylase, then there is no choice but to isolate it from a plant, but the plant can be chosen for its ready availability, high content of the enzyme, ease of extraction of proteins (Chapter 3), and low content of interfering polyphenolic compounds (Chapter 8). Of course, if

one is interested in, for example, comparative biochemistry or molecular evolution, then not only the desired protein but also its source may be completely constrained.

In general, however, plants will not be the source of choice for isolation of a protein of general occurrence and where species differences are not of interest. Microbial or fungal sources may be a better choice since they can usually be grown under defined conditions thus assuring the consistency of the starting material and, in some cases, allowing for manipulation of levels of desired proteins by control of growth media and conditions (Chapters 4 and 5). They have the disadvantage, however, of possessing tough cell walls that are difficult to break and, consequently, microorganisms are not ideal for large scale work unless the laboratory has specialized equipment needed for their disruption.

The most convenient source of proteins in most cases is animal tissue, such as heart and liver and, except for relatively small scale work, the tissues will normally be obtained from a commercial abattoir. Laboratory animals provide an alternative for smaller scale purifications. Content of a particular protein is likely to be tissue specific in which case the most abundant source will probably be the best choice. It is worth noting, however, that it is easier to isolate proteins from tissues, such as heart, than from liver (the reasons for which are outlined in Chapter 2) and hence the heart may be the better bet even if the levels of the protein are lower than in liver.

A different sort of question arises if the protein of interest exists in soluble form in a subcellular organelle, such as the mitochondrion or chloroplast. Once the source organism has been chosen, there remains the decision as to whether to carry out a total disruption of the tissue under conditions where the organelles will lyse or whether to homogenize under conditions where the organelles remain intact and can be isolated by methods such as those described in Chapters 6 and 7. The latter approach will, of course, result in a very significant initial enrichment of the protein and subsequent purification will be easier because the range and amount of contaminating proteins will be much decreased. In the case of animal tissues the decision will probably depend on the scale at which it is intended to work (assuming, of course, that access to the necessary preparative high-speed centrifuges is available). Subcellular fractionation of a few hundred grams of tissue is a realistic objective, but if it is intended to work with larger amounts, then the time required for organelle isolation probably will be prohibitive and is unlikely to

compensate for the extra work which will be involved in purification from a total cellular extract. Subcellular fractionation of plants is a much more difficult operation in most cases (*see* Chapter 7). Hence, except in the most favorable cases and for small scale work, purification from a total cellular extract will probably be the only realistic option.

In the case of membrane proteins, there again will be a considerable advantage in isolating as pure a sample of the membrane as possible before attempting purification. The ease with which this can be done depends on the organism and membrane system in question. Chapters 6 and 30 give some approaches to this problem for specific cases, but if it is intended to isolate a membrane protein from other sources, then a survey of the extensive literature on membrane purification is recommended (*see* ref. 6).

For proteins which are present in only very small quantities or which are found only in inconvenient sources, gene cloning and expression in a suitable host now provide an alternative route to purification (for a review of methods *see* ref. 7). This is, of course, a major undertaking and is likely to be used only when conventional methods are not successful. Suffice it to say that once the protein is expressed and extracted from the host cell (*see* Chapter 4 for a method of extracting recombinant proteins from bacteria), the methods of purification are the same as those for proteins from conventional sources.

1.5. Has it Been Done Before?

It is quite common to need to purify a protein whose purification has been reported previously, perhaps to use it as an analytical tool or perhaps to carry out some novel investigations on it. In this case the first approach will be to repeat the previously described procedure. The chances are, however, that it will not work exactly as described since small variations in starting material, experimental conditions, and techniques—which are inevitable between different laboratories—can have a significant effect on the behavior of a protein during purification. This should not matter too much since adjustments to the procedures should be relatively easy to make once a little experience has been gained of the behavior of the protein. One pitfall to watch out for is the conviction that there ought to be a better way of doing it. It is possible to spend a great deal of time trying to improve on a published procedure often to little avail.

Even if the particular protein of interest has not been isolated previously, it may be that a related molecule has been, for example, the same

protein but from a different organism or a member of a closely related class of proteins. In the former case, particularly if the organisms are closely related, then the properties of the proteins should be quite similar and only minor variations in procedures, for example, the pH used for an ion-exchange step, might be required. Even if the family relationships are more distant, significant clues might still be available, such as the fact that the target is a glycoprotein, which will provide valuable approaches to purification (*see* Section 2.4.). Much time and wasted effort can be saved by using information in the literature rather than trying to re-invent the wheel.

2. Exploiting Differences

Protein purification involves the separation of one species from perhaps a thousand or more species of essentially the same general characteristics (they are all proteins!) in a mixture of which it may constitute a small fraction of 1% of the total. It is, therefore, necessary to exploit to the full those properties in which proteins differ from one another in devising a purification schedule. The following lists the most important of those properties and outlines the techniques that make use of them with comments on their practical application. More details on each technique will be found in the chapters that follow.

2.1. Solubility

Proteins differ in the balance of charged, polar, and hydrophobic amino acids that they display on their surfaces and hence in their solubilities under a particular set of conditions. In particular, they tend to precipitate differentially from solution on addition of species such as neutral salts or organic solvents and this provides a route to purification (*see* Chapter 13). It is however, a rather gross procedure since precipitation will occur over a range of solute concentrations and those ranges necessarily overlap for different proteins. It is not to be expected, therefore, that a high degree of purification can be achieved by such methods (perhaps two- to threefold in most circumstances), but the yield should be high and, most importantly, fractional precipitation can be carried out easily on a large scale provided only that a suitable centrifuge is available. It is, therefore, very common for this technique to be used at the stage immediately following extraction when working on a moderate to large scale. An important added advantage is that a substantial degree of concentration of the extract can be obtained at the same time which, considering that water is

the major single contaminant in a protein solution, is a considerable added benefit.

2.2. Charge

Proteins differ from one another in the proportions of the charged amino acids (aspartic and glutamic acids, lysine, arginine, and histidine) that they contain. Hence they will differ in net charge at a particular pH or, another manifestation of them same difference, in the pH at which the net charge is zero (the isoelectric point). The first of these differences is exploited in ion-exchange chromatography which is perhaps the single most powerful weapon in the protein purifier's armory (Chapter 14). This makes use of the binding of proteins carrying a net charge of one sign onto a solid supporting material bearing charged groups of the opposite sign; the strength of binding will depend on the magnitude of the charge on the particular protein. Proteins may then be eluted from the matrix in exchange for ions of the opposite charge with the concentration of the ionic species required being determined by the magnitude of the charge on the protein.

Ion-exchange chromatography is a technique of moderate to high resolution depending on the way in which it is implemented. For large-scale work (around 100 g of protein), use is generally made of fibrous cellulose-based resins that give good flow rates with large bed volumes but not particularly high resolution; this would normally be done at an early stage in a purification. Better resolution is available with the more advanced Sepharose-based materials but generally on a smaller scale. For small quantities (≤ 10 mg), the technique of fast protein liquid chromatography (Chapter 26) is available which makes use of packing materials with very small diameters and correspondingly high resolving power; this, however, requires specialized equipment that may not be available in all laboratories. Because of the small scale, this method would usually be used at a late stage for final clean-up of the product. It should be born in mind that two proteins which carry the same charge at a particular pH might well differ in charge at a different pH. Hence it is quite common for a purification procedure to contain two or more ion-exchange steps either using the same resin at different pH values or perhaps using two resins of opposite charge characteristics (e.g., one carrying the negatively charged carboxymethyl [CM] group and the other the positively charged diethylaminoethyl [DEAE] group).

There are two main ways of exploiting differences in isoelectric points between proteins. Chromatofocusing is essentially an ion-exchange technique in which the proteins are bound to an anion exchanger and then eluted by a continuous decrease of the buffer pH so that proteins elute in order of their isoelectric points (*see* Chapter 24). It is a method of moderately high resolving power and capacity and is hence best used to further separate partially purified mixtures. The other technique is isoelectric focusing (Chapter 23) in which proteins are caused to migrate in an electric field through a system containing a stable pH gradient. At the pH at which a particular protein has no net charge (the isoelectric point), it will cease to move; if it diffuses away from that point, then it will regain a charge and migrate back again. This method, although of low capacity, is capable of very high resolution and is frequently used to separate mixtures of proteins which are otherwise difficult to fractionate.

2.3. Size

This property is exploited directly in the techniques of size exclusion chromatography (Chapter 25) and ultrafiltration (Chapter 12). In the former, the protein solution is passed through a column of porous beads, the pore sizes being such that large proteins do not have access to the internal space, small proteins have free access to it, and intermediate sized proteins have partial access; a range of these materials with different pore sizes is available. Clearly, large proteins will pass through the column most rapidly and small proteins most slowly with a range of behavior in between. The method is of limited resolving power but is useful in some circumstances particularly when the protein of interest is at one of the extremes of size. The capacity is low because of the need to keep the volume of solution applied to the column as small as possible.

In ultrafiltration, liquid is forced through a membrane with pores of a controlled size such that small solutes can pass through but larger ones cannot. It, therefore, can be used to obtain a separation between large and small protein molecules and also has the advantage that it is not limited by scale. Use of the method for protein fractionation is, however, restricted to a few special cases (*see* Chapter 12) and the principal value of the technique is for concentration of protein solutions.

A completely different approach to the use of size differences to effect protein separation is SDS-PAGE. In this method, the protein molecules are denatured and coated with the detergent so that they carry a large

negative charge (the inherent charge is swamped by the charge of the detergent). The proteins then migrate in gel electrophoresis on the basis of size; small proteins migrate most rapidly and large ones slowly because of the sieving effect of the gel. The method has enormously high resolving power and its use in various forms for analytical purposes is one of the most important techniques in analytical protein chemistry (3). The development of methods for recovery of the protein bands from the gel after electrophoresis (*see* Chapters 33 and 34) has enabled this resolving power to be exploited for purification purposes. Obviously the scale of separation is small and the product is obtained in a denatured state, but a sufficient amount often can be obtained from very complex mixtures for the purposes of further investigation (*see* Section 1.2.).

2.4. Specific Binding

Most proteins exert their biological functions by binding to some other component in the living system. For example, enzymes bind to substrates and sometimes to activators or inhibitors, hormones bind to receptors, antibodies bind to antigens, and so on. These binding phenomena can be exploited to effect purification of proteins usually by attaching the ligand to a solid support and using this as a chromatographic medium. An extract or partially purified sample containing the target protein is then passed through this column to which the protein binds by virtue of its affinity for the ligand. Elution is achieved by varying the solvent conditions or introducing a solute that binds strongly either to the ligand or to the protein itself.

Various types of affinity chromatography, as the method is called, are described in detail in Chapters 16–20. Immunoaffinity chromatography in particular is capable of very high selectivity because of the extreme specificity of antibody–antigen interactions. As mentioned above and dealt with in more detail in Chapters 16 and 19, the most common problem with this technique is to effect elution of the target protein under conditions that retain biological activity. Lectin-affinity chromatography (Chapter 18) exploits the selective binding between members of this class of plant proteins and particular carbohydrates. It, therefore, has found widespread use both in the isolation of glycoproteins and in removal of glycoprotein contaminants from other proteins, and is also capable of high specificity.

Affinity methods that rely on interactions of the target protein with low-mol-wt compounds (e.g., enzymes with substrates or substrate ana-

logs) are frequently less specific because the ligand may bind to several proteins in a mixture. For example, immobilized NAD^+ will bind to many dehydrogenases and benzamidine will bind to most serine proteases; thus a group of related enzymes rather than individual species may be isolated using these ligands. A novel application of affinity methods is provided by the use of bifunctional NAD^+ derivatives to selectively precipitate dehydrogenases from solution (Chapter 22).

The use of organic dyes as affinity ligands (Chapter 17) is interesting because these molecules seem to bind fairly specifically to nucleotide-binding enzymes, although from their structures it is not at all clear why they should do so; it is likely that hydrophobic interactions between the dye and protein also contribute to binding. Use of the latter interaction has led to development of a specific form of chromatography which uses hydrophobic stationary phases (Chapter 15); this method has elements of biospecificity in that some proteins have binding sites for natural hydrophobic ligands, but in the general case it relies on the fact that all proteins have hydrophobic surface regions to a greater or lesser extent.

Finally, many proteins are known that bind metal ions with varying degrees of specificity and this forms the basis of immobilized metal ion affinity chromatography (Chapter 20). Specific affinity of proteins for calcium ions may also be the basis, in part, for binding to hydroxyapatite but ion-exchange effects are probably also involved (Chapter 21).

In summary there is a variety of affinity methods available, ranging from medium to very high selectivity, and in favorable cases affinity chromatography can be used to obtain a single-step purification of a protein from an initial extract. Generally, however, the capacities of affinity media are not high and the materials can be very expensive thus rendering their use on a large scale unrealistic. For these reasons affinity methods are usually used at a late stage in a purification schedule.

2.5. Special Properties

In a sense the specific binding properties discussed in the previous section are "special," but that is not what is meant here. Some proteins have, for example, the property of greater than normal heat stability and in those circumstances it may be possible to obtain substantial purification by heating a crude extract at a temperature where the target protein is stable but contaminants are denatured and precipitate from solution (*see ref. 8 for an example of the use of this method*). It is not likely, of

course, that this approach will be useful in purification of proteins from thermophilic organisms since all or most of the proteins present would be expected to share the property of thermostability. Another possibility is that the protein of interest may be particularly stable at one or other of the extremes of pH; in this case incubation of an extract at low or high pH might well lead to selective precipitation of contaminants. It is always worthwhile carrying out some preliminary experiments with an unknown protein to see if it possesses special properties of this kind that would assist in its purification.

Finally mention should be made of the fact that it is now feasible, if the need is sufficiently great, to engineer special properties into proteins to assist in their purification. Typical examples include addition of polyarginine or polylysine tails to improve behavior on ion-exchange chromatography, or of polyhistidine tails to introduce affinity on immobilized metal affinity chromatography (9). It is, however, likely that these techniques would be used as a last resort if all other attempts to purify the protein failed unless recombinant DNA technology had been selected as the route to protein production and purification in the first place (*see* Section 1.4.).

3. Documenting the Purification

It is vitally important to keep an inventory at each stage of a purification of volumes of fractions, total protein content, and content of the protein of interest. The last of these is particularly important since otherwise it is very easy to end up with a vanishingly small yield of target protein and not to know at which step the protein was lost. If the protein has a measurable activity then it is equally important to monitor this since it is also possible to end up with a protein sample which is inactive if one or more steps in the purification involves conditions under which the protein is unstable.

Measurement of total protein content of fractions presents no problems. At early stages of a purification it is usually sufficient to determine the absorbance of the solution at 280 nm (making sure that it is optically clear to avoid errors owing to light scattering) and to use the rough approximation that $A_{280\text{ nm}}^{1\%} = 10$. At later stages one of the more accurate methods, such as the Bradford procedure (10) or the bicinchoninic acid assay (11), should be used unless the absorbance/dry weight correlation for the target protein happens to be known.

Measurement of the amount and/or activity of the protein of interest may or may not be straightforward. For example, many enzymes can be

assayed using simple and rapid spectrophotometric methods. For other proteins the assay may be more difficult and time consuming, such as bioassay or immunoassay (it should also be recognized that these are not necessarily the same thing; immunoassay frequently will not distinguish between inactive and active molecules so care must be taken in the interpretation of results using this method). In other situations the protein of interest may have no measurable biological activity; in such cases immunoassay can be used or, more commonly, quantitation of the appropriate band after separation of the protein on polyacrylamide gels (12). Indeed it may be that the target protein will only have been identified as a spot on two-dimensional polyacrylamide gels (13) and purification is being attempted as a preliminary to determining its biological activity.

Obviously it is not possible to be prescriptive here about what methods of analysis and quantitation to use in any specific case. What must be said, however, is that it is very unwise to embark on an attempted purification without first devising a method for quantitation of the protein of interest. Not to do so is courting failure.

4. An Example

To give the newcomer to protein purification a “feel” for what the process might look like in practice, Table 1 shows the fully documented results of the isolation of a particular enzyme starting from 5 kg of pig liver. All techniques used are described in detail in subsequent chapters and are only summarized here.

The strategy was to start by totally homogenizing the tissue in 10 L of buffer and, after removal of cell debris by centrifugation, to carry out an initial crude purification by fractional precipitation with ammonium sulphate. This had the added advantages of removing residual insoluble material from the extract (this precipitated in the first ammonium sulphate fraction) and achieving a very large reduction in volume of the active fraction. Ammonium sulphate was removed from the active fraction by dialysis.

Because of the large amount of protein remaining in the active fraction, the next step was a relatively crude ion-exchange separation using a large column (7 × 50 cm) of CM-cellulose CM23 (this has a high capacity and good flow rates but is of only moderate resolving power). Conditions were chosen so that the enzyme was absorbed onto the column and then, after washing off unbound contaminants, it was eluted with a single stepwise increase in ionic strength to 0.1M using sodium chloride.

Table 1
Example Protein Purification Schedule

Fraction	Volume, mL	Protein concentration, mg/mL	Total protein, mg	Activity, ^a U/mL	Total activity, U	Specific activity, U/mg	Purification factor ^b	Overall yield, ^c %
Homogenate	8500	40	340,000	1.8	15,300	0.045	1	100
45–70% (NH ₄) ₂ SO ₄	530	194	103,000	23.3	12,350	0.12	2.7	81
CM-cellulose	420	19.5	8190	25	10,500	1.28	28.4	69
Affinity chromatography	48	2.2	105.6	198	9500	88.4	1964	62
DEAE-Sepharose	12	2.3	27.6	633	7600	275	6110	50

^aThe unit of enzyme activity is defined as that amount which produces 1 μ mol of product per minute under standard assay conditions

^bDefined as purification factor = (specific activity of fraction/specific activity of homogenate)

^cDefined as overall yield = (total activity of fraction/total activity of homogenate)

Previous trial experiments had shown that the enzyme bound to an affinity matrix in a buffer at the same pH and salt content as that with which it was eluted from CM-cellulose, and so affinity chromatography was used for the next step without changing the buffer and without prior concentration. The enzyme was eluted by applying a linear salt gradient up to a concentration of 1M.

At this stage, electrophoresis of the active fraction under nondenaturing conditions showed the presence of two major contaminants, both of them more basic than the protein of interest. Hence, the sample was applied to a column of DEAE-Sephadex under conditions where the target protein was absorbed but the majority of the contaminating protein was not; the sample was equilibrated in starting buffer by dialysis before application to the column. The target protein was eluted from the column using a linear salt gradient and was found to be homogeneous by the usual techniques (*see* Section 1.3.).

The results in Table 1 show that the purification procedure was quite successful in that a high yield (50% overall) of enzyme activity was obtained; this was achieved by using a small number of steps each of which gave a good step yield. There will inevitably be losses on any purification step and the important point is that these and the number of steps should be kept as low as possible (a 5-step schedule in which the yield from each step is 50% will give an overall yield of 3%; a 10-step schedule with 80% step yield will give a final yield of 11%). It can also be seen from the final purification factor that the amount of this particular enzyme in the liver was low (about 0.016% of soluble protein) and hence a relatively large amount of tissue had to be used to obtain the required amount of product. This was an important factor in deciding the first two steps in the schedule (*see* Section 1.1.).

The purification in its final form can be completed in 5–6 working days. It must be born in mind, however, that each step has been optimized and that development of the procedure took several months of work. This is common when working out a new purification schedule and it is always necessary to be conscious of the time commitment when deciding to embark on purifying a protein.

References

1. Asenjo, J. A. and Patrick, I (1990) Large-scale protein purification, in *Protein Purification Applications A Practical Approach* (Harris, E L V and Angal, S , eds), IRL, Oxford, UK, pp. 1–28.

- 2 Bristow, A F (1990) Purification of proteins for therapeutic use, in *Protein Purification Applications A Practical Approach* (Harris, E L V and Angal, S , eds), IRL, Oxford, UK, pp. 29–44
- 3 Smith, B J (1994) SDS polyacrylamide gel electrophoresis of proteins, in *Methods in Molecular Biology, Vol 32 Basic Protein and Peptide Protocols* (Walker, J M., ed), Humana, Totowa, NJ, pp 23–34
- 4 Walker, J. M. (1994) Nondenaturing polyacrylamide gel electrophoresis of proteins, in *Methods in Molecular Biology, Vol 32 Basic Protein and Peptide Protocols* (Walker, J M , ed), Humana, Totowa, NJ, pp. 17–22
- 5 Dunn, M J. and Crisp, S J (1994) Detection of proteins in polyacrylamide gels using an ultrasensitive silver staining technique, in *Methods in Molecular Biology, Vol 32 Basic Protein and Peptide Protocols* (Walker, J M., ed.), Humana, Totowa, NJ, pp 113–118
6. Graham, J M. and Higgins, J A. (eds.) (1993) *Methods in Molecular Biology, Vol 19 Biomembrane Protocols I Isolation and Analysis*, Humana, Totowa, NJ
- 7 Murray, E J (ed) (1991) *Methods in Molecular Biology, Vol 7 Gene Transfer and Expression Protocols*, Humana, Totowa, NJ
- 8 Banks, B E C , Doonan, S , Lawrence, A J., and Vernon, C. A (1968) The molecular weight and other properties of aspartate aminotransferase from pig heart muscle *Eur J Biochem* 5, 528–539.
- 9 Brewer, S J. and Sassenfeld, H. M. (1990) Engineering proteins for purification, in *Protein Purification Applications A Practical Approach* (Harris, E. L V and Angal, S., eds), IRL, Oxford, UK, pp 91–111
- 10 Kruger, N. J (1994) The Bradford method for protein quantitation, in *Methods in Molecular Biology, Vol 32. Basic Protein and Peptide Protocols* (Walker, J M , ed), Humana, Totowa, NJ, pp 9–15
- 11 Walker, J. M (1994) The Bicinchoninic acid (BCA) assay for protein quantitation, in *Methods in Molecular Biology, Vol 32 Basic Protein and Peptide Protocols* (Walker, J M , ed), Humana, Totowa, NJ, pp 5–8
- 12 Smith, B J. (1994) Quantification of proteins on polyacrylamide gels (nonradioactive), in *Methods in Molecular Biology, Vol 32 Basic Protein and Peptide Protocols* (Walker, J M , ed), Humana, Totowa, NJ, pp 107–111
13. Pollard, J W. (1994) Two-dimentional polyacrylamide gel electrophoresis of proteins, in *Methods in Molecular Biology, Vol. 32 Basic Protein and Peptide Protocols* (Walker, J M., ed), Humana, Totowa, NJ, pp 73–85

CHAPTER 2

Preparation of Extracts from Animal Tissues

Shawn Doonan

1. Introduction

Extraction of soluble proteins from most animal tissues is relatively straightforward, since the cell membranes are weak and easily ruptured by a combination of mechanical and osmotic forces. The tissue is disrupted in a suitable buffer using a homogenizer and the homogenate then clarified by centrifugation. Extraction of membrane proteins requires more specialized methods, and these are dealt with in Chapter 28.

Thought needs to be given to the tissue to be used and the source of that tissue. These will sometimes be dictated by the particular project in hand. For example, the protein of interest may be restricted to a particular organ, and the objective may be to compare the structure of that protein as expressed in the rat to that expressed in some other animal. In this case the parameters are obviously fixed. In other cases, the protein may be of general occurrence and the animal of origin of no particular interest, e.g., if the objective is to study the mechanism of action of an enzyme. In such cases, availability of the tissue, amount of soluble protein in the tissue, and the ease of obtaining clarified extracts become the important criteria.

The cheapest and most convenient source of large quantities of animal tissue (pig, sheep, cattle) will normally be a commercial abattoir. These companies will usually allow personal collection of tissues, which can be packed in ice for transport back to the laboratory; tissues may be stored frozen at -20°C for later use, but this is not to be generally recommended.

From *Methods in Molecular Biology, Vol 59 Protein Purification Protocols*
Edited by S Doonan Humana Press Inc , Totowa, NJ

Alternatively, smaller animals (rats, mice, rabbits, and so forth) bred in animal facilities can be used, but this is a more expensive option and will require a large number of animals to be sacrificed for a large-scale preparation. The choice of tissue is also important. Heart is ideal if the protein is present in a reasonable quantity since it is very easy to obtain clarified homogenates from this tissue. Liver is less desirable, since it contains large quantities of nucleic acids and of fats, which make clarification of extracts more difficult. It also contains much larger quantities of soluble protein than does heart, and this increases the problems of subsequent purification.

Clearly, the tissue distributions of proteins vary enormously, and it is always advisable to carry out preliminary experiments to measure the relative content of the protein of interest in the various tissues. The final choice of starting material will then be determined by the balance between content, ease of obtaining the tissue, and the various technical issues described.

A final consideration is the subcellular distribution of the protein. If it is localized to a specific organelle, such as the mitochondrion, then it may be advantageous to carry out a subcellular fractionation (*see* Chapter 6) and extract the protein from the isolated organelles. This will clearly simplify subsequent purification, but will necessarily restrict the amount of protein obtainable since subcellular fractionation can only be carried out on a small scale. If large amounts of protein are required, then the only practical approach is to homogenize the tissue in the conventional manner, under which conditions subcellular organelles will lyse, and to purify the protein from the total extract. An example of a large-scale purification of a mitochondrial enzyme using this approach is given in ref. 1.

The method of extraction described herein is of general applicability, but there are other specialized procedures that may be of use in particular cases. Comprehensive coverage of these procedures is given in ref. 2, which, though now old, is still a valuable survey of methods.

2. Materials

In addition to normal laboratory glassware and equipment, the following are necessary:

1. Waring blender These are available from general laboratory equipment suppliers and come with vessels ranging from about 10 mL to 4 L in capacity. The vessels may be of glass or stainless steel; the latter is to be pre-

ferred, since they may be cooled more readily. An alternative for large-scale work is a domestic food blender.

2. Refrigerated centrifuge: Again, several models are available, for example, the MSE Mistral or the Sorvall RC2-B. The choice of rotors will depend on the scale of purification work envisaged. Generally useful is a six-place angle rotor for 250- or 300-mL tubes. For larger scale work, a six-place, 1000-mL swing-out rotor is useful.
3. Polypropylene screw-cap centrifuge tubes for the rotors to be used.

3. Methods

- 1 Trim fat and connective tissue from the tissue and cut into pieces of a few grams each.
2. Place the tissue in the precooled blender vessel (*see* Note 1) and add cold extraction buffer using 2 vol of buffer by weight of tissue (*see* Note 2). Use a blender vessel that has a capacity approximately that of the volume of buffer plus tissue so that the air space is minimized; this will reduce aerosol formation.
3. Homogenize at full speed for 1–3 min depending on the toughness of the tissue (or until the tissue is fully disrupted as judged by eye). For long periods of homogenization, it is best to blend in 1-min bursts with a few minutes in between to avoid excessive heating.
- 4 Pour the homogenate into a glass beaker, place on ice, and stir for 15–30 min to ensure full extraction
5. Remove cell debris and other particulate matter from the homogenate by centrifugation at 4°C For large-scale work use a 6 × 1000-mL swing-out rotor operated at about 2000–3000*g* for 30 min. For smaller scale work (up to 3 L of homogenate) a 6 × 250-mL angle rotor operated at about 5000*g* is to be preferred (*see* Note 3).
6. Pour off the supernatant carefully to avoid disturbing sedimented material. Any fatty material that has floated to the top of the tube should be removed by filtering the extract through cheesecloth or through a plug of glass wool in a filter funnel. The pellet may be re-extracted with more buffer to increase the yield (*see* Note 4).

The extract obtained at this stage may require further treatment to remove insoluble material before fractionation, depending mainly on the tissue that has been used and on the planned next stage in the purification (*see* Note 5).

4. Notes

1. Such tissues as liver, kidney, brain, and heart are easily homogenized in a blender. Such tissues as skeletal muscle, lung, and so on, are tougher, and in these cases, it is advisable to grind them using a domestic meat mincer before homogenization. Very fibrous tissues (e.g., mammary glands) should

be frozen before homogenization to facilitate disruption. Small amounts of soft tissues (1–5 g) and animal cells obtained from culture may be more conveniently homogenized using a handheld Dounce homogenizer (3)

- 2 The choice of buffer may be dictated by the particular problem in hand, but generally a buffer of moderate ionic strength at a neutral pH (e.g., 0.1M phosphate, pH 7.0) is used. Some proteins tend to attach electrostatically to tissue debris and/or to membrane fragments in which case addition of 0.1M NaCl to the extraction buffer will increase yield; this can easily be established in trial experiments. A major consideration is whether to add protease inhibitors to the extraction buffer (*see* Chapter 9 for a discussion of the problem of proteolysis). In general, this is not essential because the protein content of the extract will be high and the bulk protein lends protection to the particular protein of interest. Some proteins are, however, more susceptible to proteases than others, and some tissues (e.g., liver) have much higher protease levels than others (e.g., heart), so proteolysis may be a problem depending on the protein of interest and the source tissue. Use of protease inhibitors, particularly at the stage of extraction when volumes are large, is expensive and hence best avoided if possible. Ideally, a trial experiment should be carried out in which the level of the protein of interest is measured in an extract over a period of a few hours to see if measurable losses occur; if so, proteolysis should be suspected and steps taken to avoid it.

Some proteins are susceptible to oxidation, in which case addition of dithiothreitol (1 mM) or β -mercaptoethanol (0.1M) to the extraction buffer is advised. Similarly, if the protein of interest is inhibited by heavy metals, then EDTA (0.1M) should be added. In cases where the protein of interest has a known cofactor, it is sometimes worth including the cofactor in the homogenization buffer to prevent dissociation of the cofactor with possible consequent decrease in stability of the protein. In all these situations, trial experiments should be done to see if any advantage is obtained by potential additives.

3. Great care must be taken when using centrifuges and, in particular, to ensure that the tubes are balanced across the central axis (i.e., in a six-place rotor, tubes 1 and 4, 2 and 5, and 3 and 6 contain the same weight to within 1 g of suspensions of the same density). Hence, if only 250 mL of homogenate are available, this should not be balanced against a tube containing an equal weight of water—the densities will be different and the centers of gravity of the two tubes will differ. In this case, the homogenate should be split between two tubes. It is also unwise to fill the tubes completely when using an angle rotor. This is because the neck of the tube may distort slightly at high rotor speeds, allowing leakage of liquid; in addition to being potentially

dangerous, this could lead to damage to the centrifuge rotor. Ideally, the tubes should be filled to such an extent that when placed in the rotor, the liquid level is just below the screw-cap at its lowest point.

Conditions for centrifugation are always (or should always be) quoted in terms of the average centrifugal field, i.e., the field at the midpoint of the tube, rather than as revolutions per min (rpm). Technical data supplied with centrifuge rotors will invariably include a chart for conversion between the two. It is, however, easy to calculate the *g*-value from the dimensions of the rotor and the rpm using the relationship:

$$\text{Relative centrifugal field} = 1.118 \times 10^{-5} \times n^2 \times r$$

where *n* is the number of revolutions per minute and *r* is the distance in cm from the center of rotation to the chosen point in the centrifuge tube.

4. Using a buffer-to-tissue ratio of 2:1, a quarter to a third of the tube after centrifugation will be occupied by the pellet of cell debris, depending on the particular tissue used. This pellet will, of course, contain a substantial amount of buffer, and hence, of soluble protein. Yields can be increased by resuspending the pellet in buffer and centrifuging again, but this will obviously result in dilution of the final extract, which is undesirable since water is a major contaminant that has to be disposed of at some stage. In general, it is better to sacrifice some yield of protein in the interest of keeping the volume down, unless the tissue being used is scarce, in which case re-extraction should be done.
5. With tissues such as heart the procedure given will result in an extract of sufficient clarity to proceed to the next step in purification. With such tissues as liver, kidney, and brain, this is not so, and further steps need to be taken to remove particulate matter before proceeding, particularly if the next step is to be column chromatography where particulate matter can cause serious clogging problems.

An obvious approach is to use much higher centrifugal fields for longer periods. With a 6 × 250-mL rotor, values of up to 6000*g* can be obtained, but this does not always solve the problem; use of high centrifugal fields is also impractical when large volumes are to be processed. The problem arises mainly from the release of large quantities of nucleic acids and nucleoprotein into the homogenate; this increases the viscosity, and hence, decreases the rate of sedimentation. The best way of overcoming this difficulty is to add a polyamine, such as protamine sulfate, to the partially clarified homogenate, stir for some time, and then recentrifuge. The polyamine causes aggregation of the nucleic acids and nucleoproteins, which now sediment much more easily. A concentration of 0.1% (w/v) is usually adequate to produce a clear supernatant, but trial experiments may be required

to find optimal conditions. It is also necessary, of course, to confirm that the protein of interest is not precipitated or inactivated by this procedure.

A very effective solution to the problem, but one that is useful in only a limited number of cases is heat precipitation. This requires that the protein of interest be stable to heating at 70°C for 10–15 min. Under these conditions, many proteins are denatured and form readily sedimentable aggregates along with other suspended matter. Not only is it then easy to clarify the extract, but then there will also be a substantial purification of the protein of interest. An example where this method has been successfully used is given in ref. 4. In practice, it is necessary to heat the clarified extract rapidly to the desired temperature (usually 70°C) by putting it in a large round-bottomed glass flask, immersing the flask in a water bath at least 10°C hotter than the target temperature, and swirling constantly until the target temperature is reached. After the required time at this temperature, the flask and contents are rapidly cooled under running tap water and precipitated material removed by centrifugation.

In practice, it may be possible to ignore the problem if the first step in the proposed purification scheme is to be fractional precipitation by salt or by organic solvents (*see* Chapter 13) and if the protein of interest does not precipitate in the first fraction taken. In the case of precipitation with organic solvents, addition of the solvent decreases both the density and the viscosity of the solution, and hence, sedimentation of particulate matter will occur readily in the first fraction taken. During salt fractionation, particulate material tends to aggregate and again will usually sediment in the first fraction. Clearly, if the protein of interest precipitates early, the particulate matter will be carried with it and the problem compounded.

References

- 1 Barra, D., Bossa, F., Doonan, S., Fahmy, H. M. A., Martini, F., and Huges, G. J. (1976) Large-scale purification and some properties of the mitochondrial aspartate aminotransferase from pig heart *Eur J Biochem* **64**, 519–526
- 2 Morton, R. K. (1955) Methods of extraction of enzymes from animal tissues *Methods Enzymol* **1**, 25–51
- 3 Dignam, J. D. (1990) Preparation of extracts from higher eukaryotes *Methods Enzymol* **182**, 194–203
- 4 Banks, B. E. C., Doonan, S., Lawrence, A. J., and Vernon, C. A. (1968) The molecular weight and other properties of aspartate aminotransferase from pig heart muscle *Eur J Biochem* **5**, 528–539

CHAPTER 3

Protein Extraction from Plant Tissues

Peter R. Shewry and Roger J. Fido

1. Introduction

Plant tissues contain a wide range of proteins, which vary greatly in their properties, and require specific conditions for their extraction and purification. It is therefore not possible to recommend a single protocol for extraction of all plant proteins. Plant tissues do pose specific problems that must be taken into account when developing protocols for extraction. The first is the presence of a rigid cellulosic cell wall, which must be sheared to release the cell contents. The second is the presence of specific contaminating compounds that may result in protein degradation or modification and, where the protein of interest is an enzyme, subsequent loss of catalytic activity. Such compounds include phenolics and a range of proteases. Although it is sometimes possible to avoid these problems or partially control them by using a specific tissue or plant species, in other cases (for example, enzymes involved in secondary product synthesis), this is not possible and the biochemist must find ways to remove or inactivate active contaminants. The removal of phenolics is dealt with elsewhere in this volume (Chapter 8). Because many plant proteases are of the serine type, it is often convenient to include the serine protease inhibitor phenylmethylsulfonylfluoride (PMSF) in extraction buffers on a routine basis (*see* Chapter 9 for a general discussion of protease inhibition).

Animals have many highly specialized tissues (e.g., liver, muscle, brain) that are rich sources of specific enzymes, thus facilitating their purification. This is not usually the case with plant enzymes, which may

From *Methods in Molecular Biology*, Vol. 59 *Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

be present at low levels in highly complex protein mixtures. An exception to this is storage organs, such as seeds, tubers, and tap roots. These organs contain high levels of specific proteins whose role is to act as a store of nitrogen, sulfur, and carbon. These storage proteins are among the most widely studied proteins of plant origin, because of their abundance and ease of purification, and their economic and nutritional importance as food, feed for livestock, and raw material in the food and other industries. Indeed, seed proteins were among the earliest of all proteins to be studied in detail, with wheat gluten having been isolated in 1745 (1), the Brazil nut globulin, edestin, crystallized in 1859 (2) and a range of globulin storage proteins being subjected to ultracentrifugation analysis by Danielsson in 1949 (3).

Comparative studies of the extraction and solubility of plant proteins also formed the basis for the first systematic attempt to classify proteins. T. B. Osborne, working at the Carnegie Institute of Washington between about 1880 and 1930, compared and characterized proteins from a range of plant sources, including the major storage proteins of cereal and legume seeds (4). He defined four groups that were extracted sequentially in water (albumins), dilute salt solutions (globulins), alcohol/water mixtures (prolamins), and dilute acid or alkali (glutelins). These "Osborne groups" still form the basis for studies of seed storage proteins, whereas the terms albumin and globulin have become accepted into the general vocabulary of protein chemists.

Three detailed protein extraction protocols are given. The first two are for the extraction of the enzymically active proteins, ribulose 1,5-bisphosphate carboxylase/oxygenase (Rubisco) (EC 4.1.1.39) and nitrate reductase (EC 1.6.6.1), from vegetative tissues. Rubisco is a hexadecameric protein (eight subunits of approx 50–60 kDa and eight subunits of 12–20 kDa) with an M_r of 500,000, which catalyzes the fixation of carbon in the chloroplast stroma. It often represents more than 50% of the total chloroplast protein and is recognized as the most abundant protein in the world. In contrast, the complex enzyme nitrate reductase (NR), which has a M_r of approx 200,000, is present in plant tissues at <5 mg/kg fresh weight (5). This low abundance, combined with susceptibility to proteolysis and loss of functional prosthetic groups during extraction and purification, often leads to a very low recovery of the enzyme. The third protocol is a specialized procedure for the extraction of seed proteins from cereals, based on the classical Osborne fractionation.

Finally, two rapid methods are described for the extraction of leaf and seed proteins for SDS-PAGE analysis. These are suitable for monitoring the expression of transgenes in engineered plants.

2. Materials

1. Buffer A (Rubisco). 20 mM Tris-HCl, pH 8.0, 10 mM NaHCO₃, 10 mM MgCl₂, 1 mM EDTA, 5 mM DTT, 0.002% Hibitane, and 1% insoluble PVP.
2. Buffer B (Nitrate Reductase [NR]): 0.5M Tris-HCl, pH 8.6, 1 mM EDTA, 5 μ M Na₂MoO₄, 25 μ M FAD, 5 mM PMSF, 5 μ g/mL pepstatin, 10 μ M antipain, and 3% BSA.
3. Buffer C: 0.0625M Tris-HCl, pH 6.8, 2% (w/v) SDS, 5% (v/v) 2-mercaptoethanol or 1.5% (w/v) dithiothreitol, 10% (w/v) glycerol, 0.002% (w/v) bromophenol blue.
4. Buffer D: 0.1M Tris-HCl, pH 8.0, 0.01M MgCl₂, 18% (w/v) sucrose, 40 mM 2-mercaptoethanol.

3. Methods

3.1. Extraction of Enzymically Active Preparations from Leaf Tissues

All procedures are carried out at 0–4°C with precooled reagents and apparatus. Tissue can be used fresh, or after rapid freezing using liquid nitrogen, and stored at –20 to –80°C or under liquid nitrogen. Tissue homogenization can be made in pestle and mortar or a ground glass homogenizer (for small volumes), or a Waring blender or Polytron for larger initial weights.

The method for the extraction of Rubisco from wheat leaves is taken from Keys and Parry (6). It is reported that the extraction procedure and extraction buffers used are important in affecting the initial rate and total activities of the enzyme (*see* Note 1). It is also important for initial activity measurements to maintain the extract at a temperature of 2°C.

1. Cut 3-wk-old wheat leaves into 1-cm lengths and homogenize in an ice-cold buffer using a ratio of 6 l.
2. Filter the homogenate through four layers of muslin, and then add sufficient solid (NH₄)₂SO₄ to give 35% saturation (*see* Chapter 13).
3. After 20 min, centrifuge the suspension at 20,000g for 15 min. Discard the pellet.
4. Add further solid (NH₄)₂SO₄ to give 55% saturation. After centrifugation dissolve the pellet in 20 mM Tris-HCl containing 1 mM DTT, 1 mM MgCl₂, and 0.002% Hibitane (*see* Note 2) at pH 8.0. After clarification, the Rubisco can then be fractionated by sucrose-density centrifugation.

The method for NR extraction, using a complex extraction buffer (*see* buffer B), is taken from Somers et al. (7), who attempted to identify whether barley NR was regulated by enzyme synthesis and degradation or by an activation–inactivation mechanism.

1. Both root and shoot tissues were excised at different ages (days), weighed, frozen in liquid nitrogen and stored at -80°C .
2. Pulverize the frozen tissue in a pestle and mortar under liquid nitrogen. Extract with 1 mL/g fresh weight of buffer B (*see* Note 3)
3. Filter the homogenate through two layers of cheesecloth, and centrifuge at 30,000g to clarify. The supernatant can be used directly for enzyme activity measurements (*see* Note 4).

3.2. Extraction of Cereal Seed Proteins Using a Modified Osborne Procedure

The procedure is based on Shewry et al. (8) Air-dry grain ($\approx 14\%$ water) is milled to pass a 0.5-mm mesh sieve. The meal is then extracted by stirring (*see* Note 5) with the following series of solvents: 10 mL of solvent is used/g meal and each extraction is for 1 h. Extractions are carried out at 20°C and repeated as stated.

1. Water-saturated 1-butanol (twice) to remove lipids.
2. 0.5M NaCl to extract salt-soluble proteins (albumins and globulins) and nonprotein components (twice) (*see* Note 6)
3. Distilled water to remove residual NaCl.
4. 50% (v/v) 1-Propanol containing 2% (v/v) 2-mercaptoethanol (or 1% [w/v] dithiothreitol) and 1% (v/v) acetic acid (three times) to extract prolamins (*see* Note 7).
5. 0.05M borate buffer, pH 10, containing 1% (v/v) 2-mercaptoethanol and 1% (w/v) sodium dodecyl sulfate to extract residual proteins (glutelins) (*see* Note 8).

The supernatants are separated by centrifugation (20 min at 10,000g) and treated as follows:

6. Supernatants 2 and 3 are combined and dialyzed against several changes of distilled water at 4°C over 48 h. Centrifugation removes the globulins, allowing the soluble albumins to be recovered by lyophilization.
7. Supernatants from 4 are combined, and the prolamins recovered after precipitation, either by dialysis against distilled water or addition of 2 vol of 1.5M NaCl, followed by standing overnight at 4°C
8. Supernatants from 5 are combined and the glutelins recovered by dialysis against distilled water at 4°C followed by lyophilization (*see* Note 9). SDS can be removed from the protein using standard procedures

3.3. Extraction of Proteins for SDS-PAGE Analysis

The methods described in Sections 3.1. and 3.2. are suitable for the bulk extraction of proteins for purification of individual components. However, in some situations, for example, analysis of transgenic plants or studies of seed protein genetics, it is advantageous to extract total proteins for direct analysis by SDS-PAGE. The following methods are specially designed for this purpose.

3.3.1. Extraction of Leaf Tissues

The method, based on Nelson et al. (9), gives good results with chlorophyllous tissues.

1. Freeze the tissue in liquid N₂.
2. Grind for about 30 s in a mortar with 3 mL of buffer D/g of tissue (*see* Note 10)
3. Filter through muslin, and centrifuge for 15 min in a microfuge.
4. Dilute to about 2 mg protein/mL, ensuring that the final solution contains about 2% (w/v) SDS, 0.002% (w/v) bromophenol blue, and at least 6% (w/v) sucrose (*see* Note 11).
5. Separate aliquots by SDS-PAGE.

3.3.2. Extraction of Seed Proteins

1. Grind in a mortar with 25 μ L of buffer C/mg meal.
2. Transfer to an Eppendorf tube, and allow to stand for 2 h.
3. Suspend in a boiling water bath for 2 min.
4. Allow to cool, and then spin in a microfuge.
5. Separate 10–20- μ L aliquots by SDS-PAGE.

4. Notes

4.1. Extraction of an Enzymically Active Preparation from Leaf Tissue

1. A wide range of buffers can be used, depending on the pH range required for optimal enzyme activity and the preference (or prejudice) of the operator. However, Tris is very widely used.
2. A range of specific additions can be made in order to help preserve the activity of the enzyme under consideration. For example, with NR, it is advantageous to add a flavin compound (i.e., FAD) in order to maintain the endogenous levels needed for catalytic activity. The inclusion of both CO₂ and Mg²⁺ ions in the extraction buffer of Rubisco has been reported to be necessary (10). There is no single simple method to guarantee activity—the operator should consult published protocols for the extraction of related enzymes and be prepared to carry out exploratory extractions using different buffer compositions.

- a. 1 mM Dithiothreitol or 10 mM 2-mercaptoethanol to preserve sulphydryl groups.
 - b. 1 mM EDTA to chelate metals, especially with phosphate buffers that commonly contain inhibitory concentrations of ferrous ions.
 - c. 50 mM Sodium fluoride to inhibit phosphatases that inactivate phosphoenzymes.
 - d. 25 g/kg Fresh weight of PVP. This is an insoluble compound that binds phenolic compounds. It forms a slurry and can be removed by centrifugation.
 - e. 0.1 mM PMSF to inhibit serine proteinases. This is readily dissolved in a small volume of 1-propanol prior to mixing with the buffer **It is highly toxic.**
 - f. Glycerol (up to 30%) or other organic alcohols (ethylene glycol, mannitol) may help to stabilize some highly labile enzymes.
 - g. The addition of exogenous proteins, e.g., casein and BSA, has been used to stabilize enzymes by preventing hydrolysis owing to protease activity
 - h. Antibacterial agents, such as Hibitane, can also be added.
3. Similar methods can be used to extract enzymes from seed tissues, either by direct homogenization or after milling. Lipid-rich tissues can either be defatted with cold (4°C) acetone or an acetone powder can be made (11). **Extreme care should be taken due to the low flash point of acetone: Operations should be carried out in a fume cupboard and electrical sparks avoided.**
4. The supernatant may be concentrated by precipitation with $(\text{NH}_4)_2\text{SO}_4$ (12) and assayed directly after desalting on a column of Sephadex G25.

4.2. Extraction of Cereal Seed Proteins Using a Modified Osborne Procedure

5. Extraction may be carried out by stirring magnetically or with a paddle—the mechanical grinding that occurs may assist extraction.
6. It may be advantageous to extract the salt-soluble proteins at 4°C and include 1.0 mM PMSF (*see* Note 2) to minimize proteolysis.
7. It is sometimes of interest to extract the prolamins in two fractions. Extraction twice with 50% (v/v) 1-propanol gives monomeric prolamins and alcohol-soluble disulfide-stabilized polymers, whereas subsequent extraction twice with 50% (v/v) 1-propanol with 2% (v/v) 2-mercaptoethanol and 1% (v/v) acetic acid gives reduced subunits derived from alcohol-insoluble disulfide-bonded polymers.
8. It is usual to determine the amounts of extracted proteins by Kjeldahl N analysis of aliquots removed from the supernatants. The values can then be multiplied by a factor of 5.7 for prolamins or 6.25 for other fractions to give the amount of protein.
9. SDS-PAGE is used to monitor the compositions of the fractions.

4.3. Extraction of Proteins for SDS-PAGE Analysis

10. Addition of 2% (w/v) SDS to buffer D allows the extraction of membrane and other insoluble proteins.
11. If required, soluble and insoluble proteins can be extracted in two sequential fractions. Soluble proteins are initially extracted in buffer D (3 mL/g) and insoluble proteins by re-extracting the pellet with 0.05 vol (relative to the original homogenate) of 2% (w/v) SDS, 6% (w/v) sucrose, and 40 mM 2-mercaptoethanol.

References

- 1 Beccari, J. B. (1745) *De Frumento De Bononiensi Scientiarum et Artium atque Academia Commentarii, Tomi Secundi*. Bononia
- 2 Matschke, O. (1859) Ueber den Bau und die Bestandtheile der Kleberblaschen in Bertholletia, deren Entwicklung in Ricinus, nebst einigen Bemerkungen über Amylonblaschen. *Botanische Zeitung* **17**, 409–447
- 3 Danielsson, C. E. (1949) Seed globulins of the Gramineae and Leguminosae. *Biochem J* **44**, 387–400
- 4 Osborne, T. B. (1924) *The Vegetable Proteins*. Longmans, Green, London
- 5 Wray, J. L. and Fido, R. J. (1990) Nitrate and nitrite reductase, in *Methods in Plant Biochemistry* vol 3 (Lea, P. J., ed.), Academic, New York, pp 241–256
- 6 Keys, A. J. and Parry, M. A. J. (1990) Ribulose biphosphate carboxylase/oxygenase and carbonic anhydrase, in *Methods in Plant Biochemistry*, vol 3 (Lea, P. J., ed.), Academic, New York, pp. 1–14.
- 7 Somers, D. A., Kuo, T.-M., Kleinhofs, A., Warner, R. L., and Oaks, A. (1983) Synthesis and degradation of barley nitrate reductase. *Plant Physiol* **72**, 949–952.
- 8 Shewry, P. R., Franklin, J., Parmar, S., Smith, S. J., and Mifflin, B. J. (1983) The effects of sulfur starvation on the amino acid and protein compositions of barley grain. *J. Cereals Sci* **1**, 21–31
- 9 Nelson, T., Harpster, M. H., Mayfield, S. P., and Taylor, W. C. (1984) Light regulated gene-expression during maize leaf development. *J. Cell Biol* **98**, 558–564.
- 10 Servaites, J. C., Parry, M. A. J., Gutteridge, S., and Keys, A. J. (1986) Species variation in the predawn inhibition of ribulose-1,5-bisphosphate carboxylase/oxygenase. *Plant Physiol* **82**, 1161–1163
- 11 Nason, A. (1955) Extraction of soluble enzymes from higher plants. *Methods Enzymol* **1**, 62,63.
- 12 Green, A. A. and Hughes, W. L. (1955) Protein fractionation on the basis of solubility in aqueous solutions of salts and organic solvents. *Methods Enzymol* **1**, 67–90

CHAPTER 4

Extraction of Recombinant Protein from Bacteria

D. Margaret Worrall

1. Introduction

The advent of recombinant DNA technology and the overexpression of heterologous proteins in bacteria have posed some unique problems not previously encountered in extraction of bacterial proteins. Some of this technology has enabled secretion of recombinant proteins by bacteria into the media, thereby eliminating the need to lyse the cells. However, most situations still require lysis of the bacterial cell wall in order to extract the recombinant protein product. A number of methods based on enzymatic and mechanical means are available for breaking open the bacterial cell wall, and the choice will depend on scale of process (1) Enzymatic methods use the activity of lysozyme, which cleaves the glucosidic linkages in the bacterial cell-wall polysaccharide. The inner cytoplasmic membrane can then be disrupted easily by detergents, osmotic pressure, or mechanical methods.

Overexpression of recombinant proteins from strong promoters on multiple-copy plasmids can result in expression levels of up to 40% of the total cell protein. However, in most cases, this results in the formation of insoluble protein aggregates known as inclusion bodies, which require further extraction (2). Inclusion bodies are cytoplasmic granules seen as phase bright under the light microscope, and can contain most or all of the protein of interest. Scanning electron micrographs of *E. coli* containing inclusion bodies and isolated inclusions are shown in Fig. 1.

From *Methods in Molecular Biology, Vol 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

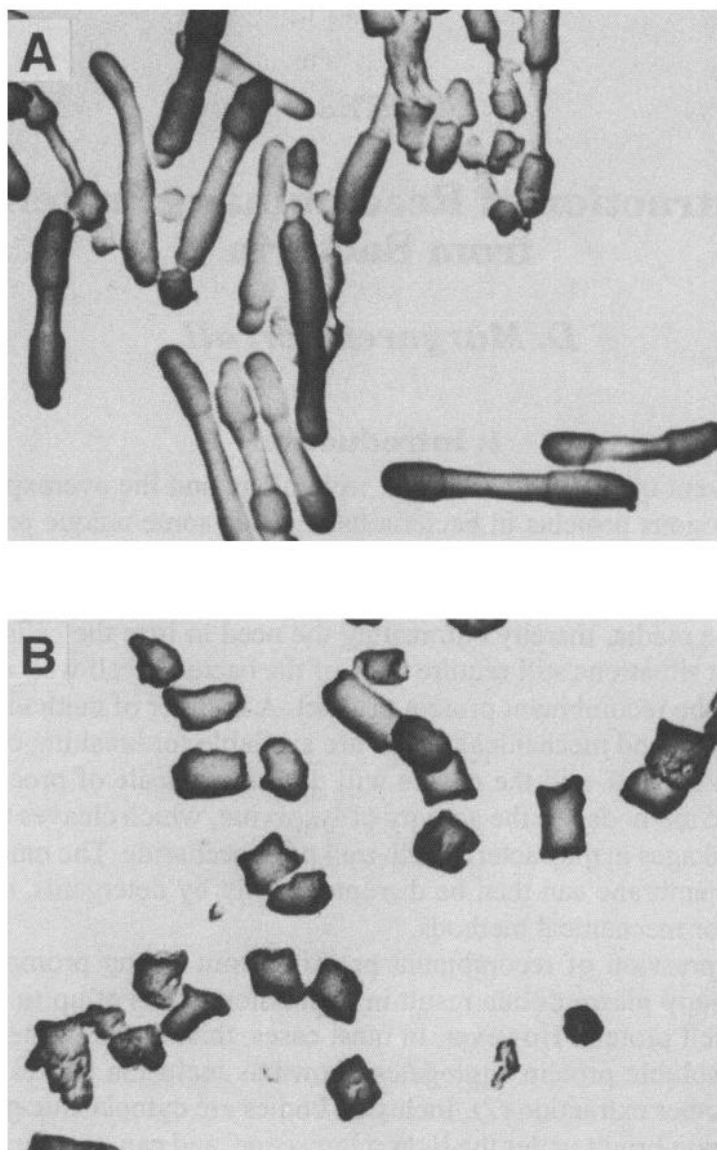


Fig. 1. (A) Scanning electron micrograph of *E. coli* containing inclusion bodies. The preparative techniques for electron micrographs have caused the cells to shrink. However, regions of the cells containing rigid inclusion bodies have not been able to shrink, showing clearly the outline of the inclusion bodies. (B) Scanning EM of isolated washed inclusion bodies. Again, it can be seen that the inclusions retain a rigid cylindrical shape.

Inclusion bodies were first reported by Williams et al. (3) on overexpression of proinsulin in *E. coli*. It is not known exactly how they are formed, but it is thought that the protein is partially or incorrectly folded. Formation of inclusion bodies is not only found on overexpression of foreign eukaryotic proteins, but is also seen on overexpression of bacterial proteins that are normally soluble (4).

A number of methods have been described for increasing the soluble fraction of the protein. Growth temperature, media composition, and host strain have all been found to affect the partitioning of the overexpressed protein between the cytosol and inclusion body fractions (5). Fusion proteins with a highly soluble protein, such as glutathione-S-transferase or thioredoxin, can also increase solubility of the protein of interest (6, 7).

The advantage of inclusion bodies is that they generally allow greater levels of expression, and they can be easily separated from a large proportion of bacterial cytoplasmic proteins by centrifugation giving an effective purification step. Some contaminating proteins are always present that may be associated with inclusion body formation.

The major disadvantage of inclusion bodies is that extraction of the protein of interest generally requires the use of denaturing agents. This can cause problems where native folded protein is required, since refolding methods are not always 100% effective and may be difficult to scale-up.

Some inclusion bodies can be solubilized by extremes of pH and temperature, but most require strong denaturing agents. Certain proteins, such as DNase 1, can be refolded after solubilization with SDS, but detergents are difficult to remove from most proteins and can interfere with subsequent refolding. The most commonly used solubilizing agents are water-soluble chaotropic agents, such as urea and guanidinium hydrochloride, which are more compatible with protein refolding. Most inclusions will be soluble in 8M urea, and a reducing agent, such as DTT, is generally required in order to prevent the formation of disulfide bonds between aggregates or denatured polypeptide chains.

2. Materials

All reagents are analytical grade.

1. Prepare a stock solution of 100 mM PMSF in isopropanol, and store at -20°C . Add PMSF to buffers just before use. (Note—PMSF is a hazardous chemical and should be treated with caution.)

2. Lysis buffer: 50 mM Tris-HCl, pH 8.0, 1 mM EDTA, 50 mM NaCl, 1 mM PMSF (*see* Note 1).
3. Hen egg lysozyme (Sigma, Poole, Dorset, UK): 10 mg/mL stock solution
4. DNase 1 (Boehringer Mannheim, Lewes, East Sussex, UK): 1 mg/mL stock solution (*see* Note 2).
5. Sodium deoxycholate (Sigma): 20 mg/mL stock solution.
6. Solubilization buffer: 50 mM Tris-HCl, pH 8.0, 8M urea, 1 mM DTT.

Urea solutions should be used within 1 wk of preparation and stored at 4°C, in order to reduce the formation of cyanate ions, which can react with protein amino groups forming carbamylated derivatives.

3. Methods

3.1. *Enzymatic Lysis of E. coli*

1. Harvest the bacterial cells by centrifugation at 1000g for 15 min at 4°C, and pour off the supernatant. Weigh the wet pellet. This is easiest to do if you have preweighed the centrifuge tubes which can then be deducted from the total weight.
2. Add approx 3 mL of lysis buffer for each wet gram of bacterial cell pellet and resuspend. Add lysozyme to a concentration of 300 µg/mL and stir the suspension for 30 min at 4°C (*see* Notes 3 and 4).
3. Add deoxycholate to a concentration of 1 mg/mL while stirring.
4. Place at room temperature, and add DNase1 to a concentration of 10 mg/mL and MgCl₂ to 10 mM. Stir suspension for a further 15 min to remove the viscous nucleic acid (*see* Note 2).
5. Centrifuge the suspension at 10,000g for 15 min at 4°C. Resuspend the pellet in lysis buffer to the same volume as the supernatant, and analyze aliquots of both for the protein of interest on SDS-PAGE. If the bulk of a normally soluble protein is found in the insoluble pellet fraction, then inclusion bodies are likely to have formed.

3.2. *Washing of Inclusion Bodies*

Washing of inclusion bodies prior to solubilization can remove further contaminant proteins, and using solutions other than water or buffer can increase the purification obtained (8). It is advisable to carry out a small scale trial to optimise the buffer and to ensure that the protein of interest is not solubilized.

1. Centrifuge 200-µL aliquots of the resuspended cell pellet in microfuge tubes at 12,000g for 10 min at 4°C. Resuspend the pellets in a range of test solutions. Lysis buffer containing 1, 2, 3, and 4M urea and 0.5% Triton X-100 are suggested. Mix and incubate for 10 min at room temperature. Centrifuge as before in a microfuge, and resuspend pellets in 200 µL H₂O.

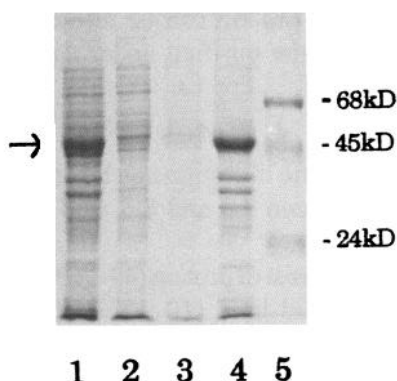


Fig. 2. Washing and solubilization of inclusion bodies containing recombinant plasminogen activator inhibitor-2 (45-kDa protein indicated). Lane 1: insoluble *E. coli* fraction; lane 2: first 2M urea wash; lane 3: second 2M urea wash; lane 4: inclusion body pellet solubilized in 8M urea.

2. Take equal volumes of the supernatant and the resuspended pellet and add to SDS boiling buffer. Analyze samples for the protein of interest on SDS-PAGE. The best washing buffer will contain the most contaminant proteins and little or none of the protein of interest.
3. Scale this procedure up, and wash the inclusion bodies twice with the optimum buffer. An example of the purification achieved on washing of inclusion bodies of plasminogen activator inhibitor-2 (PAI-2) is shown in Fig. 2.

3.3. Solubilization of Recombinant Protein from Inclusion Bodies

It is also important to optimize the solubilization solution, since a number of factors will affect solubility depending on the nature of the protein of interest. These include the nature and strength of the solubilization agent, the temperature and time taken to obtain efficient solubilization, protein concentration, and the presence or absence of reducing agents.

1. Resuspend the washed inclusion bodies in the solubilization buffer. Stir this suspension for 1 h at room temperature to ensure complete solubilization.
2. Centrifuge the solution at 20,000g for 20 min at 4°C to remove any remaining insoluble material. Check this pellet for the protein of interest on SDS-PAGE. If a substantial porportion of the protein has remained insoluble, then increase the incubation time with the solubilization buffer or try a different agent to solubilize the inclusions.

3. The extracted recombinant protein can be refolded from the urea at this stage (*see* Note 6), or may be purified under denaturing conditions (*see* Notes 5 and 7).

4. Notes

1. The lysis buffer for enzymatic digestion can be critical. Hen egg lysozyme has a pH optimum of between 7.0 and 8.6, and also works best in ionic strength of 0.05M.
2. Bacterial extracts roughly consist of protein (40–70%), nucleic acid (10–30%), polysaccharide (2–10%), and lipid (10–15%). The nucleic acid fraction can often cause high viscosity. In addition to DNase treatment detailed in Section 3.1., step 4, this nucleic acid can also be removed from soluble protein solutions by precipitation with positively charged compounds, such as polyethylenamine (9). Precipitation methods should not be used with inclusion body preparations, since the precipitate will cocentrifuge with the inclusions.
3. If extracting soluble proteins from bacteria, it may be desirable to use more gentle methods of enzymatic lysis where sucrose buffers are used (1). Inclusion of extra protease inhibitors, such as aprotinin or leupeptin, in the lysis buffer may also be advantageous if proteolysis of the target protein is occurring. This is generally not necessary when the protein is packaged in inclusion bodies, but inhibitors may be required in the solubilization and refolding buffers, since proteins in semifolded states are more susceptible to proteolysis.
4. Mechanical lysis is also effective in disrupting the bacterial outer cell wall, and a number of mechanical methods are available. Sonication is suitable for smaller scale purifications, but generation of heat during sonication can be difficult to control and may cause denaturation of proteins. For larger scale processing, the French press and the Mantin Gaulin press are most commonly used. These devices lyse the cells by applying pressure to the cell suspension followed by a release of pressure, which causes a liquid shear and thus cell disruption. Multiple passes of the cells through the presses are generally necessary to obtain adequate lysis (1).
5. It is often desirable to carry out some purification of the protein under denaturing conditions before refolding. Urea solutions are compatible with ion-exchange chromatography, metal ion-affinity chromatography, and gel filtration. Guanidium hydrochloride is compatible with gel filtration chromatography but owing to its charge, it cannot be used in ion-exchange purification steps.
6. Many methods have been described for refolding of proteins from urea (for reviews, *see* refs. 10 and 11). The most common techniques use dilution and dialysis to reduce the urea or guanidinium concentration gradually,

and the presence of thiol reagents is generally important for correct formation of disulfide bonds.

7. Proteins used for animal immunization purposes often do not require refolding unless antibodies to tertiary epitopes are required. It is also possible to use inclusion bodies directly injected, since particulate antigens are highly immunogenic. Sonication of the inclusions into smaller particles is recommended prior to injection (12).

References

- 1 Cull, M and McHenry, C. S. (1990) Preparation of extracts from prokaryotes. *Methods Enzymol* **182**, 147–153.
- 2 Kane, J. F and Hartley, D. L. (1989) Formation of recombinant protein inclusions in *Escherichia coli* *Tibtech* **6**, 95–101.
3. Williams, D. C., Van Frank, R. M., Muth, W. L., and Burnett, J. P. (1982) Cytoplasmic inclusion bodies in *E. coli* producing biosynthetic human insulin proteins *Science* **215**, 687–689
- 4 Gribbskov, M and Burgess, R. R. (1983) Overexpression and purification of the sigma subunit of *Escherichia coli* RNA polymerase. *Gene* **26**, 109–118
- 5 Schein, C. H and Noteborn, M. H. M. (1988) Formation of soluble recombinant proteins is favoured by lower growth temperature. *BioTechnology* **6**, 291–294.
- 6 Smith, D. B. and Johnson, K. S. (1988) Single step purification of polypeptides expressed in *E. coli* as fusions with glutathione-S-transferase *Gene* **67**, 31–40
- 7 LaVallie, E. R., DiBlasio, E. A., Kovacic, S., Grant, K. L., Schendel, P. F., and McCoy, J. M. (1993) A thioredoxin gene fusion expression system that circumvents inclusion body formation in the *E. coli* cytoplasm. *BioTechnology* **11**, 187–193
8. Schoner, R. G., Ellis, L. F., and Schoner, B. E. (1985) Isolation and purification of protein granules from *Escherichia coli*, cells overproducing bovine growth hormone *BioTechnology* **3**, 151–154.
9. Burgess, R. and Jendrisak, J. (1975) A procedure for the rapid large-scale purification of *E. coli* DNA dependant RNA polymerase involving Polymyxin P precipitation and DNA cellulose chromatography *Biochemistry* **14**, 4634–4645
- 10 Marston, F. A. O. (1986) The purification of eukaryotic polypeptides synthesized in *Escherichia coli* *Biochem J* **240**, 1–12
- 11 Kohno, T., Carmichael, D. F., Sommer, A., and Thompson, R. C. (1990) Refolding of recombinant proteins. *Methods Enzymol* **185**, 187–195.
12. Harlow, E. and Lane, D. (1988) *Antibodies. A Laboratory Manual* Cold Spring Harbor Laboratory, Cold Spring Harbor, New York pp 88–91.

CHAPTER 5

Protein Extraction from Fungi

Paul Bridge

1. Introduction

In order to study proteins from yeasts and filamentous fungi, it is important to consider a number of basic features of the organisms. First, filamentous fungi undergo a growth cycle that includes differentiation and compartmentalization. In addition, both the filamentous fungi and yeasts will age during growth, and older cultures will undergo autolysis. As a result, particular proteins may only be associated with one part of the growth cycle, such as sporulation or autolysis, and this must be taken into account in determining growth conditions and sampling times.

Second, many of the enzymes produced during the growth period are sequential and may either be subject to significant repression or require induction by a substrate or substrate component. Examples of this include the requirement for chitin or chitin-like components to induce chitinases (1), and the repression of some fungal proteases by glucose (2).

Third, fungi possess rigid cell walls and complex cell-wall/membrane systems (3). It is therefore important to ascertain the potential location of proteins prior to their extraction, since cell-wall-associated and extracellular proteins will be lost during intracellular extractions. An example of this is the utilization of many of the traditional fungal nutrients, such as cellulose and lignin, where significant levels of extracellular enzymes will be produced. Most of these extracellular enzymes can be produced in sufficient concentrations for them to be purified and characterized directly from the spent growth medium (4,5).

A simple growth and extraction procedure is described here. This is a standard regime that will allow the extraction of intracellular proteins

From *Methods in Molecular Biology*, Vol 59 *Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

from a wide range of filamentous fungi, and has been used successfully with many fungal genera, including *Fusarium*, *Ganoderma*, *Aspergillus*, *Colletotrichum*, *Beauveria*, *Phoma*, *Verticillium*, and *Metarhizium* (6). The method has not been optimized toward any particular fungal group, and has proven suitable for filamentous ascomycetes and basidiomycetes as well as yeasts (7–9). The major variation that will be needed for different fungal groups is the growth medium and the length of the growth period (see Notes 1 and 2). Although a crude method, extracts produced in this way retain sufficient integrity and activity for enzyme assays and isoenzyme electrophoresis. An additional feature of this method is that the spent culture fluid may be retained for the detection of extracellular enzymes. Initially, this will only contain a small number of glucose-independent enzymes, but as the culture grows and the free glucose concentration decreases, further enzymes can be detected or extracted (7,10)

2. Materials

Fungal growth media and buffers should be sterilized prior to use. Growth media and buffers can routinely be sterilized at 10 psi for 10 min in a benchtop autoclave. Although the materials listed here are unaffected, it should be remembered that in complex media and buffers, individual components may break down or react during autoclaving, and so may need to be individually filter sterilized.

- 1 Malt extract agar (MEA). 20 g Malt extract (Oxoid, Basingstoke, UK), 1 g peptone (Oxoid; Bacteriological), 20 g glucose, 15 g agar, 1 L distilled water (11)
- 2 Glucose yeast medium (GYM). 1 g $\text{NH}_4\text{H}_2\text{PO}_4$, 0.2 g KCl, 0.2 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 10 g glucose, 1 mL 0.5% aqueous $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 1 mL 1% aqueous $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, distilled water to 1 L (7)
- 3 Pectin broth: 0.9 g $\text{NH}_4\text{H}_2\text{PO}_4$, 2 g $(\text{NH}_4)_2\text{HPO}_4$, 0.1 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.5 g KCl, 10 g citrus pectin (Sigma, Poole, UK), 1 L distilled water.
- 4 Tris-glycine buffer: 3 g Trizma (Sigma, Poole, UK), 14.4 g glycine, 1 L deionized water, pH 8.3.
- 5 Pectinase gel: 0.2 g Citrus pectin, 10 g acrylamide, 0.25 g methylene-bis-acrylamide, 0.1 mL TEMED (*N,N,N',N'*-tetramethylethylenediamine), 0.1 g ammonium persulfate, 100 mL gel buffer.
- 6 Gel buffer: 0.525 g Citric acid monohydrate, 4.598 g Tris, 1 L deionized water.
7. Electrode buffer: 7.22 g Boric acid, 15.75 g sodium tetraborate decahydrate, 1 L deionized water
- 8 0.1M Malic acid.

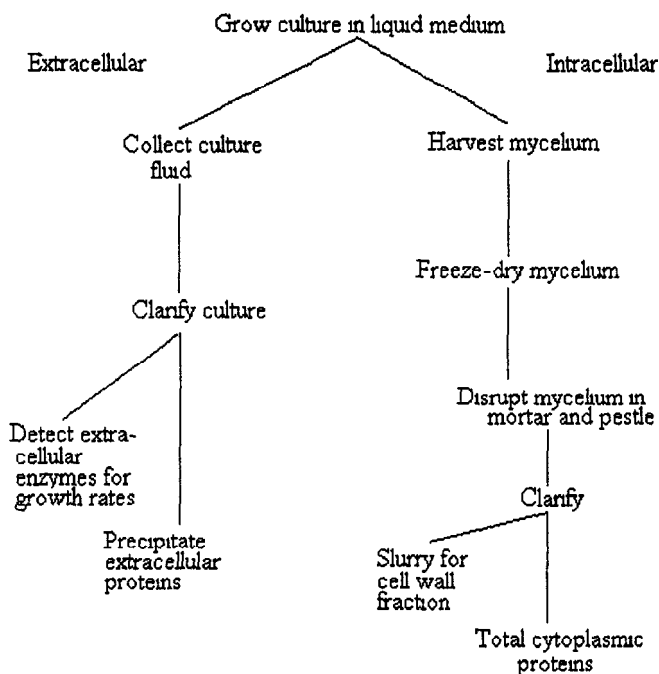


Fig. 1. Schematic diagram of the complete protocol.

- 9 0.015% Aqueous ruthenium red.
10. Sterile deionized water.

3. Methods

The method presented here will enable the extraction of intracellular proteins and electrophoresis of extracellular enzymes from filamentous fungi and yeasts (Fig. 1). The Notes section details further considerations that may be needed for specific organisms or extractions

3.1. Extraction of Intracellular Proteins from *Metarhizium*

The following protocol describes the extraction of aqueous intracellular proteins from the filamentous fungus *Metarhizium anisopliae*. Further details regarding growth and extraction conditions for other filamentous fungi are given in the Notes 2–7.

1. Grow *Metarhizium* culture on malt extract agar for 7 d at 25–28°C.
2. Remove a plug (approx 0.5-cm diameter) of culture from the agar plate with a flamed cork borer or scalpel. Cut the plug into at least 10 smaller

pieces with a flamed scalpel and inoculate into 10-mL sterile GYM in a 28-mL Universal bottle or small flask (*see* Note 3).

3. The 10-mL culture is a starter culture for the subsequent main growth period and is required to ensure an actively growing inoculum. Incubate the starter culture at 25–28°C in an incubator, with shaking, for 60–72 h.
4. Aseptically transfer the starter culture into a 250-mL conical flask containing 60 mL of fresh sterile GYM. Replace the flask in the incubator and continue incubation with shaking for a further 60–72 h (*see* Note 4).
5. Harvest the mycelium from the flask by vacuum-assisted filtration. A Buchner funnel containing Whatman No. 3 filter paper is suitable for this purpose (*see* Note 5). Take care to ensure that any aerosols that may be formed from the filtration or the vacuum pump are minimized and contained.
6. Wash the mycelium once in the Buchner funnel with sterile deionized water, and transfer the harvested mycelium from the filter paper to a plastic Petri dish with a flamed or alcohol-sterilized spatula.
7. Freeze the mycelium at –20°C for storage prior to extraction.
8. Prepare the mycelium for extraction by freeze-drying for 24 h. This is one of several possible methods of obtaining cell breakage (*see* Notes 6 and 7).
9. Disrupt the freeze-dried mycelium by briefly grinding it in a mortar and pestle (*see* Note 6). The ground mycelium should be collected in sterile 1.5-mL microcentrifuge tubes in approx 500-mg amounts. This is roughly equivalent to the conical portion of the tube. These tubes of ground mycelium can be stored at –20°C for at least 3 mo.
10. Rehydrate 500 mg of ground mycelium in 1 mL of Tris-glycine buffer (*see* Note 8). This will need mixing with a micropipet tip or Pasteur pipet to form an even slurry.
11. Clarify the slurry by centrifugation at 12,500g for 40 min at 4°C. After centrifugation, collect the supernatant into another sterile microcentrifuge tube (*see* Note 9).
12. The collected supernatant will contain the total cytoplasmic proteins. The samples as prepared here typically contain 15–100 mg/mL protein (*see* Note 10). The extracts can be used directly in gel electrophoresis or column chromatography. Alternatively, standard precipitation techniques can be used to purify specific proteins further (*see* Note 9).

3.2. Electrophoresis of Extracellular Enzymes

The following method describes the induction and electrophoresis of extracellular pectinases. Pectin is used, both in the production medium and in the polyacrylamide gel, to induce the enzymes and to avoid the necessity for subsequent overlays for staining. The method given here was originally described by Cruickshank and Wade (4), and is given here as in

Paterson and Bridge (6). The method has successfully been used without modifications with *Fusarium*, *Penicillium*, *Colletotrichum*, *Ganoderma*, and *Rhizoctonia*. A horizontal gel system is required, and self-adhesive vinyl tape (Dymo) is used to produce wells in a simple gel mold. Sephadex is used in the sample wells to make more well-defined enzyme patterns (see Note 11). Bromophenol blue is not used to measure the buffer front, since it can be seen adequately without the dye. This type of pectin/acrylamide gel gives enzyme patterns that are asymmetrical, unlike conventional isozyme bands. This is probably because of more definite chromatographic adsorption effects from charged pectin molecules, which are dissolved in the gel, i.e., the protein molecules are interacting with the gel.

1. Inoculate cultures into 2 mL pectin broth (PB) in a 5 mL Bijoux bottle.
2. When the cultures have grown sufficiently (usually 7 d), remove 200 μ L of the culture fluid, and suspend in superfine (G-25) Sephadex at a concentration of 20 mg/200 μ L.
3. Prepare a pectin-acrylamide gel for horizontal gel electrophoresis (see Note 11). Use three pieces of Dymo tape (2.5×5.0 mm) to form a well with a volume of 20 μ L in the center of a gel mold. Similar pieces of tape are placed 1.25 cm apart across the mold to form the other wells.
4. Set up the horizontal electrophoresis with anode and cathode buffer compartments containing gel buffer. Use filter paper wicks to form the "contact" between the buffer and the gel.
5. Place the negative electrode 3.5 cm below the sample wells to allow adequate migration of any positively charged pectinase component.
6. After electrophoresis, submerge gels in 0.1M malic acid for 1 h, rinse in distilled water, and stain overnight in 0.015% (w/v) ruthenium red at 4°C. Rinse the gel for 1 h with three changes of distilled water.
7. Enzyme patterns are recorded by:
 - a. Tracing the gel and enzyme patterns onto vinyl transparencies and photocopying the tracing;
 - b. Photographic recording; and/or
 - c. Direct comparison from dried gels.
8. Polygalacturonase (PG) activity gives transparent zones, pectin esterase (PE) activity gives dark zones, and pectin lyase (PL) gives yellow zones.

4. Notes

1. The growth medium selected will vary depending on the requirements of the fungus under study. The GYM medium given in the Section 2. is a general-purpose medium suitable for the growth of many filamentous ascomycetes and basidiomycetes. For organisms that may be more fastidious, a

richer medium, such as peptone yeast extract glucose (12), can be very useful (see ref. 6). Oomycetes, such as *Pythium* and *Phytophthora*, may require either a source of sterol or a more complex organic medium, such as V8 medium (11). Organic components in growth media can vary between suppliers, and this variation may affect the biochemical properties of the fungus (13). It is therefore important to standardize organic components, such as yeast extract, by only using a single source of supply.

2. Growth conditions are important in obtaining a reliable and constant source of protein, and it is usually desirable to extract from cultures consisting of material of a constant age. This is obtained by first inoculating fungal cultures into small starter cultures of the main growth medium. These starter cultures are incubated, usually as shaken liquids, and provide the inoculum for the main growth phase. Although the exact conditions will vary depending on the organism studied, a typical starter culture would consist of 10 mL of GYM medium in a 25–30 mL vessel, incubated on an orbital shaker at 25°C for 60 h. The entire starter culture is then used as inoculum for 60 mL of GYM medium in 250-mL conical flasks, which is then incubated at 25°C on the shaker for 3–5 d. Growth periods and temperatures will depend on the rate of growth and differentiation of the fungus studied. Most yeast cultures will produce sufficient suitable biomass after incubation at 30°C under a growth regime of 24 h in a starter culture and 24 h in a shaken flask. Many filamentous fungi however require 60 h in a starter culture followed by 3–5 d in shaken flasks at 25°C. In order to maintain reproducibility, incubation temperatures should be kept constant.
3. The inoculum used for the initial starter cultures is important and should contain a large number of potential “growing” points. That is, the inoculum should ideally be particulate so that mycelial or yeast growth can occur at many different points. This presents no problems if the culture is naturally particulate, such as yeast cells, or if conidial suspensions can be used (c.g., ref. 5). However, if the fungus grows only as a mycelial mat, an inoculum made from broken mycelia will give a more homogenous and greater biomass than one derived from plugs or “lumps” of culture.
4. In many cases, the proteins of interest will be associated with active growth of the fungus. It is therefore often necessary to establish some form of growth curve for the organism prior to harvesting the growth. The growth rate of yeasts, and fungi growing in a yeast-like phase, can be estimated by sampling and measuring the turbidity of the growth medium over a period of time (14). However, with filamentous organisms, this becomes impractical. One alternative is to assay the culture medium for the extracellular enzymes β -glucosidase, β -galactosidase, and diacetyl-chitobiosidase. In many fungi, these enzymes are each associated with particular features of the growth

cycle when grown in glucose-containing media. β -glucosidase is generally produced in the early phases of growth, β -galactosidase is not produced until the glucose concentration has been reduced, and diacetyl-chitobiosidase is generally produced during autolysis. These activities can be readily screened directly from culture fluids with the substrates 4-methylumbelliferyl β -D-glucoside, 4MU β -D-galactoside, and 4MU β -D-N,N'-diacetyl-chitobioside (6,10). The substrates can be kept as 50-mM solutions in dimethylformamide, and for use, the stock solutions are diluted 0.15:9.85 mL in 0.05M sodium acetate, pH 5.4. Activity is determined by adding 50 μ L of culture fluid to 50 μ L of each substrate and incubating these for 4 h at 37°C. Positive enzyme activity can then be detected as fluorescence under UV light after the addition of 50 mL of saturated aqueous sodium bicarbonate.

5. When the required phase of growth has been reached, the fungal biomass must be harvested from the growth medium. Again methods differ depending on the organism, but in general, yeasts can be harvested by centrifugation at 3–7000g, whereas filamentous fungi are better harvested by vacuum filtration onto Whatman No. 3 filter papers in a Buchner funnel. In both cases, the resulting biomass should be washed in sterile distilled water to remove any residues from the media. If extracellular proteins, such as extracellular enzymes, are required, these can be extracted directly from the culture fluid after the mycelium has been harvested. Typically, the culture fluid is clarified by centrifugation and proteins extracted by one of the general protein extraction techniques, such as precipitation with acetone or methanol/ammonium or by immunoprecipitation (15,16).
6. Efficient cell breakage is necessary to ensure a good recovery of proteins from filamentous fungi. A wide variety of cell breakage methods have been reported in the literature, varying from enzymatic digestion of the cell wall to physical disruption procedures, such as grinding mycelium in a mortar and pestle (6) or homogenizing thick cell pastes (17). Enzymatic digestion has been used to generate protoplasts, which can be harvested from the remaining cell debris. The protoplast preparations can then be lysed to release intracellular proteins (18,19). Many fungi produce active intra- and extracellular proteinases, and these are usually active under the conditions used for enzymatic digestion of cell walls. As a result, physical disruption of an "inert" sample is generally preferred. Protease activity has been suppressed by phenylmethylsulfonylfluoride (PMSF) in some cases (20). Fungal cell walls can be disrupted by briefly grinding freeze-dried mycelium in a mortar and pestle. Additional abrasives are not usually required, although carborundum may be added if required. This procedure should not be performed on an open bench owing to the possible hazardous nature of the dust produced. A single flask of 60 mL actively growing culture will

give about 500–2000 mg of freeze-dried material. The method of grinding a freeze-dried culture as described here is quick and reliable, but requires access to freeze-drying equipment. A commonly used alternative is to grind the harvested mycelium in liquid nitrogen (15). However, appropriate safety measures should be considered to avoid any potential contact or splashing from the liquid nitrogen

7. Yeast cells can be disrupted by passage through a pressure cell, such as a French Press (21), although three or more passages may be necessary to achieve 70% breakage. Yeast cells can also be broken by shaking frozen cultures with glass beads. The glass beads used are generally larger than the Ballotini beads used for bacteria and 2–2.5-mm beads can produce adequate disruption after 10–15 min of shaking. This method can generate sufficient local heating to denature some proteins, so effective cooling of the system is required
8. The ground mycelium is rehydrated in the method at 500 mg in 1 mL of Tris-glycine buffer. This will need mixing to form an even slurry. In some cases, detergent-containing buffers have been shown to give higher yields of total protein (22,23), but the simple Tris-glycine buffer used here is adequate for general purposes.
9. The total aqueous extract will contain the cytoplasmic proteins. Normally, the pellet of unbroken cells and debris is discarded although this can be saved if the cell-wall fraction is required. This extract will, however, contain many other components, including polysaccharides. The degree of further purification necessary will depend on the protein and level of activity required, as well as the level of polysaccharide contamination. As mentioned, the supernatant contains the total cytoplasmic proteins and may be used directly. Proteins in these crude extracts can be readily separated by polyacrylamide gel electrophoresis (7,8), and this technique can be used to provide cell-free total protein patterns or, with specific stains, to demonstrate particular enzymes. Alternatively, it may be necessary to clarify the supernatant further by a second centrifugation or filtration through a 0.45- μ m filter. Specific proteins can then be purified through standard precipitation techniques as mentioned in Note 5. Further clarification and separation, generally by differential centrifugation in sucrose density gradients will be required to extract the cell-wall fraction (23)
10. The protein content of the supernatant should be estimated by one of the standard protein determination methods, such as Lowry's determination (24)
11. Under the conditions used, pectinase isoenzymes may be both positively and negatively charged. It is therefore very important to undertake the electrophoresis in a horizontal gel system that allows for the construction of central sample wells

References

- 1 St. Leger, R. J., Cooper, R. M., and Charnley, A. K. (1986) Cuticle-degrading enzymes of entomopathogenic fungi: regulation of production of chitinolytic enzymes. *J. Gen. Microbiol* **132**, 1509–1517
2. Clarkson, J. H. (1992) Molecular biology of filamentous fungi used for biological control, in *Applied Molecular Genetics of Filamentous Fungi* (Kinghorn, J. R. and Turner, G., eds.), Blackie Academic and Professional, Glasgow, Scotland, pp 175–190
- 3 Peberdy, J. F. (1990) Fungal cell walls—a review, in *Biochemistry of Cell Walls and Membranes in Fungi* (Kuhn, P. J., Trinci, A. P. J., Jung, M. J., Goosey, M. W., and Copping, L. G., eds.), Springer-Verlag, Berlin, Germany, pp 5–30
4. Cruickshank, R. H. and Wade, G. C. (1980) Detection of pectin enzymes in pectin acrylamide gels *Analyt Biochem* **107**, 177–181.
- 5 Elad, Y., Chet, I., and Henis, Y. (1982) Degradation of plant pathogenic fungi by *Trichoderma harzianum* *Can J Microbiol* **28**, 719–725
6. Paterson, R. R. M. and Bridge, P. D. (1994) *Biochemical Techniques for Filamentous Fungi*. CAB International, Wallingford, UK.
7. Mugnai, L., Bridge, P. D., and Evans, H. C. (1989) A chemotaxonomic evaluation of the genus *Beauveria* *Mycological Res.* **92**, 199–209
8. Jun, Y., Bridge, P. D., and Evans, H. C. (1991) An integrated approach to the taxonomy of the genus *Verticillium*. *J. Gen. Microbiol.* **137**, 1437–1444.
- 9 Monte, E., Bridge, P. D., and Sutton, B. C. (1990) Physiological and biochemical studies in *Coelomycetes* *Phoma Studies Mycology* **32**, 21–28
10. Barth, M. G. and Bridge, P. D. (1989) 4-Methylumbelliferyl substituted compounds as fluorogenic substrates for fungal extracellular enzymes. *Lett Appl Microbiol* **9**, 177–179.
- 11 Smith, D. and Onions, A. H. S. (1994) *The Preservation and Maintenance of Living Fungi*, 2nd ed. CAB International, Wallingford, UK
12. Conti, S. F. and Naylor, H. B. (1959) Electron microscopy of ultrathin sections of *Schizosaccharomyces octosporus* I Cell division. *J. Bacteriol* **78**, 868–877
- 13 Filtenborg, O., Frisvad, J. C., and Thrane, U. (1990) The significance of yeast extract composition on secondary metabolite production in *Penicillium*, in *Modern Concepts in Penicillium and Aspergillus Classification* (Samson, R. A. and Pitt, J. I., eds.), Plenum, New York, pp 433–441
- 14 Barnett, J. A., Payne, R. W., and Yarrow, D. (1990) *Yeasts. Characteristics and Identification*, 2nd ed. Cambridge University Press, Cambridge, UK
15. St. Leger, R. J., Staples, R. C., and Roberts, D. W. (1991) Changes in translatable mRNA species associated with nutrient deprivation and protease synthesis in *Metarhizium anisopliae* *J Gen Microbiol* **137**, 807–815
16. Kim, K. K., Fravel, D. R., and Papavizas, G. C. (1990) Production, purification and properties of glucose oxidase from the biocontrol fungus *Talaromyces flavus* *Can. J Microbiol* **36**, 199–205
- 17 Hien, N. H. and Fleet, G. H. (1983) Separation and characterization of six (1→3)- β -glucanases from *Saccharomyces cerevisiae* *J Bacteriol* **156**, 1204–1213
- 18 Sambrook, J., Fritsch, E. F., and Maniatis, T. (1989) *Molecular Cloning: A Laboratory Manual*, vol 3 Cold Spring Harbor Laboratory, Cold Spring Harbor, New York.

- 19 Messner, R and Kubicek, C P (1990) Synthesis of cell wall glucan, chitin and protein by regenerating protoplasts and mycelia of *Trichoderma reesei* *Can J Microbiol* **36**, 211–217
- 20 Kim, W K. and Howes, N K. (1987) Localization of glycopeptides and race-variable polypeptides in uredosporling walls of *Puccinia graminis tritici*, affinity to concanavalin A, soybean agglutinin, and *Lotus* lectin *Can J Botany* **65**, 1785–1791
21. Schnaitman, C A. (1981) Cell fractionation, in *Manual of Methods for General Bacteriology* (Gerhardt, P., Murray, R G. E., Costilow, R N., Nester, E W., Wood, W. A., Krieg, N. R., and Briggs Phillips, G , eds), American Society for Microbiology, Washington, DC, pp. 52–61
- 22 Kim, W K , Rohringer, R., and Chong, J. (1982) Sugar and amino acid composition of macromolecular constituents released from walls of uredosporlings of *Puccinia graminis tritici* *Can J Plant Pathol* **4**, 317–327
- 23 Fèvre, M (1979) Glucanase, glucan synthases and wall growth in *Saprolegnia monoica*, in *Fungal Walls and Hyphal Growth*, British Mycological Society Symposium 2 (Burnett, J H and Trinci, A P. J., eds.), Cambridge University Press, Cambridge, UK, pp 225–263.
- 24 Lowry, O. H , Rosebrough, N J , Farr, A L., and Randall, R J (1951) Protein measurement with the folin phenol reagent *J Biol Chem* **193**, 265–275

CHAPTER 6

Subcellular Fractionation of Animal Tissues

Norma M. Ryan

1. Introduction

A number of techniques exist which exploit various physical parameters or biological properties of cells as means of investigating the complexity of cellular organelles and membranes. Subcellular fractionation using the centrifuge is the basis of traditional methods for separating cellular components. Even when other methods are used to effect the separation, it is often the case that better results will be obtained if the material is first purified or partially purified by centrifugal methods (*see refs. 1 and 2 for a recently published discussion of the theory and applications of the use of the centrifuge in preparative procedures*). In this chapter some of the basic centrifugal techniques used to fractionate a typical animal cell are described.

Subcellular fractionation involves three successive steps (for a detailed discussion of the principles involved *see refs. 3 and 4*). The first step converts a tissue or cell suspension into a homogenate. The second step reintroduces a new kind of order into the system by grouping together, in separate fractions, those components of the homogenate of which certain physical properties, such as density or sedimentation coefficient, fall between certain limits set by the investigator. The third step consists of the analysis of the isolated fractions. Interpretation of the results involves a retracing of these three steps. It is essential to perform all steps of this process, regardless of whether the objective of the work is to study the spatial organization of components within a cell (i.e., subcellular fractionation itself) or

From *Methods in Molecular Biology, Vol. 59 Protein Purification Protocols*
Edited by S. Doonan, Humana Press Inc., Totowa, NJ

to use the procedures as a means of obtaining a partially purified component which then can be more readily purified using other procedures.

Tissue disruption or homogenization may be accomplished by a variety of means such as grinding, ultrasonic vibrations, and by making use of the osmotic properties of cells. The grinding of tissues to form homogenates is usually accomplished with the Potter–Elvehjem homogenizer, which is highly effective with soft tissues, such as liver and kidney. Tough tissues, such as muscle, generally require the use of a mincer and blender of which a common example is the Waring Blender. In recent years the Chaikoff Press, which entails the forcing of the material through fine holes under high pressure, has found much use. Osmotic methods of tissue disruption are most useful when employed in connection with red blood cells and reticulocytes. The use of ultrasonic vibrations is useful for the disruption of bacterial cells and some animal cells. In such systems cooling tends to cause a problem because very high temperatures are generated at the point of disintegration. Homogenization should be carried out so as to leave the subcellular organelles virtually intact, although somewhat distorted, and yet shear the plasma membrane, endoplasmic reticulum, and other endomembranous systems into fragments which form spherical vesicles.

The suspension so prepared contains whole cells, partially broken cells, nuclei, mitochondria, and so on. Separation of the various components, in theory, can be achieved by the application of any method that exploits the differences between the physical and/or chemical properties of the constituents. Methods such as electrophoresis, counter-current distribution, and centrifugation are but a few of the methods available. The method most routinely used is centrifugation in one form or another, both because of the wide availability of the equipment necessary to carry out the procedures and also the facility of fractionating relatively large amounts of material with easy recovery of sample when fractionation is complete.

The behavior of particles in a centrifugal field and the various factors that affect their rate of sedimentation have been studied in great detail (5,6). The rate of movement of an ideal (spherical) particle in a centrifugal field is described by the following equation:

$$V = dr/dt = \{[a^2(Dp - Dm)]/18\eta\} \times \omega^2 r \quad (1)$$

where V is the velocity of particle in cm/s = dr/dt , a is the particle diameter; Dp is the particle density; Dm is the density of the medium; ω is the

angular velocity in radian/s, which is equal to the number of revolutions per second $\times 2\pi$; r is the distance between the particle and the center of rotation in centimeters; and η is the viscosity of the medium.

The centrifugal force (CF) on a particle in a spinning rotor is given by:

$$CF = m\omega^2 r \quad (2)$$

where m is the mass of the particle, r is the distance between the particle and the center of rotation in centimeters, and ω is the angular velocity in radian/s.

When centrifugation conditions are reported it is a common practice to list the relative centrifugal force (RCF). RCF is the force on a particle at some r value (generally for a point midway down the centrifuge tube, r_{av}) divided by the force on that particle in the earth's gravitational field:

$$RCF = \omega^2 r / g \quad (3)$$

where g is the gravitational constant, 980 cm/s.

Thus:

$$RCF = \omega^2 r / 980 \quad (4)$$

Since temperature affects both the density and viscosity of the medium, it thus affects the rate of sedimentation of the particles.

In this chapter the subcellular fractionation of rat liver tissue is described as a fairly typical example of a fractionation procedure; ref. 7 should be consulted for information on variations in procedures appropriate for other tissues. The separation of the various constituents of an homogenate formed from rat liver tissue, using differential centrifugation, is described. The method described here does not yield an absolute purification of the individual components within a rat liver cell, but rather a partial purification of the cellular components results which will enable alternative/subsequent purification procedures to be carried out with a far higher chance of success.

2. Materials

All reagents are of Analar grade.

1. Homogenization buffer medium: 0.25M sucrose, 5 mM imidazole-HCl, pH 7.4 (see Notes 1 and 2).
2. 0.9% Saline for perfusion of the liver.
3. Resuspension buffer medium: 0.25M sucrose, 5 mM imidazole-HCl, pH 7.4 (see Note 3)

4. Refrigerated laboratory centrifuge, such as Sorvall RC5B fitted with an 8×50 mL rotor (*see* Note 2).
5. Potter–Elvehjem homogenizer.
6. Dounce homogenizer.
7. Beckman L65 ultracentrifuge fitted with an 8×35 mL rotor or equivalent (*see* Note 2)

3. Methods

3.1. Preparation of the Homogenate

1. Remove the liver from a rat large enough to yield approx 10 g liver (wet weight), having first perfused the liver with 0.9% saline. The perfusion should be carried out immediately after the sacrifice of the rat by appropriate means (*see* Note 4)
2. Wash the liver free of blood, hairs, and so on, by suspending it in ice-cold homogenizing medium.
3. Blot the tissue lightly with filter paper to dry it
4. Mince the liver finely in a preweighed, chilled beaker and weigh again (*see* Note 5).
5. Homogenize the liver in homogenizing medium to form a 25% (w/v) homogenate, using six passes of a Potter–Elvehjem homogenizer (*see* Note 6)
6. Dilute homogenate to 12.5% (w/v).

3.2. Subcellular Fractionation (*see* Note 7)

1. Retain a small portion of the 12.5% (w/v) homogenate for analysis and centrifuge the remainder at 4°C at a speed of 600g for 10 min in a refrigerated laboratory centrifuge (*see* Note 8)
2. Resuspend the pellet in the same volume of homogenization medium as previously and centrifuge again at 600g for an additional 10 min (*see* Note 9).
3. Combine the post-600 g-minute supernatants and centrifuge at 15,000g for 10 min in order to prepare a fraction rich in mitochondria and lysosomes.
4. Resuspend the resulting pellet in the same volume of homogenization medium as previously and centrifuge again at 15,000g for another 10 min (*see* Note 10).
5. At the end of the washing steps pool all the supernatants and use in subsequent procedures (*see* Note 11). If the material retained in the pellet is required for further studies, resuspend it in homogenizing medium or other appropriate medium (*see* Note 3)
6. Pool the post- 15×10^4 g-minute supernatants and centrifuge at 100,000g for 60 min in an ultracentrifuge at 4°C . The resulting complex microsomal pellet contains vesicles derived from the plasma membrane, endoplasmic reticulum—some containing ribosomes and some ribosome-free—peroxisomes, polysomes, endosomes, Golgi stacks, and other such membranous systems from within the cell, while the supernatant will contain the remain-

der of the cellular components, that is, the soluble components and smaller elements, such as free ribosomes (*see* Note 12).

7. Resuspend each of the pellets in a volume of 0.25M sucrose, 5 mM imidazole-HCl, pH 7.4 using a loose Dounce homogenizer. It may be more appropriate to use a different resuspension medium depending on the ultimate objectives (*see* Note 3).
8. Store all fractions in a deep freeze (*see* Note 13) until required for analysis (*see* Note 14) or further purification (*see* Note 15).

4. Notes

1. Homogenization is the first essential step in any fractionation procedure. It involves the disruption of an ordered system and results in a loss of some morphological information, but homogenization is necessary in order to apply the techniques of biochemistry to a study of subcellular components. The choice of an adequate homogenization medium is a critical one. The homogenization medium most suited to the biological material involved can only be elucidated by a process of trial and error. Sucrose is very widely used, but such details as concentration of sucrose, pH of the buffer, traces of specific cations, and so on, vary considerably. The best homogenate is the one which lends itself most successfully to fractionation and will result in a satisfactory resolution of the disrupted components of the homogenate. It is also the one which enables the next experiment to be carried out.
2. For best results the rat liver and every preparation made from the tissue must be kept cold (0–4°C) from the moment the organ is removed from the rat. Thus, all solutions must be prechilled to 0°C before addition to tissue, as should all glassware which comes in contact with the preparations. Centrifugation must be carried out in refrigerated centrifuges at 0–4°C, ensuring that the rotors are prechilled.
3. Depending on the final objectives of the purification procedures and the restrictions on the techniques following the differential centrifugation, it may be more appropriate to use a different resuspension medium than the one described here. Sucrose interferes with many assays and is not always easy to eliminate from preparations.
4. The method chosen for sacrifice of the rat will depend on the aims of the experiments. Most often this will be cervical dislocation. Anesthetics are occasionally used but these could have undesirable side effects on the tissue of interest, and therefore should only be employed if it is certain that the side effects will not affect the subcellular component/protein of interest.
5. Heavy metal ions are powerful enzyme inhibitors, so avoid sticking scissors, forceps, spatulas, and so on, into the tissue preparations. Manipulations, such as stirring of solutions or resuspension of pellets, should be carried out using glass rods.

6. This step is critical and the clearance of the homogenizer should be considered with care. Once again it is a matter of trial and error until precisely the right conditions are determined to suit the purposes of the experiment.
7. Subcellular fractionation is accomplished by the stepwise process of differential centrifugation, which separates particles from a supernatant in the form of a pellet. Differential pelleting is the simplest method for obtaining a crude separation which exploits the mass of the major organelles and membrane systems. All steps are carried out in the temperature range 0–4°C. A measured portion of the original homogenate is retained for further analysis. It is extremely important to keep an accurate record of all volumes and weights used in preparing the fractions. Otherwise it will be impossible to interpret the results in a meaningful way.
8. This first centrifugation step is designed to remove all nuclei, whole cells, partially intact cells, and plasma membrane sheets, and is very effective in this.
9. The objective of this washing step is to reduce contamination of the fraction by membrane components.
10. This washing step can be repeated two or three times, depending on how critical the degree of contamination by other membranous components is to the objectives of the experiment.
11. This fraction is not a pure preparation of mitochondria and lysosomes, but rather it is enriched in mitochondria and lysosomes with a reduction in the level of contaminating membranous components.
12. The ultimate location of the Golgi Apparatus and vesicles derived from the Golgi stacks and associated “trafficking” vesicles will be determined by the extent to which the homogenization procedures disrupt the networks of the Golgi. Sometimes these will be found in the post- 6×10^3 g-minute supernatant, but more often they are detected primarily in the post- 15×10^4 supernatant. The ribosomes may also be pelleted by subjecting the supernatant to sedimentation at 100,000g at 4°C in an ultracentrifuge for a minimum of 3 h.
13. In normal procedures, aging studies to determine the lability of the enzyme markers to be studied should be carried out. Freezing may damage some particles resulting in a loss of activity or of a constituent. The interference of components of the media used in the fractionation process with the assay of the markers should also be thoroughly assessed before proceeding with the analysis and interpretation.
14. After fractionation of the tissue each fraction should be assayed, in addition to the original homogenate, for selected markers for the purpose of following each constituent throughout the fractionation procedure. de Duve (2,8) laid down two criteria for the selection of an enzyme as a marker which can also be applied to a chemical constituent:

- a. The enzyme must have a specific location, that is, be present only in one type of particle; and
- b. That all subunits of a given population have the same enzyme content as related to their mass or total protein.

The second criterion is not absolutely essential for a marker. Known markers for specific organelles are used to appreciate the efficiency of the fractionation procedure and thereby the quality of the homogenate. If two markers show the same distribution pattern it is an indication, but not proof, that they may be associated with the same particle. By a comparison of the specific activity of the markers in the different fractions, an estimation of the contamination or degree of purity of the particles may be obtained.

The markers selected are usually biochemical (chemical constituents, enzyme activities, immunological), but morphological and sedimentation coefficient analysis can also prove very informative. A list of some of the classical enzymic markers normally employed in fractionation studies is provided in ref. 7. The amount of biochemical constituents, such as DNA, RNA, cholesterol, total protein, and lipids should be assayed and calculated in each of the fractions. The absolute activities and specific activities of each of the relevant enzymes should be assayed. The concentration of the components or enzyme activities should be expressed as a percentage of the total constituent or enzyme activity found in the homogenate, that is, the percent recovery of each marker is calculated for every fractionation experiment carried out. Quantitative recovery of each marker is important and before any interpretation of the data is made, quantitative recovery of any enzyme activity/constituent must be established. The results of the analysis are presented as distribution patterns or as frequency histograms if possible.

It is essential to prepare a balance sheet of the constituents and enzyme activities in the differing fractions compared with those in the original homogenate. This is achieved by:

- a. Retaining a portion of the original homogenate;
- b. Recording accurately the volume of the homogenate used for centrifugation and also the volumes of buffered medium in which the particles are resuspended; and
- c. Carrying out all analyses on the homogenate as well as on the prepared fraction. Concentration of a constituent or enzyme activity should be expressed as a percentage of that which was present in the original homogenate.

A particular enzyme activity may be recovered in the fractions at values considerably less than or more than 100%. In the former case this may be caused by the handling of the materials or the removal of an influencing factor (i.e., an activator or a cofactor) which may have been separated into

another fraction. Recoveries in excess of 100% would tend to indicate the removal of some inhibiting substance.

15. The subcellular fractionation that has been described here applies principally to soft tissues, especially liver tissue. However, the properties described do apply to most animal cells, although different methods of homogenization, in particular, may be necessary to effect the desired subcellular fractionation. What has been described is the crude preparation of crude subcellular fractions which are partially purified and which may form the basis for purification of a specific membrane fraction or a particular protein by either techniques, such as density gradient centrifugation or other protein purification procedures as described in Chapters 14–30.

References

- 1 Rickwood, D (ed) (1992) *Preparative Centrifugation A Practical Approach*, IRL, Oxford, UK
- 2 de Duve, C (1967) General principles, in *Enzyme Cytology* (Roodyn, D. B , ed), Academic, London, pp 1–26
3. Birnie, G D. and Rickwood, D (eds.) (1978) *Centrifugal Separations in Molecular and Cell Biology*, Butterworths, London.
4. de Duve, C. and Beaufay, H (1981) A short history of tissue fractionation *J Cell Biol* **91**, 293s–299s
5. Schachman, H. K (1959) *Ultracentrifugation in Biochemistry*, Academic, London
6. Svedberg, T. and Pederson, K. O (1940) *The Ultracentrifuge*, Clarendon, Oxford (Johnson Reprint Corporation, New York)
- 7 Evans, W H (1992) Isolation and characterisation of membranes and cell organelles, in *Preparative Centrifugation A Practical Approach* (Rickwood, D , ed), IRL, Oxford, UK, pp 233–270
- 8 de Duve, C (1971) Tissue fractionation—past and present. *J Cell Biol* **50**, 20d–55d

CHAPTER 7

Subcellular Fractionation of Plant Tissues

Isolation of Chloroplasts and Mitochondria from Leaves

Alyson K. Tobin

1. Introduction

The successful isolation of intact, functional organelles from plant tissue is fraught with difficulties. Leaves are often covered in waxy cuticles, and frequently contain silica (as in grasses) and toxic components, such as phenolics, proteolytic enzymes, and high concentrations of acids and salts in the vacuole. In addition, all higher plants have one major barrier in common, the presence of a rigid, cellulose cell wall, which has to be broken in order to release the organelles. Mechanical isolation methods, i.e., where leaf material is macerated in a mechanical homogenizer, are likely to succeed only for a limited number of species, e.g., pea and spinach, where the leaves do not contain large amounts of tough, thickened tissue. Otherwise, the prolonged homogenization required to release significant numbers of organelles from tough leaves, e.g., of grasses, such as wheat or barley, results in most of them being broken and inactive. For this reason, the only viable method for obtaining good-quality organelles from species such as these is to isolate them from protoplasts. Although chloroplasts can be successfully isolated from protoplasts, the yield of mitochondria is so small that, unless very large-scale protoplast preparations are employed, the technique is generally inappropriate for mitochondrial isolation. This means that there are many plant species from whose leaves it has proven impossible to isolate good-quality mitochondria using existing techniques.

From *Methods in Molecular Biology, Vol 59 Protein Purification Protocols*
Edited by S Doonan Humana Press Inc , Totowa, NJ

A major factor in the success, or otherwise, of all of these methods is the quality of the plant material. Even the best technician cannot make good chloroplasts or mitochondria from poor-quality plants, and it is essential both to optimize and standardize the growing conditions if reproducible results are to be obtained.

Finally, isolated organelles are easily damaged and must be handled gently. Detergents and volatile solvents are extremely harmful, since they disrupt the lipid-rich membranes. Detergents should never be used to wash any apparatus used for organelle isolation. It is also important to work well away from anyone using volatile solvents (particularly phenol) in the laboratory, since even the vapor can be disruptive to organelles.

Clearly, there is no universally applicable technique, and it is necessary to optimize the conditions for each species. In this chapter, two different methods for isolating chloroplasts are described: from protoplasts and using a mechanical method. It is advisable always to try the mechanical method first, because this is much quicker and simpler, and to use protoplasts only if results are unsatisfactory. Two methods of protoplast and chloroplast isolation are described to illustrate the different requirements of different species, in this case wheat (*Triticum aestivum* L.) and barley (*Hordeum vulgare* L.).

2. Materials

Media 1–15 will keep for a maximum of 2 wk at 4°C providing that BSA, sodium pyrophosphate, MnSO_4 , and Percoll are omitted, and added to the stocks, where required, immediately before use.

2.1. Solutions for Wheat Protoplast and Chloroplast Preparation

1. Medium 1: 0.5M sucrose, 5 mM MES, pH 6.0, 1 mM CaCl_2 .
2. Medium 2: 0.4M sucrose, 0.1M sorbitol, 5 mM MES, pH 6.0, 1 mM CaCl_2 .
3. Medium 3: 0.5M sorbitol, 5 mM MES, pH 6.0, 1 mM CaCl_2 .
4. Medium 4: 0.4M sorbitol, 10 mM EDTA, 25 mM Tricine, pH 8.4.
5. Wheat digestion medium: 0.6 g Cellulysin (Sigma, St. Louis, MO), 6 mg Pectolyase (Sigma) in 30 mL of medium 3.

Wheat and barley digestion media are made immediately before use by adding cellulysin, pectolyase, and cellulase, as appropriate, to the surface of the medium. This is left to stand, without stirring, until the dry powders have been fully absorbed into the medium, preventing the formation of lumps of dry powder that are difficult to disperse.

2.2. Solutions for Barley Protoplast and Chloroplast Preparation

1. Medium 5: 0.4M sorbitol, 10 mM MES, pH 5.5, 1 mM CaCl₂, 1 mM MgSO₄
2. Barley digestion medium: 1.5% (w/v) cellulase (Onozuka R10) in medium 5.
3. Medium 6: 35% (v/v) Percoll in medium 8.
4. Medium 7: 25% (v/v) Percoll in medium 8
5. Medium 8: 0.4M sorbitol, 25 mM Tricine, pH 7.2, 1 mM CaCl₂, 1 mM MgSO₄.
6. Medium 9: 0.33M sorbitol, 50 mM Tricine, pH 7.8, 2 mM EDTA, 1 mM MgSO₄, 1 mM MnSO₄

2.3. Solutions for Pea Chloroplast Preparation

1. Medium 10: 0.33M sorbitol, 50 mM Tricine, pH 7.9, 2 mM EDTA, 1 mM MgCl₂, 0.1% (w/v) BSA. BSA is added to solutions in the same way as described in Section 2.1 for addition of enzymes to digestion media.
2. Medium 11: 40% (v/v) Percoll, 0.33M sorbitol, 50 mM Tricine, pH 7.9, 0.1% (w/v) BSA

2.4. Solutions for Chloroplast Assays

1. Medium 12: 0.33M sorbitol, 20 mM HEPES, pH 7.6, 10 mM EDTA, 0.2 mM KH₂PO₄, 30 mM MgCl₂.
2. Medium 13: double-strength medium 12.
3. Medium 14: 0.33M sorbitol, 50 mM Tricine, pH 8.2, 2 mM EDTA, 1 mM MgSO₄, 1 mM MnSO₄
4. Medium 15: 0.33M sorbitol, 50 mM Tricine, pH 8.2, 10 mM KCl, 5 mM sodium pyrophosphate, 2 mM EDTA, 2 mM ATP
5. 0.1M K₃Fe(CN)₆. Make in water, and store in the dark at -20°C; protect from bright light
6. 1.0M NH₄Cl.
7. 1.0M KHCO₃ freshly made in the assay buffers.
8. 1.0M NaHCO₃ freshly made in the assay buffers.

2.5. Solutions for Mitochondrial Isolation

The mitochondrial isolation medium and wash medium will both keep at 4°C for up to 2 wk if stored in the absence of PVP, BSA, and cysteine which should be added immediately before use, where required. Solution I (minus BSA) will keep for 2 wk providing BSA is added fresh on the day, and solutions II and III should be freshly made.

1. Mitochondrial isolation medium: 0.3M sucrose, 50 mM MOPS-KOH, pH 7.5, 2 mM Na₂EDTA, 1 mM MgCl₂, 1% (w/v) PVP 40 (soluble), 0.4% (w/v) BSA, 4 mM cysteine.

- 2 Mitochondrial wash medium: 0.3 M sucrose, 20 mM MOPS-KOH, pH 7.5, 0.1% (w/v) BSA.
- 3 Solution I: 0.6 M sucrose, 20 mM KH_2PO_4 , 0.2% (w/v) BSA, pH 7.5.
4. Solution II: 20 mL solution I, 11.2 mL Percoll, 8.8 mL of 40% (w/v) PVP 40
5. Solution III: 20 mL solution I, 11.2 mL Percoll, 8.8 mL H_2O

3. Methods

All procedures are carried out at 4°C, unless otherwise stated. All solutions and apparatus should be prechilled when working at this temperature.

3.1. Isolation of Chloroplasts from Protoplasts

3.1.1. Wheat Protoplast Preparation

- 1 Take primary leaves from 7–10-d-old plants (*see* Note 1). Finely chop the leaves into thin sections, approx 1–2 mm thick. Use sharp, single-sided razor blades. Spread the leaf sections onto the surface of 30 mL of wheat digestion medium (*see* Note 2) in a shallow dish that provides a large surface area of medium (a crystallizing dish is suitable). Add sufficient leaf material to cover the surface of the medium completely. Incubate for 3 h at 28°C. Do not shake or stir the medium during this period.
2. Following digestion, carefully remove the digestion medium from the dish, leaving the leaf sections in place. Add 20 mL of medium 3 to the leaf sections, and gently swirl to release the protoplasts into the medium. Filter through coarse nylon mesh, e.g., a plastic tea strainer, and retain the filtrate. Return the leaf sections to the dish, add a further 20 mL of medium 3, swirl again, filter, and combine the two filtrates.
3. Centrifuge the filtrates in a swing-out rotor at 150g for 5 min (*see* Note 4). Discard the supernatant, and gently resuspend the pellets in a total volume of 20 mL of medium 1. Divide the suspension equally between four tubes and overlay 2 mL of medium 2, followed by 1 mL of medium 3 onto each suspension to form a discontinuous gradient. Centrifuge at 250g for 5 min in a swing-out rotor.
4. Intact protoplasts collect at the interface between medium 2 and 3. Using a wide-bore Pasteur pipet, carefully remove the layer of protoplasts, transfer to a clean centrifuge tube, and add 2 vol of medium 3. Centrifuge at 150g for 5 min in a swing-out rotor. Resuspend the pellet in medium 4 to give a chlorophyll concentration of approx 1 mg/mL.

3.1.2. Wheat Chloroplast Preparation

1. Resuspend the protoplasts (prepared as in Section 3.1.1.), in 5 mL of medium 4. Pour the suspension into a modified 25-mL disposable plastic syringe (*see* Note 3).

2. Break the protoplasts by passing the suspension through 20- μ m pore size nylon mesh, and immediately centrifuge at 250g for 1–2 min in a swing-out rotor.
3. Gently resuspend the pellet in medium 4 to give a chlorophyll concentration of approx 1 mg/mL.

3.1.3. Barley Protoplast Preparation

1. Take primary leaves from 7–10-d-old barley plants (*see* Note 1). Carefully remove the lower epidermis, and lay the leaves, lower side down, onto the surface of 50 mL of barley digestion medium (*see* Note 2) in a clear plastic sandwich box. Cover the box with cling film, and incubate at 30°C for 2 h without disturbance.
2. Following digestion, harvest the protoplasts by gently swirling the leaves within the digestion medium. Decant the suspension through nylon mesh, e.g., a plastic tea strainer, into 50-mL centrifuge tubes.
3. Centrifuge at 250g for 5 min in a swing-out rotor, discard the supernatants, and resuspend the pellet in 10 mL of medium 6.
4. Divide the suspension equally between 2 $\times$ 20-mL centrifuge tubes. Overlay each with 2 mL of medium 7 followed by 1 mL of medium 8 to form a discontinuous gradient.
5. Leave the gradients to stand on ice for 1 h, and then centrifuge at 300g for 5 min in a swing-out rotor.
6. Intact protoplasts collect at the interface between media 7 and 8. Carefully remove the protoplasts using a wide-bore Pasteur pipet, transfer to a clean 20-mL centrifuge tube, and add 2 vol of medium 9.
7. Centrifuge at 150g for 5 min in a swing-out rotor, discard the supernatant, and gently resuspend the pellet in medium 9 to give a chlorophyll concentration of approx 1 mg/mL.

3.1.4. Barley Chloroplast Preparation

1. After removing the protoplasts from the gradient (as in step 6), dilute (approx fivefold) in medium 9, and transfer to a modified 25-mL disposable plastic syringe (*see* Note 3).
2. Break the protoplasts by passing the suspension through 20- μ m pore size nylon mesh, and immediately centrifuge at 150g for 2 min in a swing-out rotor.
3. Gently resuspend the pellet in medium 9 to give a chlorophyll concentration of approx 1 mg/mL (*see* Note 5).

3.2. Mechanical Isolation of Chloroplasts from Pea Leaves

1. Take 10 g (fresh weight) of pea leaves, add to 40 mL of ice-cold medium 10, and cut into small pieces using razor blades or sharp scissors.

2. Homogenize (5 s full speed.), using an Ultraturrax or Polytron homogenizer. Filter the homogenate through four layers of muslin (prewetted with medium 10)
3. Divide the filtrate equally between two 50-mL centrifuge tubes. Underlay each portion of filtrate with 10 mL of medium 11.
4. Centrifuge at 2500g for 1 min in a swing-out rotor.
5. The chloroplasts form a soft, green pellet at the bottom of the centrifuge tube. Carefully remove the supernatant layers by aspiration without disturbing the pellet. Gently resuspend the pellet (e.g., using a fine paintbrush) in 1.0 mL of medium 10

3.3. Assay of Chloroplast Intactness

Chloroplast intactness (*see* Note 6) is assayed according to the method of Lilley et al. (1). The rate of ferricyanide-dependent oxygen evolution is measured using an oxygen electrode (Hansatech, Norfolk, UK).

- 1 The assay medium, final volume 1.0 mL, is added to an oxygen electrode and consists of medium 12 and 2 mM $K_3Fe(CN)_6$ (final concentration, added from a 0.1M stock solution). Intact chloroplasts are added (final concentration of approx 50 μg chlorophyll/mL assay), and the assay is illuminated (PAR of 1000 $\mu mol/m^2/s$). Following the addition of 5 mM NH_4Cl (final concentration), the rate of O_2 evolution is measured, and this gives the "intact rate" of ferricyanide-dependent O_2 evolution.
- 2 To determine the "broken rate," chloroplasts (the same amount as used in the intact assay) are added to 0.5 mL of H_2O in the oxygen electrode. After 2 min, 0.5 mL of medium 13 is added, followed by 2 mM $K_3Fe(CN)_6$ (as in step 1). Following illumination and the addition of NH_4Cl (as in step 1), the rate of O_2 evolution is measured.
- 3 Chloroplast intactness is calculated as:

$$[(\text{Broken rate} - \text{intact rate})/\text{broken rate}] \times 100 = \% \text{ intactness} \quad (1)$$

3.4. Assay of Photosynthetic Activity (CO_2 -Dependent Oxygen Evolution) of Chloroplasts (*see* Notes 6–8 and 10)

3.4.1. Wheat

1. The assay medium, final volume 1.0 mL, is added to an oxygen electrode (Hansatech, Norfolk, UK), and consists of medium 4, 10 mM $KHCO_3$ (final concentration), 0.15 mM KH_2PO_4 (final concentration), and chloroplasts (approx 50 μg chlorophyll/mL assay).
2. The rate of oxygen evolution is measured under illumination (minimum PAR of 1000 $\mu mol/m^2/s$) at 20°C.

3.4.2. Barley

1. The assay medium, final volume 1.0 mL, is added to an oxygen electrode (Hansatech), and consists of medium 14, 20 mM KHCO_3 (final concentration, added from a 1.0M stock), 0.2 mM KH_2PO_4 (final concentration), and chloroplasts (approx 50 μg chlorophyll/mL assay).
2. Oxygen evolution is measured as for wheat chloroplasts.

3.4.3. Pea (see Note 9)

1. The assay medium, final volume 1.0 mL, is added to an oxygen electrode (Hansatech), and consists of medium 15, 10 mM NaHCO_3 (added from a 1.0M stock), and chloroplasts (approx 50 μg /assay).
2. Oxygen evolution is measured as for wheat chloroplasts.

3.5. Assay of Chloroplast Purity

Marker enzymes are assayed in aliquots of the original, crude homogenate, and in the final chloroplast preparation to determine contamination by cytosol, peroxisomes, and mitochondria, the main contaminants of concern.

3.5.1. Cytosolic Contamination

Phosphoenol pyruvate (PEP) carboxylase, exclusive to the cytosol, is assayed as follows: 50 mM HEPES, pH 8.0, 5 mM MgCl_2 , 0.08 mM NADH, 1 U malate dehydrogenase, and 10–20 μL protoplasts, chloroplasts, or crude extract are added to a cuvet, final volume 1.0 mL. The assay is carried out at 25°C, and the reaction is started with the addition of 2 mM PEP and measured as the decrease in absorbance at 340 nm owing to the oxidation of NADH. The method is modified from that of Foster et al. (2).

3.5.2. Mitochondrial Contamination

Citrate synthase, exclusive to the mitochondria, is assayed as follows: 50 mM Tricine, pH 8.0, 1.0 mM MgCl_2 , 1.0 mM oxaloacetate (OAA), 0.2 mM acetyl Coenzyme A, 1.5 mM 5,5'-dithio-bis-(2-nitrobenzoic acid; DTNB) and approx 50 μL extract are added to a cuvet, final volume 1.0 mL. The assay is carried out at 25°C, and the reaction is started with the addition of OAA and measured as the increase in absorbance at 412 nm owing to the formation of the 2-nitro-5-thiobenzoate anion. The assay is based on the method of Cooper and Beevers (3).

3.5.3. Peroxisomal Contamination

Glycolate oxidase, exclusive to the peroxisome in leaves (4), is assayed as follows: 50 mM MOPS, pH 7.8, 3 mM EDTA, 0.008% Triton X-100,

0.66 mM reduced glutathione, 0.2 mM flavin mononucleotide (FMN), 0.033% (v/v) phenylhydrazine, and 50 μ L extract are added to a cuvet in a final volume of 3.0 mL. The assay is carried out at 25°C, and the reaction is started with the addition of 5 mM sodium glycolate and measured as the increase in absorbance at 324 nm due to the formation of glyoxylate phenylhydrazine. The assay is based on the method of Behrends et al. (5)

3.6. Isolation of Mitochondria from Pea Leaves **(see Note 11)**

1. Use sharp razor blades to remove fully expanded leaves by cutting through the petiole. All subsequent steps are carried out at 4°C
2. Add 600 mL of mitochondrial isolation medium/100 g (fresh weight) of leaves.
3. Cut the leaves into approx 1-cm² pieces, using razor blades or sharp scissors
4. Homogenize (in 150-mL aliquots) using an Ultraturrax homogenizer (Orme Scientific, Manchester, UK) using 2 $\times$ 5 s bursts at full speed (mark 10).
5. Filter through four layers of muslin, prewetted with chilled isolation medium.
6. Centrifuge (*see* Note 12) at 2960g for 5 min in a fixed-angle rotor in 250-mL centrifuge tubes, and discard the pellets.
7. Centrifuge the supernatant at 17,700g for 15 min in a fixed-angle rotor in 250-mL centrifuge tubes. Gently, using a fine paint brush, resuspend the pellet in a small volume of wash medium, and dilute to a final volume of 30 mL in the same medium.
8. Centrifuge at 1940g for 10 min in a fixed-angle rotor in 50-mL centrifuge tubes. Discard the pellet. Spin the supernatant at 12,100g for 10 min in a fixed-angle rotor. Gently resuspend the pellet (as before) in a small volume (~1 mL) of wash medium.
9. Load this onto a continuous Percoll/PVP gradient made up as follows. Add 17 mL of solution III to the mixing chamber (i.e., nearest the outlet) of a gradient former and 17 mL of solution II to the left-hand chamber. Form the gradient into a 50-mL centrifuge tube. Centrifuge at 39,200g for 40 min in a fixed-angle rotor. Intact mitochondria form a straw-colored band near to the bottom of the gradient (*see* Note 13).
10. Remove the top half of the gradient, which contains broken chloroplasts (green color), and discard. Using a wide bore Pasteur pipet, carefully remove the mitochondrial fraction without disturbing the rest of the gradient.
11. Add approx 30 mL of wash medium to the mitochondria, and centrifuge at 12,100g for 10 min in a fixed-angle rotor in 50-mL centrifuge tubes.
12. Carefully remove the supernatant by aspiration, taking care not to disturb the soft mitochondrial pellet. Repeat step 11.

13. Carefully remove the supernatant as before, and gently resuspend the purified mitochondria in a minimal volume of wash medium.

3.7. Assay of Mitochondrial Activity

3.7.1. Measurement of Respiratory Control Ratios and P/O Ratios

The assay is carried out in an oxygen electrode (Rank Bros., Cambridge, UK; Hansatech) at 25°C. The reaction mixture consists of 0.3M sucrose, 10 mM MOPS, pH 7.2, 10 mM KH_2PO_4 , 2 mM MgCl_2 , 0.1% (w/v) BSA (defatted), mitochondria (0.5–1.0 mg protein/mL assay), and 10 mM substrate (e.g., malate, glycine) in a final volume of 1.0 mL. Oxygen uptake is measured continuously. The state 3 rate of respiration is the rate of oxygen uptake in the presence of ADP. This is determined by adding 100 μM ADP to the above assay and measuring oxygen uptake. The state 4 rate of oxygen uptake is the rate following state 3 when all of the ADP has been converted to ATP (i.e., state 4 is the “ADP-limited” rate). The Respiratory Control Ratio is the ratio of state 3 to state 4 rates (Fig. 1) (*see* Note 14). The ADP/O ratio is calculated from the amount of oxygen consumed during the phosphorylation of a known amount of added ADP. The ADP/O differs for different substrates. For NAD-linked substrates, the ADP/O ratio is 3.0, and for NADH or succinate it is 2.0.

3.7.2. Assay of Mitochondrial Purity and Intactness

To determine mitochondrial purity, the same marker enzymes are assayed as for chloroplasts. Although there are valid assays for mitochondrial intactness, the preferred method of determining mitochondrial quality is to measure the Respiratory Control Ratio (RCR) and ADP/O ratios. For purified leaf mitochondria, a RCR in excess of 2.0 is acceptable, whereas ADP/O ratios should approach theoretical maximum values (2.0 for NADH or succinate, 3.0 for glycine, malate, and so forth).

4. Notes

1. The age of the plant is an important factor in determining the yield of protoplasts; the yield of intact protoplasts decreases rapidly beyond the age ranges presented here. It may be necessary to vary the concentration of digestive enzymes and/or the digestion time if older or younger material is to be used.
2. Different species may well require different digestive enzymes and different periods of digestion, and it is important to optimize these conditions. A simple method is to carry out small-scale digestions with 5 mL of digestion medium. Approximate protoplast yields may be assessed by measur-

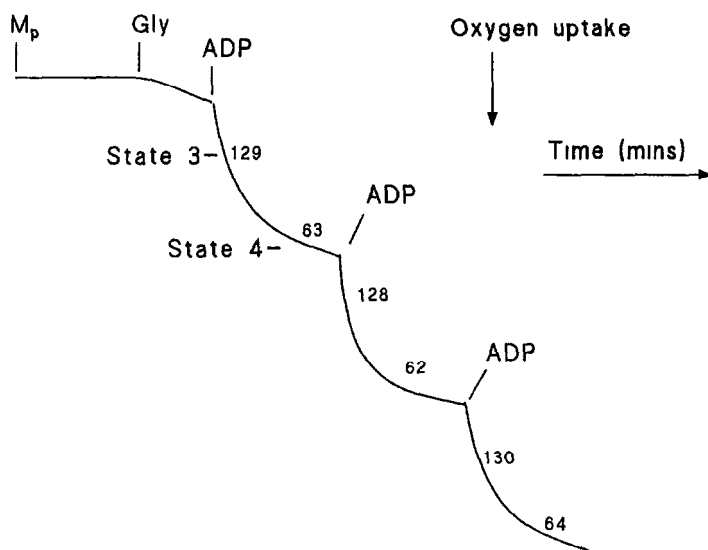


Fig. 1. Oxygen consumption by pea leaf mitochondria. The assay was carried out in an oxygen electrode as described in the text. The following additions were made: purified mitochondria (M_p , 0.4 mg protein), 10 mM glycine (gly), and 100 μ mol ADP (where indicated.). State 3 refers to the rate of oxygen uptake in the presence of saturating concentrations of ADP. State 4 occurs when ADP has been depleted. The numbers on the traces refer to the rate of oxygen uptake in nmol/min/mg mitochondrial protein. The RCR is the state 3 rate divided by the state 4 rate. The ADP/O ratio is the amount of oxygen consumed (in natoms) during the phosphorylation of a given amount of ADP (in nmol).

ing chlorophyll content of the medium after the incubation period and by visual inspection of the extract under a microscope. The chlorophyll content will increase with increasing periods of digestion, but eventually the number of intact protoplasts (readily seen as completely circular in outline under the light microscope) will decrease with prolonged digestion. Since the cellulase and cellulysin preparations are not generally very pure, it is advisable to use the minimum amount during digestion.

3. To break the protoplasts, use a 25-mL disposable syringe adapted as follows: cut off the end section to which the syringe needle is usually attached, so that the barrel becomes an open cylinder. Cover the end with a piece of nylon mesh, pore size 20 μ m. Remove the syringe plunger, and pour the protoplast suspension into the open end of the syringe barrel. Carefully replace the plunger, and displace it slowly so that the protoplasts are forced through the mesh into a collecting beaker on ice. This is usually sufficient

to completely break the protoplasts. Check by inspecting an aliquot under a light microscope. Broken protoplasts will no longer be circular in outline. If intact protoplasts remain, repeat the passage through the mesh. This procedure has to be carried out with care, since rough handling at this stage will result in chloroplast breakage.

4. All centrifugations for protoplast and chloroplast isolation are carried out with the brakes off.
5. The most serious problem during chloroplast isolation is "clumping," where the organelles aggregate into sticky lumps. Whenever this occurs, the quality of the chloroplasts is poor. One contributory factor is the age of the plant material; the older the leaves, the more likely clumping will occur. I have tried varying the EDTA and Mg ion concentration in the resuspension medium, as suggested by other authors, yet this does not always solve the problem. There also appears to be a difference between cultivars. The barley variety Klaxon is particularly problematic, whereas Maris Mink is more reliable. The peeling of the epidermis prior to digestion was found to reduce the problem of chloroplast clumping in barley significantly, and it also increased the yield of intact protoplasts.
6. Chloroplasts isolated from protoplasts should be at least 90% intact and capable of CO_2 -dependent O_2 evolution at rates comparable to those of the isolated protoplasts. For wheat and barley, depending on the growing conditions, rates of at least $40 \mu\text{mol O}_2$ evolved/h/mg chlorophyll should be achieved. Rates for pea will be higher than this, at least $50 \mu\text{mol O}_2$ evolved/h/mg chlorophyll. The rate of CO_2 -dependent O_2 evolution is a better determination of chloroplast quality than is the intactness assay, since it is possible for chloroplasts to break, release their stromal contents, and then reseal. These chloroplasts would appear to be intact, according to the ferricyanide assay, and yet would be incapable of photosynthesis.
7. The conditions required for chloroplasts to carry out CO_2 -dependent O_2 evolution are often quite critical, hence the difference in assay media for the different species. The pH, divalent cation concentration, and phosphate concentration should all be optimized for the species being studied. ATP, ADP, and pyrophosphate may also be required to varying extents.
8. The phenomenon of "induction" is frequently observed in isolated chloroplasts, where a significant lag occurs between the start of illumination and the onset of CO_2 -dependent O_2 evolution. This lag may last for several min and is owing to the depletion of Calvin cycle intermediates from the chloroplast. Preillumination of the leaves with bright light for 30 min prior to extraction will reduce the induction time if this is thought to be a problem.
9. A high stromal starch concentration may result in poor-quality chloroplasts when using mechanical isolation, since starch grains are thought to disrupt

the chloroplast envelope physically during homogenization. To minimize this effect, leaves should be harvested close to the end of the dark period, when starch content will be at a minimum

10. Isolated protoplasts are stable for many hours, whereas chloroplasts will only remain active for a short period of time. Barley and wheat chloroplasts should be used as soon as possible after isolation and will rarely remain viable for more than 2 h. Pea chloroplasts will generally last longer, but it is always good practice to carry out experiments on isolated organelles as soon as possible after isolation. If a large number of experiments is planned, it is possible to make one large protoplast preparation in the morning and to break aliquots at intervals during the day to provide freshly prepared chloroplasts. The remaining protoplasts may be kept on ice until required. It is good practice to determine the rate of CO_2 -dependent O_2 evolution at the beginning and end of the series of experiments to determine whether deterioration has occurred.
11. The method described for the isolation of mitochondria from pea leaves will also work for spinach.
12. All centrifugations for mitochondrial isolation are carried out with the brake on.
13. A major source of contamination of the mitochondria is from broken chloroplasts. Ideally the mitochondrial band that forms on the density gradient should be straw colored and not green. A high chlorophyll content is the result of thylakoid contamination, often caused by too harsh homogenization or overloading of the gradient with too much extract.
14. Isolated leaf mitochondria will only remain active for approx 2 h. It is good practice to determine the RCR at the start and end of the series of experiments to ascertain whether there has been significant deterioration.

References

1. Lilley, R. McC., Fitzgerald, M. P., Rienits, K. G., and Walker, D. A. (1975) Criteria of intactness and the photosynthetic activity of spinach chloroplast preparations. *New Phytol* **75**, 1–10.
2. Foster, J. G., Edwards, G. E., and Winter, K. (1982) Changes in levels of Phosphoenol pyruvate carboxylase with induction of Crassulacean acid metabolism in *Mesembryanthemum crystallinum* L. *Plant Cell Physiol.* **23**, 585–594.
3. Cooper, T. G. and Beevers, H. (1969) Mitochondria and glyoxysomes from castor bean endosperm. Enzyme constituents and catalytic capacity. *J Biol Chem* **244**, 3507–3513.
4. Tolbert, N. E. (1981) Microbodies-peroxisomes and glyoxysomes. *Annu Rev Plant Physiol* **26**, 45–73.
5. Behrends, W., Rausch, U., Löffler, H. G., and Kindl, H. (1982) Purification of glycolate oxidase from greening cucumber cotyledons. *Planta* **156**, 566–571.

CHAPTER 8

The Extraction of Enzymes from Plant Tissues Rich in Phenolic Compounds

William S. Pierpoint

1. Introduction

1.1. The Problems

Many enzyme extracts must, of necessity, be made from green leaves, fruits, and other vegetable tissues that contain large amounts of phenols and polyphenols, which hinder or, unless precautions are taken, completely prevent a successful extraction. In spite of this, the processes by which the polyphenols interfere are rarely studied and still very incompletely understood. Some principles are clear, mostly from work done in past decades, and are applied on an *ad hoc* basis to current problems. They can usually be adapted to devise a successful if rather complicated procedure, but there is seldom the time or interest for researchers to establish what the problems really were and if they could have been better overcome. This is unfortunate, but wholly understandable. The reactions involved are often very complex, demand specialist investigations, and may be only relevant to extraction from a particular plant species.

Part of the difficulty is caused by the vast range of phenolic compounds that may be involved. Harborne (1) estimated that several thousand structures are known, and summarized the vagaries of their distribution. Simple phenols, such as catechol, are comparatively rare or are present only in traces. Phenolic acids, such as p-OH benzoic and syringic acid are almost ubiquitous, as are the phenyl-propanoids (C_6-C_3-), including

From *Methods in Molecular Biology, Vol. 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

caffeic acid and its quinic acid-ester, chlorogenic acid. Thousands of flavonoid compounds are known, differing in hydroxylation pattern, oxidation state of the heterocyclic ring, and degree of glycosylation: some of them are restricted to particular species and particular tissues, whereas others, such as quercetin and cyanidin, are widespread. Perhaps the most notorious phenolic compounds in the present context are the polymeric, astringent tannins, whose protein binding properties have been appreciated in food technology and the leather industry for centuries. Characteristic of some "advanced" orders of cotyledonous plants are the "hydrolyzable" tannins, based on gallic acid residues linked, often as esterified chains, to glucose or some other polyhydric alcohol. More widespread are the "condensed" tannins, oligomers of the flavonoid catechin linked by C_4-C_8 interflavan bonds, which occur in most orders of the vascular plants. These polymers have many of the features of monomeric phenols that predispose them to combine with proteins, but they have more of them and often in a disposition that facilitates and strengthens this reaction.

The initial binding of phenolic compounds, both monomers and polymers, to enzymes and other proteins is via noncovalent forces, which may, initially, be reversible. It is believed that hydrophobic, ionic, and H-bonds may be involved depending on the specific phenol and protein involved. Fully methoxylated phenolics with a high content of aromatic rings are, of course, more likely to be bound hydrophobically. Free phenolic groups, especially vicinal dihydroxy groups, may form hydrogen bonds with, for instance, the CO and NH of peptide bonds. Such complexes have only been investigated thoroughly in a few simple model cases (2,3), and shown to involve a variety of these linkages and also coordination attachments involving cations. Nevertheless, Haslam and his colleagues have extrapolated from such information to produce a model for the binding of polymeric tannins to proteins. It involves two stages. In the first, the tannin is attached to hydrophobic sites on the surface via its aromatic residues. This bonding is then reinforced by the formation of H-bonds between the phenolic groups and nearby polar functions on the protein. The final product is thus a dissociable complex in which the surface of the protein is rendered less hydrophobic and more susceptible to aggregation and precipitation. If the polyphenol is large enough to interact with more than one protein molecule, the likelihood of aggregation and precipitation will, of course, be much increased. More recent developments of this work (4) have emphasized the structural fea-

tures of proteins, including those of the proline-rich, salivary proteins of herbivores, that predispose them to such coupling with tannins: for most enzymes, this complexing facilitates inactivation.

Phenolic compounds form irreversible covalent linkages with proteins mainly as a consequence of the oxidation of their vicinal dihydroxy groups to quinones or semiquinones. These oxidations may occur nonenzymically in alkaline conditions, especially in the presence of metal ions. They are, however, catalyzed by a variety of enzymes, including *o*-diphenol oxidases (polyphenoloxidases; PPO), monophenoloxidases (laccases), and by peroxidases in the presence of H_2O_2 . The resulting quinone molecules are highly reactive, not only polymerizing with each other, but oxidizing other phenolic compounds and, most relevant in the present context, combining with reactive groups in proteins causing aggregation, crosslinking, and precipitation. These reactions are described as “enzymic-browning,” for the products, although complex and poorly defined, are usually brown in color. Leaf extracts that brown rapidly are generally regarded (5) as poor sources of active enzymes. Insights on aspects of the browning reactions that affect proteins have been gained from studies of single proteins in simple, model oxidizing systems (6). They emphasize the vulnerability of nucleophilic groups, such as the NH_2 - and SH -groups in amino acid side chains to substitution reactions with quinone rings to give protein N- or S-substituted phenols. These may be re-oxidized by excess quinone to the *o*-quinone state, when they have the potential to react with other nucleophiles producing intra- or interprotein crosslinks, or, in more alkaline conditions, react with quinones giving more complex, greenish, protein N-substituted hydroxyquinone polymers. Other reactions may occur depending on the conditions and especially on the phenols being oxidized and the nature of the oxidizing system. It would take a major analytical effort to characterize satisfactorily the products formed when leaf proteins are exposed to the “natural” enzymic browning reactions of leaf extracts.

1.2. Some Solutions

In most cases, where it is required to extract enzymes from phenol-rich tissue, the specific phenolic compounds and oxidizing systems that are present are unknown or can only be guessed at. An ideal approach would be to establish their nature so that the simplest method of preventing interference could be established. Thus, Gray (7) described how the

phenolic compounds of bean leaves could be simply extracted and a suitable adsorbent for them chosen. A more usual approach is to follow a general procedure that has worked for other tissues and deals with as many eventualities as possible. Generally, these procedures involve disrupting fresh or deeply frozen (-70°C) tissue as quickly as possible in the presence of polymers that adsorb phenolic compounds, both monomers and polymers, and in conditions that minimize the oxidative reactions which produce quinonoid compounds.

Many polymers, natural and synthetic, have been used to adsorb phenols. They include albumins, hide powders, powdered nylon (ultramid), polyvinylpyrrolidones both soluble (PVP) and insoluble (polyvinylpolypyrrolidone; PVPP), relatively uncharged polystyrene and polyacrylic resins, such as Amberlites XAD-2, -4, and -7, and ion-exchange resins based on polystyrenes, both anion exchangers (Bio-Rad AG1-X8, AG2-X8, and Dowex-1) and cation exchangers (Dowex-50). These polymers are listed in the reviews and papers by Loomis (5), Loomis et al. (8), Rhodes (9), and Smith and Montgomery (10). They interact with phenolic compounds in different ways so that PVP, for example, is thought to form stable H-bonds to phenol groups via its $-\text{CO}-\text{N}<$ linkages, whereas the porous polystyrene resins present large, adsorptive, hydrophobic surfaces. As a consequence, their affinities for different phenols differ. The adsorbents may either be soluble, as are the tannin-binding proteins and PVP, or, more usually, insoluble like PVPP and the resins, so that they can be readily removed from the tissue extracts.

An obvious way of preventing the oxidative reactions is to work in anaerobic conditions, but this is both difficult and cumbersome. It is much easier to make extracts in the presence of low-mol-wt compounds that form unoxidizable complexes with phenols, or that either inhibit oxidases, trap quinones, or reduce quinones back to phenols (5,6,11). Borate and germanate have been used to complex phenols, and copper-chelating agents, such as diethyldithiocarbamate (Dieca), used to inhibit copper-dependent PPOs. Quinone trapping agents that have been used include benzene sulfinic acid, and a range of substances, including ascorbate, metabisulfite, and 2-mercaptoethanol used as quinone reductants. However, detailed studies (e.g., 12) have made it clear that the action of many of these compounds cannot be simply explained, and that they may act in more than one manner. Thus, thioglycollate both inhibits oxidases and reduces quinones, and Dieca inhibits oxidases and reacts with quino-

nes; even PVP, primarily thought of as a phenol-adsorbent, is also an oxidase inhibitor.

The complexity of leaf extracts and of the reactions that may occur in them thus makes it difficult, if not impossible, to write a procedure suitable for extracting enzymes from all phenol-rich plant tissues. The method described here is the first few stages in the procedure used for extracting the photosynthetic enzyme ribulose biphosphate carboxylase (Rubisco; EC 4.1.1.39) from green tissue (13). It has been used routinely by Keys and his colleagues for many years and, with a few adaptations, applied successfully to a wide range of plants, including some ferns and mosses. More recently, with the modification mentioned later, it has been used to extract active Rubisco from the "difficult" leaves of Mediterranean tree species (14). The subsequent purification steps specific to Rubisco have been omitted; the extracts, however, should be a suitable starting material for the purification of many other soluble enzymes by appropriate, specific procedures.

2. Materials

1. Use reagents of AR quality wherever possible, and otherwise of the highest standard available.
2. Ammonium sulfate: Use the grade (BDH-Merck, Lutterworth, Leicestershire, UK) especially low in heavy metals that is suitable for enzyme work.
3. PVPP (Polyclar AT; Sigma-Aldrich, Poole, Dorset, UK): free from metal ions and other contaminants by boiling in 10% HCl for 10 min and then washing extensively with glass-distilled water (5). Air-dry for storage and rehydrate for at least 3 h before using: Hydration increases the weight of the polymer about fivefold (8).
4. Extraction buffer: 100 mM HEPES/NaOH, 10 mM NaHCO₃, 10 mM MgCl₂, 0.1 mM EDTA, 10 mM 2-mercaptoethanol (2-Me), 10 mM sodium diethyldithiocarbamate (Dieca), 0.1% Tween 80. Prepare freshly overnight, adjust its pH to 7.0 (*see Note 4*) before the addition of NaHCO₃ and Dieca, and readjust it afterward if necessary. Add 2-Me just before use. NaHCO₃ is added specifically to protect Rubisco activity and is unnecessary in extracting other enzymes.
5. Resuspension buffer: Tris-HCl, pH 7.0, containing 1.0 mM dithiothreitol (DTT).

3. Methods

3.1. Extraction of Proteins (*see Note 1*)

1. Cut leaf material (about 100 g), without mincing, into small pieces, and immerse in 500 mL of cold extraction medium contained in the bowl of a chilled Ato-Mix (MSE) or similar blender (*see Note 2*).

2. Add insoluble PVPP (4–8 g), which has been soaked overnight in 250 mL of extraction medium, and homogenize the mixture at high speed for four periods of 15 s with a 30-s interval between the periods.
3. Filter the extract through four layers of muslin into a measuring cylinder, and allow to stand to settle any froth (*see* Note 3).
4. Treat the extract with a further addition of 4 g of PVPP, which has been soaked overnight in the extraction buffer, stir gently for 10 min, and clarify by centrifugation at 10,000g for 30 min to remove PVPP and cell residues. The resulting supernatant fluid provides a suitable starting material for purification procedures appropriate for particular enzymes (*see* Notes 5 and 6).

For enzymes like Rubisco, which can be further purified and concentrated by ammonium sulfate precipitation, this second treatment with PVPP can conveniently be included in the precipitation schedule described in Section 3.2., and the PVPP removed along with a fraction of inactive protein. The ammonium sulfate concentrations are those found to be suitable for Rubisco; other enzymes may require different concentrations.

3.2. Ammonium Sulfate Fractionation of the Extract

- 1 After the second addition of PVPP, transfer the extract into a beaker and add ammonium sulfate (20 g/100 mL, 35% saturation) in portions with continuous stirring until it has dissolved.
2. Transfer the extract to centrifuge bottles, allow to stand for 30 min, and centrifuge at 25,000g for 15 min.
3. Discard the sediment containing PVPP and inactive protein. Filter the supernatant through eight layers of muslin, treat with a further 12 g/100 mL of ammonium sulfate, allow it to stand for 30 min, and centrifuge as before.
4. Resuspend this protein fraction, precipitated between 35 and 55% saturation, in 25 mL of 20 mM Tris-HCl, pH 7.0, containing 10 mM DTT, and treat for a third time with 3 g of PVPP that has been soaked in this resuspension buffer overnight.
5. After allowing it to stand for 10 min with occasional stirring, centrifuge the suspension at 100,000g for 30 min, and filter the supernatant liquid, containing active Rubisco, through a 50- μ m mesh nylon gauze. This extract can be used for further stages of Rubisco's purification (*see* Note 7).

4. Notes

- 1 All operations were performed as far as possible at 5°C either in a cold room or on a cold bench. Apparatus and solutions were cooled overnight. Operations were performed as quickly as possible to minimize exposure of enzymes to phenolic compounds.

2. The use of relatively large amounts of extraction liquid, usually five to seven times the weight of the leaf material, ensures that cells are disrupted while they are submerged, that the concentration of liberated phenols and oxidative enzymes is immediately diluted, and that there is an adequate supply of phenol adsorbent and oxidase inhibitors. The volume of the medium should be such that the speed of blender blades, which is adequate to disrupt the leaves, does not produce undue vortexing and the sparging of air into the homogenate.
3. The original extraction medium contained 10 times as much Tween 80, but this often gave rise to excessive frothing.
4. The pH of the extraction medium will affect some interactions of phenols with proteins and adsorbents. Hydrophobic interactions will presumably be unaffected. Ionization of phenolic groups in alkaline conditions will prevent the formation of H-bonds with proteins, but it will also prevent the formation of H-bonds with PVPP. Alkaline conditions (pH > 7.5) will increase the auto-oxidation of dihydroxy phenols to quinonoids, and also the auto-oxidation of protective thiol reagents. In the instances cited by Loomis (5), enzyme extraction was generally optimal at pH values between 7.0 and 6.5. The optimal pH to extract an uncharacterized enzyme from an unfamiliar, phenol-containing plant, is a matter for experimentation.
5. Choice of polyphenol-adsorbents:
 - a. Insoluble adsorbents: PVPP was chosen as it is an excellent adsorbent of preformed tannins as well as many monomeric phenols. Its affinity for different classes of compounds is illustrated in Fig. 1 taken from the experiments of Gray (7). It is less effective in absorbing hydroxycinnamic acids, such as chlorogenic acid, that are common, widespread substrates for the browning reaction. In this respect, ion-exchange resins, such as the anion-exchanger Dowex-1 (Fig. 1) or the uncharged polystyrene Amberlites XAD-2 and XAD-4 (Rohm and Haas Co., Philadelphia, PA) are more effective. Dowex-1, X10 (Bio-Rad, Hercules, CA) was the preferred adsorbent used for the extraction of mevalonate kinase from the leaves of French beans (*Phaseolus vulgaris*, 7); it was pre-equilibrated with the buffered (sodium phosphate; 50 mM, pH 7.5) extraction medium, and used at the rate of 20 g/100 mL of extraction medium. The Amberlite resins, especially XAD-4, appear to have been used more widely, presumably because of their great porosity and hydrophobic surface area and relative freedom from charged groups. They require extensive washing with water and methanol following manufacturer's instructions (4), but this can be done in bulk so that they may be stored, ready for use, in a moist condition. They have been used in specific extraction procedures in quantities ranging from 0.1 g (15) to 1–2 g (8) hydrated weight/g of fresh tissue. It must be remembered

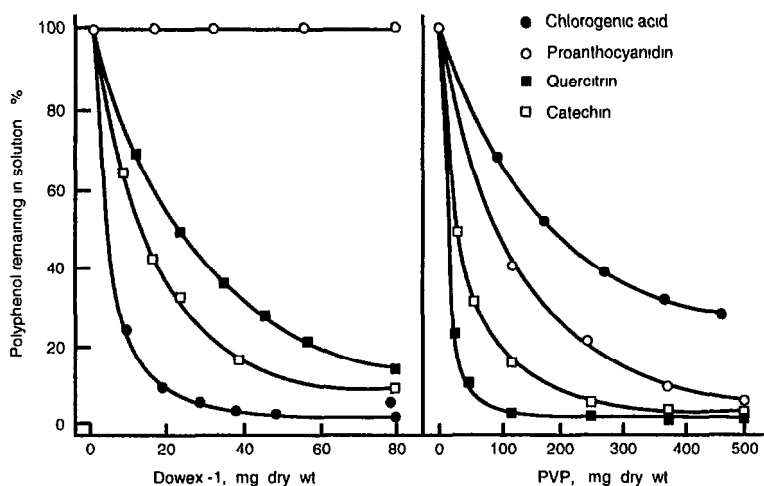


Fig. 1. Adsorption of different types of polyphenols by Dowex-1 (left) and PVPP (labeled PVP, right). Two milligrams of polyphenol in 3 mL phosphate buffer (50 mM; pH 7.5) were blended for 30 s with different amounts of hydrated adsorbents. The mixture was centrifuged and unadsorbed polyphenol in the supernatant estimated from $A_{280 \text{ nm}}$. Reprinted from Gray (7) by kind permission of the author and the editor on behalf of Pergamon Press.

that these resins may absorb protein as well as polyphenols, although tests using bovine serum albumin (BSA) and plant peroxidases suggest that this is negligible with the Amberlite resins at broadly neutral pH values (8). The ability of these resins to adsorb leaf components other than polyphenols, which interfere with enzyme extraction, such as the monoterpenes from peppermint leaves and the isothiocyanates of horseradish (8), are additional advantages of their use.

A possible disadvantage of the presence of either PVPP or polystyrene polymers during leaf disruption, especially when blenders are used, is a small degree of breakdown. Thus, PVPP is reported to produce soluble PVPP-protein complexes in the extract, and the resins release UV-absorbing compounds, such as benzoic acid, following blending. The adsorbents could, of course, be added after leaf disintegration or used as columns through which the extracts could be filtered. Any advantages of these procedures would be offset by the disadvantage of not having adsorbents present during leaf disruption when cell proteins are first exposed to liberated polyphenols.

Because the adsorptive abilities of PVPP and polystyrene resins are different and to some extent complementary (e.g., Fig. 1), they have

often been used together when extracting refractory plant tissues. Thus, PVPP and XAD-4 (5 and 1 g hydrated weight, respectively, per g tissue) have been used successfully in extracting active enzymes from such unpromising material as apple fruits, peppermint leaves, and even the hulls of walnuts (8). More recently, a mixture of PVPP and XAD-4 was shown to be most effective in extracting polyphenoloxidases from strawberry fruits (16).

- b. Soluble polyphenol absorbents: A recent procedure for extracting Rubisco from the leaves of Mediterranean trees uses casein (2% [w/v]) as well as PVPP (1% [w/v]) in the extraction medium (14). A more usual protein additive is BSA used at about 1%. These proteins are known to react with tannins as well as with enzyme-generated quinones (17). BSA is especially helpful in the extraction of organelles, such as mitochondria (*see ref. 5*), when the protein, along with adsorbed and reacted polyphenols, is effectively removed during centrifugal sedimentation of the organelles. The separation of added casein from Rubisco is, apparently, satisfactorily accomplished in the subsequent chromatographic stages of Rubisco purification. Rubisco, however, is the major protein component of leaves. Researchers attempting to purify enzymes present in very small amounts are often reluctant to add large amounts of protein to their extracts.

Soluble forms of PVP have been used, and indeed were the first forms of this type of polymer to be used as polyphenol adsorbents. They adsorb phenolic compounds, but are unlikely to react with generated quinones. They are commercially available in three forms, with molecular weights about 10,000, 40,000, and 360,000, respectively, so that a size suitable for separation by gel filtration from a specific enzyme can be conveniently chosen. Earlier publications (5,8) emphasized the necessity of using only pharmaceutical grades of these chemicals: the grades supplied by a major supplier in the UK (Sigma) are presumed to be satisfactory, as they have been tested for suitability in such demanding processes as plant tissue culture and nucleic acid manipulation.

6. Choice of oxidase inhibitor/antioxidant: The procedure described uses both 2-ME (10 mM) and Dica (10 mM) to prevent the browning reactions. They both inhibit PPO, and this may be the principal cause of their effectiveness, although they also combine with quinones, either reducing them, complexing them, or both. A more recent modification of the procedure (14) omits the Dica and increases the concentration of 2-Me to 100 mM. This modification also includes casein (2% [w/v]) in the extraction medium and this, as mentioned, also reacts with quinones.

A brief survey of extraction procedures described in the then current volumes of *Phytochemistry* (6), suggested that 2-Me was a common choice of antioxidant, as were dithiothreitol (DTT) and dithioerythritol (DTE), the two stereoisomers of Cleland's reagent. Ascorbate was used less commonly. DTT, like 2-Me, appears to act as an enzyme inhibitor and also as a quinone scavenger, but the action of neither DTT nor DTE has been studied in detail. Ascorbate probably acts principally by reducing quinones back to re-oxidizable phenols, and so is readily removed by low concentrations of enzyme-produced quinones (12). It should therefore be used in relatively high (>50 mM) concentrations. The thiol reagents, while they may be very effective in many, if not most cases, may be much less so in others. 2-Me activates unwanted proteases in extracts of tobacco leaves (6), and the technical literature on DTT and DTE lists a number of enzymes that are inhibited by these reagents.

Many other compounds have proven useful as inhibitors/antioxidants, but have been less generally used. Mercaptobenzothiazole, a copper chelator that powerfully inhibits PPO, has been advocated by Palmer (cited in ref. 5), and at low concentrations (0.1 mM) aids in extracting enzymes from tobacco leaves and mitochondria from potato tubers. Inorganic sulfites, especially metabisulfite, are effective in the extraction of enzymes from many leaves (9,18), although their potential to cleave bonds in proteins demands a careful choice of concentration (approx 4 mM). Benzene sulfinic acid is a milder reagent that is thought principally to be a quinone scavenger, and is effective in extracting acylases from tobacco leaves (19). KCN powerfully inhibits some oxidases, but is, understandably, now seldom used for this purpose. At 15 mM, it is reported to be a less effective inhibitor of the PPO from apple peel than Dieca, potassium metabisulfite, 2-Me, or ascorbic acid. It was used (1 mM), in conjunction with sodium metabisulfite (10 mM), in successfully extracting mevalonic acid kinase from the phenol-rich leaves of French beans (20).

7. All the extracts, including the final one are likely to contain active proteinases, which may modify or inactivate sensitive enzymes. Rubisco in extracts from young green tissues is apparently insensitive to them. Another photosynthetic enzyme, the phosphatase that hydrolyzes 2'-carboxy-D arabinitol 1-phosphate, an inhibitor of Rubisco that is induced in leaves during darkness, is very sensitive. During its purification from the leaves of French beans, using a method similar to that described in Section 3. (21), it is necessary to include the proteinase inhibitors phenylmethylsulfonyl fluoride (PMSF; 1 mM) and either benzamidine hydrochloride (1 mM) or p-aminobenzamidine dihydrochloride (1 mM) in the extraction buffer, in the resuspension buffer, and also in the liquid (1 L) against which the resuspended

ammonium sulfate precipitate is dialyzed overnight before chromatography on an affinity column. These inhibitors are added to buffer solutions and dialysis water along with 2-Me (50 mM) just before use. PMSF is usually dissolved in the minimal amount of ethanol before addition to the liquids.

Acknowledgment

The author acknowledges the help and encouragement of A. J. Keys and M. A. J. Parry.

References

1. Harborne, J. B (1980) Plant phenolics, in *Encyclopedia of Plant Physiology* vol 8 (Bell, E. A and Charlwood, B V, eds), Springer-Verlag, Berlin, Germany, pp 329–402
2. Spencer, C M, Cai, Y, Martin, R, Gaffney, S H., Goulding, P N, Magnolato, D., Lilley, T. H., and Haslam, E. (1988) Polyphenol complexation—some thoughts and observations *Phytochemistry* **27**, 2397–2409
3. Haslam, E (1989) *Plant Polyphenols Vegetable Tannins Revisited* Cambridge University Press, Cambridge, UK.
4. Luck, G, Liao, H, Murray, N J, Grimmer, H R, Warminski, E E, Williamson, M P, Lilley, T. H., and Haslam, E (1994) Polyphenols, astringency and proline-rich proteins. *Phytochemistry* **37**, 357–371.
5. Loomis, W D (1974) Overcoming problems of phenolics and quinones in the isolation of plant enzymes and organelles *Methods Enzymol* **31**, 528–544
6. Jervis, L. and Pierpoint, W S (1989) Purification technologies for plant proteins *J Biotechnol* **11**, 161–198
7. Gray, J C (1978) Absorption of polyphenols by polyvinylpyrrolidone and polystyrene resins *Phytochemistry* **17**, 495–497.
8. Loomis, W D, Lile, J D, Sandstrom, R P, and Burbott, A J. (1979) Adsorbent polystyrene as an aid in plant enzyme isolation. *Phytochemistry* **18**, 1049–1054
9. Rhodes, M. J C (1977) The extraction and purification of enzymes from plant tissue, in *Regulation of Enzyme Synthesis and Activity in Higher Plants* (Smith, H, ed.), Academic, London, UK, *Proc Phytochem Soc Eur* **14**, 245–269
10. Smith, D. M and Montgomery, M W. (1985) Improved methods for the extraction of polyphenol oxidase from d'Anjou pears *Phytochemistry* **24**, 901–904
11. Anderson, J W (1968) Extraction of enzymes and subcellular organelles from plant tissues. *Phytochemistry* **7**, 1973–1988.
12. Pierpoint, W S (1966) The enzymic oxidation of chlorogenic acid and some reactions of the quinone produced *Biochem J* **98**, 567–580.
13. Bird, I. F., Cornelius, M J, and Keys, A. J (1982) Affinity of RuBP carboxylases for carbon dioxide and inhibition of the enzymes by oxygen *J Exp Bot* **33**, 1004–1013
14. Delgado, E., Medrano, H, Keys, A. J, and Parry, M A. J (1995) Species variation in Rubisco specificity factor *J Exp Bot* **46**, 1775–1777
15. Hallahan, D L., Dawson, G W., West, J M, and Wallsgrave, R M (1992) Cytochrome P-450 catalysed monoterpene hydroxylation in *Nepeta mussini* *Plant Physiol Biochem* **30**, 435–443

- 16 Wesche-Ebeling, P. and Montgomery, M. W. (1990) Strawberry polyphenol-oxidase: extraction and partial characterization. *J Food Sci* **55**, 1320–1324
- 17 Pierpoint, W. S. (1969) *o*-Quinones formed in plant extracts: their reaction with bovine serum albumin. *Biochem J* **112**, 619–629.
- 18 Anderson, J. W. and Rowan, K. S. (1967) Extraction of soluble leaf enzymes with thiols and other reducing agents. *Phytochemistry* **6**, 1047–1056
- 19 Pierpoint, W. S. (1973) An *N*-acylamino acid acylase from *Nicotiana tabacum* leaves. *Phytochemistry* **12**, 2359–2364
- 20 Gray, J. C. and Kekwick, R. G. O. (1973) Mevalonate kinase in green leaves and etiolated cotyledons of the French bean *Phaseolus vulgaris*. *Biochem J* **133**, 335–347
- 21 Kingston-Smith, A. H., Major, I., Parry, M. A. J., and Keys, A. J. (1992) Purification and properties of a phosphatase in French bean (*Phaseolus vulgaris*) leaves that hydrolyzes 2'-carboxy-D-arabinitol 1-phosphate. *Biochem J* **287**, 821–825

CHAPTER 9

Avoidance of Proteolysis in Extracts

Robert J. Beynon and Simon Oliver

1. Introduction

Proteolytic enzymes (or proteases, peptidases, or proteinases) hydrolyze the peptide bond in proteins and peptides. The nomenclature is imprecise, but there is a broad acceptance that endopeptidases break bonds that are “internal” in the primary sequences, whereas exopeptidases trim one, two, or perhaps three amino acids from the amino or carboxy terminus of the substrate. Every cell and subcellular compartment has its own complement of proteolytic enzymes, and in normal circumstances, the activities of the proteolytic enzymes are well regulated. When a tissue is disrupted, however, this control is lost, and the proteinases may then attack proteins at a rate that leads to a loss of those proteins within the time scale of the study. Adventitious proteolysis is a technical problem that may require modification to methodology to minimize the assault on the protein of interest (1–4).

The isolation of almost every protein will incur losses, and in some circumstances, those losses become unacceptable. Exactly what constitutes “unacceptable” will be dictated by individual needs, and will depend on downstream requirements and time scale. Few purification protocols generate pure protein, but rather, reduce contaminant proteins to the level where they cannot be detected by routine methods—consider the difference between a Coomassie-blue-stained gel and a silver-stained gel—the additional sensitivity of the latter will usually identify many contaminants in a preparation that looked homogenous by the former. Even apparently pure proteins may contain low levels of contaminating

From *Methods in Molecular Biology, Vol. 59 Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

peptidases, which may explain the tendency to assume that any loss of protein, or of biological activity associated with that protein is owing to proteolytic destruction. Loss of material or activity can, of course, be owing to other, nonproteolytic processes—a strategy based on suppression of proteolysis will only be effective if proteolysis is really the culprit. In this chapter, therefore, we advocate the use of methods to prove that proteolysis is a problem and to measure proteolytic activity directly in the samples, as a prerequisite to prevention.

As mentioned previously, loss of material, as protein, or as a biological activity has many potential causes. Eliminate the possibility of, for example, thermal denaturation, oxidative damage, adsorption onto surfaces, or persistent binding to the column matrix. More subtle reasons for losses of activity might be complexation with an inhibitor in a time-dependent process, or loss of an essential cofactor or accessory protein as a consequence of a purification step. In crude extracts, and particularly when the protein has yet to be purified, discriminating between proteolysis and other events is difficult, but it is as well to be aware of the range of traumas that a protein can undergo after it has been removed from the normal cellular milieu, substantially diluted and removed from the protective effects of other proteins and low-mol wt ligands, such as natural reductants. Even if recoveries are initially good, proteolysis can become a problem during storage. Slow proteolytic degradation might destroy the material, or worse, allow it to retain some biological functions, but change others. Awareness of this possibility is the first step in its control.

Proteolysis of a protein may also occur *in vivo* before the cells are disrupted. This is sometimes evident in heterologous expression systems, where a protein is exposed to different proteolytic systems in the host cell. Strategies for prevention of this type of intracellular degradation are very different from those for prevention of postextraction degradation and are not discussed here.

1.1. Recognition of Proteolytic Attack

1.1.1. Why are Proteins Susceptible to Digestion?

The classical studies of Linderström-Lang and colleagues, over 40 years ago, opened the way to an understanding of the attack by proteinases on native structures (5,6). Two types of proteolysis were defined: a “zipper” mechanism wherein the proteinase “nicks” one substrate mol-

ecule, and then attacks a further molecule, often at the same site, or, alternatively, “all or none” proteolysis in which the initial proteolytic attack renders the products much more likely to be further digested, such that they are quickly and completely digested to limit peptides. In the first mode, the products are resistant to further digestion, and accumulate as partially degraded proteins that may retain some biological function—the Klenow fragment of DNA polymerase, for example. In the second mode, the products of the first proteolysis are destabilized and become more susceptible to further digestion, such that transient intermediate products are unlikely to be observed.

Compared to unfolded polypeptides, most proteins are in fact relatively resistant to attack by endo- and exopeptidases. A denatured protein is usually much easier to digest than the native counterpart—denaturation and enhanced digestibility are in part the function of the low pH of the stomach and the lysosome. Folding of a protein chain into a compact tertiary or quaternary structure protects many peptide bonds because they are internalized in the protein. Moreover, there is accumulating evidence that proteolytic susceptibility can, in part, be attributed to the ability of the whole or a segment of a protein chain to be flexible (7,8). The more flexible a loop, the more likely it is to be accommodated in the active site of the proteinases in a productive interaction. Thus, proteolytic attack can be dependent on the ability to cut a specific site on the protein that is flexible and susceptible. After the first nick, the behavior of the protein can adopt either of the two extreme models described by Linderstrom-Lang. Treatments that diminish this flexibility will protect the protein, and thus achieve the same goal as inhibition or removal of the proteinases.

1.1.2. Experimental Evidence for Proteolytic Degradation

Evidence for proteolytic degradation of a protein in a crude extract or during the course of a protein purification comes from several sources. A few guidelines may be provided herein, but we stress that each case must be considered unique, and that there is no general solution that will apply in all cases.

1. Loss of biological activity: If the only parameter that can be measured is biological activity, proving that proteolysis is responsible may be difficult. As discussed in Section 1., there are many reasons why the activity of a protein may be diminished. It may be necessary to conduct some analyses of the time-course of inactivation. Is the inactivation first order? Is the rate of inactivation reduced by addition of stabilizers, such as glycerol or non-

ionic detergents? Can inactivation be enhanced or prevented by addition of proteinase activators or inhibitors (*see* Section 1.3.)?

2. Loss of protein. Loss of the protein of interest is harder to assess, particularly if the protein is in low abundance in a crude mixture, such as a cell extract. If it is naturally abundant, or is expressed heterologously at high levels, it may be possible to identify the protein on 1-D gel electrophoresis. Under circumstances in which, for example, 50% of the biological activity is lost, it ought to be possible to *see* a similar loss of material on the gel. Of course, this strategy cannot discriminate proteolytic destruction of the protein from, for example, absorption to glass or plastic surfaces. Experiments should be conducted with siliconized glassware, in the presence of a protective protein of different size, such as albumin, or in the presence of a nonionic detergent, such as Tween-80 at 0.1% (w/v).

Monitoring of proteolysis is simpler if an antibody to the protein can be used for Western blotting. The protein can be identified at low concentrations, and degradation will be manifest as loss of the band of the correct molecular weight, possibly with the appearance of degradation products. Reducing and nonreducing gels may give different results, depending on the relative disposition of nicksites and disulfide-linked cysteine residues, and both types of gels should be used if the protein is suspected or known to contain disulfide bonds

3. Microheterogeneity: Proteolysis can manifest itself as a degree of heterogeneity in a finished product. Multiply sized bands on SDS-PAGE, or on isoelectric focusing may indicate limited proteolytic attack. A blotting antibody can help to confirm that the heterogeneity is owing to the protein of interest.

If sufficient material is available for electrospray or MALDI-TOF mass spectrometry, then the precise mass of the isolated protein and the extent of heterogeneity can be checked directly. It will be important to know the extent of posttranslational modification that the protein might have undergone

1.2. Strategies for Prevention of Proteolysis

Although individual cases will require unique solutions, there are some general principles that can be addressed. First, cells differ in the intracellular concentrations of proteinases, and artifactual proteolysis can be diminished by the use of a cell/tissue in which proteinase levels are low. For example, liver and kidney contain much higher amounts of proteolytic activity than skeletal or cardiac muscle. In many single-cell systems, mutant strains, defective in the expression of one or several proteinases, have been developed.

Many proteinases coexist with naturally occurring inhibitors. Complexation between proteinase and inhibitor may reduce the proteolytic

activity in crude broken cell preparations, and this can be advantageous. However, in a subsequent purification step, the proteinase and inhibitor may be separated, and the proteolytic degradation may manifest itself late in the purification protocol.

Proteins offer protection to each other by competing for the active site of the proteinase. The early stages of a protein purification, in which the goal is to eliminate much of the bulk, unwanted protein, may therefore have the effect of rendering a protein more readily digested by a contaminating proteinase. Proteolytic losses can therefore, paradoxically, become more of a problem as the protein is purified.

Unlike many procedures for purification of nucleic acids, which use potent denaturants to destroy nucleases, the need to preserve the native structure of proteins means that such denaturants cannot be used. Also, many proteinases are more stable to a range of denaturants than their substrates. A denaturing step, which has the aim of inactivating the proteinase to diminish proteolysis, may in fact have the opposite effect, and render the protein more susceptible to digestion. This behavior is often seen in sample preparation for SDS-PAGE—when the sample is heated in SDS containing buffer, the protein is unfolded and undergoes rapid destruction, with apparent loss of material from the expected position on the gel. We routinely precipitate samples with trichloroacetic acid (TCA) before SDS-PAGE. This concentrates the samples, eliminates a great deal of buffer variability, and denatures proteinases and substrates in a rapid step that does not allow time for extensive artifactual proteolysis. Typically, a final concentration of TCA of 5% (w/v) at 4°C for 10 min will be adequate. The precipitated proteins are recovered by brief centrifugation (10,000g for 1 min, in a benchtop microcentrifuge), and the TCA is poured away. Residual TCA, which would otherwise change the pH of the sample buffer, is removed by repeated gentle washing with acetone or diethyl ether, and after three washes, the protein pellet is redissolved in SDS-PAGE sample buffer.

Routine additions to buffers can modify the activity of proteinases in the preparation. For example, reducing agents, such as 2-mercaptoethanol, activate cysteine proteinases, but inhibit metalloproteinases. Chelators also inhibit metalloproteinases. Some proteinases are activated by the presence of calcium ions. Thus, the choice of buffer may influence proteinase activity.

Thus, judicious choice of appropriate methodology may help to diminish the risk of proteolysis. However, there are many circumstances in

which it is not possible to separate proteins and potentially damaging proteinases. Under these circumstances, the only real option is to inhibit the proteinases *in situ*.

1.3. Proteinases and Their Inhibition

A strategy of prevention of proteinase activity is usually based on a combination of two approaches: inhibition of the proteinases *in situ* and separation of the proteinases from the protein of interest. Proteinases bring about the hydrolysis of peptide bonds by several catalytic mechanisms, and inhibition of their action will be different for each mechanistic class (9). Mechanism-based inhibitors will comprise a reactive group that is susceptible to attack by the active site residues, and high-affinity interaction of the reactive group with the enzyme can be achieved by additional functional groups that bind, for example, to the specificity pocket or the extended subsites of the proteinases. Most of these inhibitors bring about covalent modification of the active site residues, and are thus irreversible. Once the proteinase has been inhibited, it should not be necessary to add further inhibitor.

Another class of inhibitors capitalizes on the affinity of the proteinases for a substrate analog, without covalent interaction between them. These noncovalent inhibitors can be proteins or low-mol wt peptide analogs. Such inhibitors may also be usefully immobilized on columns for removal of the proteinases from the protein solution, although this is rarely used as an option. The complex formed between the proteinase and the inhibitor is reversible, and depends on the concentration of inhibitor in solution. If that inhibitor concentration is reduced by, for example, inactivation or dialysis, then the proteinase will recover activity. Strategies that add inhibitor to a crude preparation, which then undergoes precipitation (by ammonium sulfate or polyethylene glycol), column chromatography, or dialysis, must ensure that fresh inhibitors are added when appropriate.

In a few instances, the noncovalent binding is so tight that to all practical intents and purposes, the inhibitor can be considered as practically irreversible, and it will only be necessary to sustain low concentrations of inhibitor in buffers.

Serine proteinases use a nucleophilic serine residue to attack the carbonyl carbon of the scissile bond. The serine residue is a much stronger nucleophile than other serine residues in proteins, and this extreme prop-

erty is facilitated by input of electrons from an aspartate residue via a histidine residue—the charge relay system. The serine and the histidine residues are both targets for inhibition. The inhibitors used most commonly are sulfonyl fluorides of which the most common is phenylmethyl sulfonyl fluoride (PMSF). All sulfonyl fluorides are unstable in solution, and a common mistake is to prepare stock solutions of PMSF in buffer or to assume that the inhibitor continues to work for extended times. At neutral pH values and 25°C, PMSF has a half-life of <1 h. Provided all of the serine proteinases are inhibited within this time scale, PMSF is effective, but exposure of cryptic proteinase activity later in a purification would need PMSF to be added afresh. Stock solutions are normally stored at –20°C in dry organic solvents, in which state PMSF is stable for weeks. Sulfonyl fluorides are also potent inhibitors of acetylcholinesterase and are thus toxic. A second class of proteinase inhibitors are derived from coumarins, of which 3,4 dichloroisocoumarin (3,4-DCI) is best known. Unlike PMSF, 3,4-DCI does not react quickly with acetylcholinesterase and is relatively nontoxic. Although 3,4-DCI is inactivated quite quickly in aqueous buffers (half-life about 20 min at neutral pH values), stock solutions in organic solvents are relatively stable.

Cysteine proteinases use a cysteine residue as the nucleophile to attack the scissile bond, and most inhibitors target this cysteine residue. The reactivity of the cysteine residue means that it reacts well with general-purpose —SH reactive reagents, such as iodoacetamide or iodoacetic acid, but these also modify free —SH groups in other proteins. If a cysteine proteinase is suspected, it may be preferable to use the epoxide inhibitor E-64 (L-trans-epoxysuccinyl-leucylamide-[4-guanidino]-butane, *N*-[*N*-L-3-transcarboxyrane-2-carbonyl]-L-leucyl-*α*-methyl-L-glutamate). This is a potent irreversible inhibitor of cysteine proteases that does not react with other —SH groups in proteins and that will not react with, and be consumed by, other low-mol wt thiols. Stock solutions are stable for days in aqueous buffers.

Some peptide aldehydes, such as leupeptin, chymostatin, and elastatinal, have been used to inhibit serine or cysteine proteases. These are reversible inhibitors and will need to be maintained at a reasonable working concentration in the buffers. Also, these peptides are prone to inactivation, and it is preferable to develop strategies based on irreversible inhibition.

Aspartic proteinases use a pair of aspartic residues to polarize the scissile bond. In general, aspartic proteinases are less of a problem than other proteinases, and they often have acidic pH optima. The nearest to a gen-

eral-purpose inhibitor of aspartic proteinases is pepstatin, a reversible, but tight-binding inhibitor.

Metalloproteinases use a bound zinc ion as an electrophile to polarize a water molecule, which then attacks the scissile bond. The most general inhibitors of metalloproteinases are chelators, such as EDTA or 1,10 phenanthroline. If chelators cannot be included in the sample buffers, more complex strategies, based on some awareness of the type of metalloproteinase, may dictate an inhibition strategy. For example, some metalloproteinases are inhibited by phosphoramidon.

Method 1 provides a recipe for an inhibitor cocktail that ought to prevent proteolysis under many circumstances. However, it should be stressed that such cocktails are not guaranteed to work under all conditions. Detailed literature, the mechanism of action, and manipulation of a range of commercially available proteinase inhibitors can be found elsewhere (*see, for example, 3,4,9*).

1.4. Assay of Endopeptidase Activity

There is some virtue in adopting a simple, yet sensitive assay for proteolytic activity, for tracking proteolytic activity during sample work-up, or for monitoring the efficiency of inhibition strategies. In a previous publication (3), the use of a radioiodinated peptide was advocated, but it is appreciated that few laboratories would be willing to take on this method unless they were already equipped for this type of radiochemical work. Accordingly, we present here a different assay, using a protein labeled with fluorescein isothiocyanate (FITC) based on a previously published protocol (10,11) (Methods 2 and 3). The fluorescent labeled protein is digested with a proteinase containing sample, and the undigested and large FITC-peptides are separated from small FITC products by precipitation with TCA. The FITC peptides in the supernatant fraction are monitored in a fluorimeter.

A second method (Method 4), which does not work for all endopeptidases, is zymography. In this method, a proteinase sample is electrophoresed in an SDS-PAGE into which has been copolymerized a protein, such as gelatin or casein. After the gel has been run, it is incubated in a nonionic detergent to remove the SDS, and then in a "refolding" buffer that may allow the proteinases to recover their structure and activity. If the proteinase is active, the gelatin is digested and converted to low-mol wt peptides that are washed out of the gel. Subsequent staining with

Coomassie blue indicates the zone of lysis as a clear region on a uniform blue background.

Zymography is quite sensitive, and a visible zone of lysis can be seen when nanogram amounts of enzyme are loaded. If excess proteinase is loaded, the zone of lysis can become large and diffuse. Also, if the proteinase is small, excess loading can show up as an unstructured zone of staining because the proteinase can diffuse out of the gel and digest the gelatin over all or part of the surface. In general, zymography does not work well if the gel or sample is reduced, possibly because of the opportunities for incorrect disulfide pairing, and the role of preformed disulfide bonds in directing folding. Not all proteinases will refold correctly and, thus, will not be detectable by this method.

2. Materials

2.1. Method 1: Stock Inhibitor Solutions

1. PMSF stock solution: 0.2M in dry methanol or propanol. Dissolve 38 mg ($M = 174.2$) of PMSF in 1.0 mL of solvent. PMSF is toxic! Weigh this compound in a fume hood, and wear disposable gloves and a mask. Store at -20°C .
2. 3,4-DCI stock solution: 10 mM in DMSO. Dissolve 2.2 mg ($M = 215$) of 3,4-DCI in 1.0 mL of DMSO. Store at -20°C .
3. Iodoacetic acid stock solution: 200 mM in water. Dissolve 42 mg ($M = 207.9$) of sodium iodoacetate in 1.0 mL of water. Use immediately.
4. E64-c stock solution: 5 mM in water. Dissolve 1.8 mg of E64-c ($M = 357.4$) in 1.0 mL water. Store at -20°C .
5. 1,10 phenanthroline stock solution: 100 mM in methanol. Dissolve 19.8 mg 1,10 phenanthroline ($M = 198.2$) in 1.0 mL of methanol. Store tightly capped at room temperature or 4°C .
6. EDTA stock solution: 0.5M in water. Dissolve 18.6 g EDTA (disodium salt, dihydrate, $M = 372.2$) into 70 mL water, titrate to pH 7.0 or 8.5, and make up to 100 mL. Stable at room temperature or 4°C .
7. Pepstatin stock solution: 10 mM in DMSO. Dissolve 6.9 mg pepstatin ($M = 685.9$) in 1.0 mL of DMSO. Store at -20°C .

2.2. Method 2: Preparation of FITC-Casein Substrate

1. Casein (purified powder, Sigma [St. Louis, MO] cat. no. C-5890).
2. Fluorescein isothiocyanate (Sigma cat. no. F-7250)
3. 0.05M Sodium carbonate, 0.15M NaCl, pH 9.5.
4. 0.05M Tris-HCl, pH 8.6.
5. Sephadex G-25 column, equilibrated with assay buffer.

2.3. Method 3: Assay of Enzyme Activity with FITC-Casein

1. Stock assay buffer (5X final concentration), e.g., 0.1M HEPES, pH 7.5.
2. FITC-casein: 1 mg/mL, in water or buffer, or casein fluorescein isothiocyanate (Sigma cat. no. C-0528).
3. TCA: 5% (w/v) in water.
4. Neutralizing buffer: 0.5M Tris-HCl, pH 8.6

2.4. Method 4: Zymography

1. Stock gelatin solution: 1.2% (w/v) in water. Store at 4°C and microwave gently on the defrost setting to melt before use. Use gelatin electrophoresis-grade, Type A, Sigma cat. no. G-8150.
2. Triton-X100: 2.5% (w/v) in water.
3. 10X Refolding buffer: 0.5M Tris-HCl, 2M NaCl, 5% CaCl₂, 0.67% (w/v) Brij35, pH 7.6. Adjust pH before adding Brij35.

3. Methods

3.1. Method 1: Working Inhibitor Cocktails

1. From the stock solutions described in Section 2.1., make a working inhibitor cocktail in water (not buffer, since some buffer compounds accelerate decomposition of the inhibitors). For 1.0 mL of working solution, and for each class of proteinases, proceed as follows:
 - a. Serine: 200 μ L PMSF (20 mM final) or 200 μ L 3,4-DCI (2 mM final),
 - b. Cysteine: 200 μ L iodoacetate (40 mM final) or 200 μ L E64c (1 mM final),
 - c. Metallo: 100 μ L 1,10 phenanthroline (10 mM final) or 100 μ L EDTA (50 mM final); and optionally
 - d. Aspartic: 100 μ L pepstatin (1 mM final).

Make up the final volume to 1.0 mL with water.

2. Use the working cocktail within 1 h of preparation. Dilute this by 20-fold into the sample (see Notes 1 and 2).

3.2. Method 2: Preparation of FITC-Casein Substrate

1. Dissolve 2 g of casein in 100 mL of sodium carbonate buffer, pH 9.5. Add 100 mg of FITC. Mix gently for 1 h at room temperature.
2. Dialyze the FITC-casein against several changes of 2 L 0.05M Tris-HCl, pH 8.5, followed by the buffer that is preferred for routine assay. Alternatively, dialyze against Tris-HCl, and then distilled water to exchange the substrate into any buffer in the future. The Tris buffer is used to consume excess reagent.
3. Determine the protein concentration by standard procedures. If needed, calculate the number of FITC residues/casein molecule from the $A_{490\text{nm}}$ of

the conjugate at pH 8.6. The extinction coefficient is 61,000/M/cm. The conjugate can now be frozen at -20°C .

4. To change the buffer in which the substrate is dissolved and to remove residual unbound FITC, apply the conjugate to a Sephadex G-25 column, equilibrated and eluted with a suitable assay buffer. Typically, the column volume should be about 10 times greater than the volume of FITC-casein for good buffer exchange. Monitor the elution profile at 280 nm, and collect the protein peak.
5. This substrate, approx 5 mg/mL, can be diluted to 1 mg/mL, and stored in aliquots at -20°C .

3.3. Method 3: Assay of Enzyme Activity with FITC-Casein

1. Combine 10–50 μL of enzyme sample with 10 μL FITC-casein in a microfuge tube. Add further assay buffer to make a reaction volume of 100 μL . Include a blank consisting of the buffer only.
2. Incubate at the desired temperature for 1–24 h.
3. Stop the reaction by adding 200 μL of 5% (w/v) TCA with mixing. Incubate the tubes at 4°C for 1 h to allow proteins to flocculate. Sediment the precipitated proteins by centrifugation at approx 10,000g for 10 min in a benchtop centrifuge at room temperature
4. Add 100 μL of the supernatant to 2.9 mL of 0.5M Tris-HCl, pH 8.6. This strong buffer neutralizes the TCA, because the fluorescence of FITC is pH dependent.
5. Mix thoroughly, and measure the fluorescence with the excitation wavelength set at 490 nm and the emission wavelength set at 525 nm. Slit widths of 10 nm or less should be used.
6. The fluorescence of a sample, incubated in the absence of a proteinase and substituting water for the TCA, but otherwise processed identically, gives the total fluorescence of the substrate and allows quantitation of the casein digestion as weight solubilized/time (see Notes 3–5).

3.4. Method 4: Zymography

1. If necessary, desalt samples into a low salt buffer (e.g., 0.02M HEPES, pH 7.5) using spun columns. This will improve the sharpness of the bands.
2. Add 5 μL of nonreducing SDS-PAGE sample buffer to 20 μL of the enzyme sample, and incubate at 37°C for 1 h. Do not heat the sample in a boiling water bath.
3. While the sample is incubating, cast the gel in a standard kit, including gelatin (see Note 6) at a final concentration of 0.6% (w/v).
4. Typically, load between 1 and 20 μL of each sample. In early experiments, two different loadings (e.g., 1 and 15 μL) could usefully be included.

5. Run the electrophoresis as usual. There may need to be some adjustment of running time. At the end of the run, remove the gel without touching the surface.
6. Soak the gel in 2.5% Triton-X 100 for 1 h at room temperature to wash out SDS
7. Rinse the gels in deionized water until foaming ceases (at least three times)
8. Incubate the gels in 1X refolding buffer, overnight at 37°C (*see* Notes 7 and 8)
9. Rinse the gels three times in deionized water.
10. Stain the gels with Coomassie brilliant blue for 1 h.
11. Destain the gels (several hours of destaining may be needed before the bands can be seen) (*see* Note 9)

4. Notes

1. The inhibitor cocktail can introduce salt (NaEDTA) and organic solvents to the sample. The 20-fold dilution is designed to minimize the effects of these constituents, but there may be circumstances in which this could still be problematic.
2. Although the serine and cysteine proteinase inhibitors are irreversible, the aspartic and metalloproteinase inhibitors are reversible. Even if the inhibitors are added at an early stage, they may be lost during purification. Alternatively, cryptic proteinases may be exposed at a later step by, for example, zymogen activation or dissociation of an endogenous inhibitor. Be prepared to add further inhibitor cocktail throughout the purification scheme.
3. The FITC-casein assay is very sensitive (ng to sub-ng) amounts of proteinase. The substrate should be stored at -20°C, because even a low level of bacterial contamination will give higher blank values. Always include a blank sample in FITC assays. If a standard is needed, a protease solution of 20 ng/mL trypsin will give a strong fluorescence in 2–3 h.
4. FITC-casein, like casein, is not very soluble at pH values below 4.0. This assay is not suitable for proteinases that have pH optima at low pH values.
5. This assay uses small amounts of substrate, which are usually far below saturating concentrations for the enzyme. As such, linearity over time will be lost if more than 10–20% of the substrate is digested. If digestion is limited, then fluorescence signal is proportional to the amount of enzyme added.
6. For zymography, other proteins, such as fibronectin or casein, can be copolymerized into the gel. The presence of the protein in the gel may alter the mobility of the proteinase, and a mol-wt estimate obtained by zymography should be carefully checked. It is possible to include a lane of mol-wt markers at sufficiently high a concentration that they can be seen as even darker bands on the uniform blue background.

7. The "refolding" step seems to be important, and a temperature of 37°C gives much improved recovery of activity over an incubation at room temperature. For metalloproteinases, there is apparently no necessity to add zinc ions to the refolding buffer, and this should be discouraged because many metalloproteinases are inhibited by excess zinc.
8. Zymography is not routinely used for analysis of cysteine proteases because of the need to add reducing agents to activate the enzyme. It might be worthwhile adding a reducing agent (10 mM dithiothreitol) for the last few hours of incubation in refolding buffer.
9. Because zymography has the potential to separate noncovalently bound inhibitors from proteinases, it may indicate proteolytic activity when none is apparent in soluble extracts.

References

1. Pringle, J. R. (1975) Methods for avoiding proteolytic artifacts in studies of enzymes and other proteins from yeast, in *Methods in Cell Biology*, vol 12 (Prescott, D. M., ed.), Academic, New York, pp. 149–184.
2. Pringle, J. R. (1978) Proteolytic artifacts in biochemistry, in *Limited Proteolysis in Microorganisms?* (Cohen, G. N. and Holzer, H., eds.), US Department of Health, Washington, DC, pp. 191–196.
3. Beynon, R. J. (1988) Prevention of unwanted proteolysis, in *Methods in Molecular Biology, Vol 3 New Protein Techniques* (Walker, J., ed.), Humana, Clifton, NJ, pp. 1–23.
4. North, M. J. (1989) Prevention of unwanted proteolysis, in *Proteolytic Enzymes—A Practical Approach* (Beynon, R. J. and Bond, J. S., eds.), IRL, Oxford, pp. 105–124.
5. Neurath, H. (1978) Limited proteolysis—an overview, in *Limited Proteolysis in Microorganisms* (Cohen, G. N. and Holzer, H. eds.), US Department of Health, Washington, DC, pp. 191–196.
6. Price, N. C. and Johnson, C. M. (1989) Proteinases as probes of conformation of soluble proteins, in *Proteolytic Enzymes—A Practical Approach* (Beynon, R. J. and Bond, J. S., eds.), IRL, Oxford, pp. 163–191.
7. Hubbard, S. J., Campbell, S. F., and Thornton J. M. (1991) Molecular recognition. conformational analysis of limited proteolytic sites and serine proteinase inhibitors *J Mol Biol.* **220**, 507–530.
8. Hubbard, S. J., Eisenmenger, F., and Thornton, J. M. (1994) Modeling studies of the change in conformation required for cleavage of limited proteolytic sites *Protein Sci* **3**, 757–768.
9. Salvesen, G. and Nagase, H. (1989) Inhibition of proteolytic enzymes, in *Proteolytic Enzymes—A Practical Approach* (Beynon, R. J. and Bond, J. S., eds.), IRL, Oxford, pp. 25–55.
10. Sarath, G., De La Motte, R. S., and Wagner, F. W. (1989) Protease assay methods, in *Proteolytic Enzymes—A Practical Approach* (Beynon, R. J. and Bond, J. S., eds.), IRL, Oxford, pp. 83–104.
11. Twining, S. S. (1984) Fluorescein isothiocyanate-labeled casein assay for proteolytic enzymes. *Anal Biochem* **143**, 30–34.

CHAPTER 10

Concentration of Extracts

Shawn Doonan

1. Introduction

In a typical purification starting from 1 kg of tissue, the volume of the initial homogenate might be 2 L, and the final product of the procedure might be 2 mL of pure protein solution; that is, a volume decrease of 1000-fold is required. This emphasizes the fact that water is a major contaminant during protein purification, and at several steps during the procedure the need will arise for concentration of the active extract or fraction. Concentration may occur as a concomitant of a step in purification, for example, fractional precipitation (*see* Chapter 13) or ion-exchange chromatography (*see* Chapter 14); in the latter case, protein from a dilute solution may be absorbed onto the resin and subsequently eluted in a smaller volume by application of a salt gradient. In general, however, one or more steps specifically aimed at concentration of the protein solution will need to be done.

For concentration of initial cell extracts, particularly when working on a large scale (1–5 L of homogenate), the method of choice is precipitation by ammonium sulfate. Virtually all proteins are precipitated from solution at sufficiently high salt concentrations. This arises from the fact that protein surfaces tend to have hydrophobic patches, which, in solution, are surrounded by ordered water molecules. When salt is added to the protein solution, water is recruited to solvate the ions of the dissociated salt, thus progressively exposing the hydrophobic regions of the protein surface. At some point, these patches will start to interact, leading to aggregation and ultimately to precipitation of the proteins from

From *Methods in Molecular Biology*, Vol 59 *Protein Purification Protocols*
Edited by S Doonan Humana Press Inc., Totowa, NJ

solution (for a more detailed discussion, *see* ref. 1). The requirements for the salt are that it be highly soluble in water, that its component ions be innocuous to proteins, and that it has a low heat of solution; ammonium sulfate satisfies these criteria most completely. The technique is, then, to precipitate the proteins from solution by addition of ammonium sulfate, recover the proteins by centrifugation, resuspend in the minimum amount of water or buffer, and remove residual ammonium sulfate by dialysis.

A second technique for protein concentration that is also applicable to relatively large volumes, but that is usually used at later stages in a purification protocol (e.g., after column chromatography) is forced dialysis or ultrafiltration. In this use is made of semipermeable membranes, which allow passage of water and other small molecules, but not of proteins. If a protein solution is placed in a bag of such material and pressure is applied, then small molecules will be forced out and the protein molecules retained. The rate of passage of small molecules will, of course, be severely reduced by precipitation of protein onto the walls of the dialysis bag; this is quite likely to occur with crude protein solutions, and it is for this reason that the method is not generally used at early stages in a purification procedure.

The particular version of membrane filtration described in the present chapter requires a minimal amount of specialized equipment and can be used to concentrate relatively large volumes of solution in an overnight experiment. A variant of the method that requires the use of commercial pressure cells, but has the advantage that it can be used for fractionation as well as for concentration, is described in Chapter 12. For small protein samples, drying by lyophilization is an alternative means of removal of water, but has the disadvantage that buffer ions (unless volatile) will remain in the sample; this technique is described in Chapter 31.

2. Materials

2.1. Precipitation with Ammonium Sulfate

1. Ammonium sulfate (preferably Analar grade).
2. Electric paddle stirrer.
3. Refrigerated centrifuge and rotor, e.g., a 6 × 250 mL angle rotor.
4. Screw-top plastic tubes for the rotor to be used.
5. Visking dialysis tubing (14- or 19-mm inflated diameter; 26- or 31-mm flat width). This is available from most suppliers of laboratory materials (e.g., Fisons, Crawley, UK).

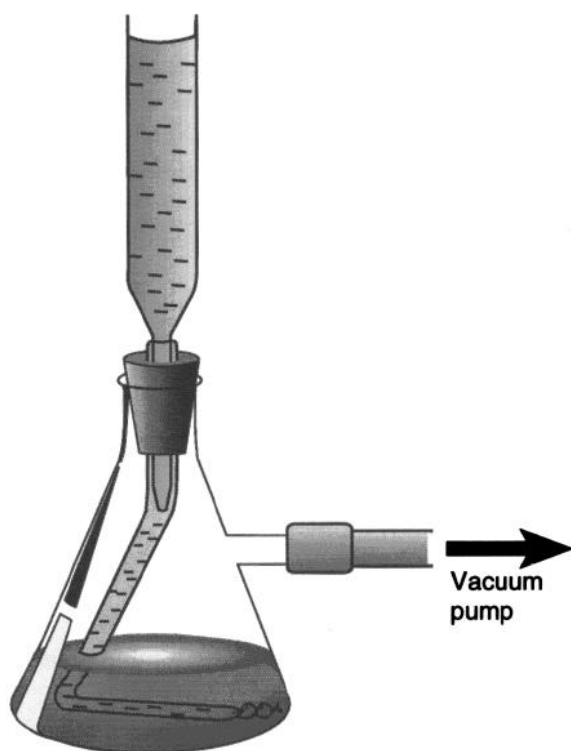


Fig. 1. Experimental arrangement for concentration by forced dialysis. The dialysis tubing should extend up the stem of the funnel so that it is above the top of the bung. In this way, it will be gripped tightly and will not slip off when the vacuum is applied.

2.2. Forced Dialysis

1. Heavy-walled Buchner flask (in the range 500-mL to 5-L capacity depending on the volume of solution to be concentrated—see Note 6).
2. Rubber bung to fit the flask and bored to accommodate the stem of a glass funnel.
3. Glass funnel (see Fig. 1 and Note 8).
4. Visking dialysis tubing (6-mm inflated diameter; 10-mm flat width).
5. Efficient water vacuum pump.

3. Methods

3.1. Precipitation with Ammonium Sulfate

1. Measure the volume of the protein solution to be concentrated, and pour it into a glass beaker of capacity about twice the measured volume of solution.

- 2 Weigh out 0.5 g of solid ammonium sulfate for every 1 mL of protein solution. If the ammonium sulfate contains lumps, then these should be broken up using a pestle and mortar (*see* Note 1).
- 3 Place the beaker containing the protein on ice, and stir either manually or with a paddle stirrer. A slow rate of stirring should be used to avoid foaming
- 4 Add the ammonium sulfate to the protein solution in small batches over a period of several minutes ensuring that one lot of ammonium sulfate has dissolved before adding the next.
5. After addition is complete, leave to stand for 10 min to ensure complete precipitation and then recover precipitated protein by centrifugation at about 5000g for 30 min using screw-top plastic tubes (*see* Note 2).
- 6 Decant off the supernatant solution from each tube, and suspend the protein pellets in the minimum volume of water or of an appropriate buffer (*see* Note 3).
7. Transfer the protein suspension to a dialysis sack and dialyze against at least two changes of buffer using 100 times the volume of the sample and allowing 3–4 h for equilibration (*see* Note 4).
- 8 After dialysis, remove any precipitated material by centrifugation

3.2. Forced Dialysis

- 1 Cut a length of the 6-mm diameter dialysis tubing (*see* Note 5) about twice as long as the height of the flask (*see* Note 6).
2. Soak the dialysis tubing in the buffer to be used for a few minutes (*see* Note 7). Tie two firm knots in one end. Open the other end by rubbing gently between the fingers, and then push the open end onto the stem of the funnel (*see* Note 8) so that the tubing extends 3–4 cm up the stem. The stem of the funnel should be moistened with buffer to facilitate this and a towel used to grip the dialysis tubing. Care must be taken not to tear the dialysis tubing.
- 3 Pour buffer (*see* Note 9) into the Buchner flask to a depth of about 5 cm. Thread the knotted end of the dialysis tubing through the hole in the rubber bung, pull through, and then push the stem of the funnel through the hole until the end has projected through. Place the bung in the flask (*see* Fig. 1).
- 4 Evacuate the flask using a water pump (*see* Note 10), seal the outlet, and test for leaks in the system by rotating and inverting the flask so that all parts of the dialysis tubing are sequentially immersed in the buffer. Pin-hole leaks will be apparent from a stream of air bubbles emerging from the tubing. If any are present, reject the piece of tubing and start again.
5. Release the pressure, and pour protein solution into the funnel. Air will usually be trapped in the tubing. To remove this, take the bung out of the flask, and gently squeeze the tubing to expel air bubbles.
6. Replace the tubing and bung in the flask, evacuate, and then seal off the outlet. Place the flask in a cold room overnight or until the volume of solution has decreased to the desired extent (*see* Note 11).

7. Release the vacuum, cut off the dialysis tubing from the funnel, and transfer the concentrated solution to an appropriate container (*see* Note 12).

4. Notes

1. This amount of ammonium sulfate gives a concentration of about 75% saturation at 0°C, which should be high enough to precipitate most proteins. An initial experiment should be carried out to confirm that the protein of interest is indeed precipitated at this concentration. Higher values can be used (up to a maximum of 0.7 g/mL, which corresponds to 100% saturation), but this increases the density of the solution and makes it more difficult to sediment the protein.

Note that strong solutions of ammonium sulfate are acidic, so it is important that the protein solution be buffered at a neutral pH and a buffer concentration of 0.05–0.1 M. Note also that ammonium sulfate (even Analar grade) contains traces of metal ions; hence, if the protein of interest is metal-sensitive, EDTA (10 mM) should be added to the protein solution

2. Care must be taken with this step to protect both the centrifuge and the operator! The tubes must be balanced to within 0.1–0.2 g across the rotor axis (*see* Note 3; Chapter 2), and it is particularly important not to counter-balance a tube full of 75% ammonium sulfate with a tube full of water because of the large difference in density. Either the ammonium sulfate/protein suspension should be divided between two tubes or the suspension should be balanced using a solution of ammonium sulfate of the same concentration. It is also most important to avoid spillage of ammonium sulfate solutions into the centrifuge head. Such solutions are extremely corrosive to the materials of which rotors are constructed and can lead to irreversible damage rendering the rotors unsafe to use. After centrifugation of ammonium sulfate (or other salt) solutions, rotors should always be removed and washed in warm water

Note that proteins precipitated in 70% ammonium sulfate are usually stable, and hence it is often convenient to store the suspension overnight at 4°C in this form before proceeding with the purification schedule. Indeed, pure proteins are often stored at –20°C in 70% ammonium sulfate for long periods.

3. This step is crucial to achieving a satisfactory degree of concentration. Add a very small volume of water or buffer (about 10 mL if using a 250-mL tube), and resuspend the protein pellet using a glass rod. If several centrifuge tubes have been used, then transfer the protein suspension from tube to tube, resuspending the pellet each time; only add more water or buffer if the suspension becomes too thick to transfer readily. After transfer of the final suspension to the dialysis bag, a further small aliquot of water or buffer can be used to rinse out the tubes and the rinsings combined with the

suspension. A common error is to attempt to redissolve the protein pellets after centrifugation; since the pellets contain considerable quantities of ammonium sulfate, this takes a large volume of buffer, and it is easy to end up with a volume comparable to that at the outset!

4. Visking tubing comes in several sizes. For a given volume of solution, the larger the diameter of the dialysis tubing used, the shorter the piece that will be required; attainment of equilibrium, however, will be slower with short, fat bags than with long, thin ones. The sizes recommended are a compromise between these two factors.

For most applications, it is only necessary to soak the dry tubing in water or in the buffer and it is ready to be used. The tubing does, however, contain significant quantities of sulfur compounds and of heavy metal ions; the latter may be a problem if the protein of interest is metal-sensitive. They may be removed by boiling the dialysis tubing in 2% (w/v) sodium bicarbonate 0.05% (w/v) EDTA for about 15 min, washing with distilled water, and then boiling in distilled water twice for 15-min periods. Prepared tubing can be stored indefinitely at 4°C in water or buffer containing 0.1% (w/v) sodium azide.

For most applications, Visking tubing is perfectly adequate, but problems will arise if the protein of interest is small. The pores in this type of tubing are of such a size that the nominal mol-wt cutoff (NMWC) is about 15,000, although larger proteins may still pass through the pores if they have an elongated shape. As a rule of thumb, Visking tubing can be used with confidence if $M_r > 20,000$. For dialysis of smaller proteins, Spectropor tubings with a range of NMWC values starting at 1000 are available (e.g., from Pierce and Warriner, Chester, UK), and the appropriate tubing should be selected. These tubings are much more expensive than Visking tubing, and the rate of dialysis decreases with pore size; hence, they should only be used if strictly necessary.

To remove ammonium sulfate from the protein suspension, a piece of dialysis tubing with a volume about twice that of the suspension is taken and securely closed with a double knot at one end. The suspension is then poured into the bag using a funnel, air is removed from the top part of the bag by running it between the fingers, and the top secured with a double knot. It is very important to have this space in the bag to allow for expansion, since water will flow in while the internal salt concentration is high. If insufficient space is left, the bag can become very tight owing to this inflow. Bags rarely burst since the membranes are quite strong, but they are tricky to open in this state, the best way is to insert one end into a measuring cylinder and then prick the bag with a scalpel

Equilibrium will be reached in about 3–4 h, but only if the system is stirred; otherwise about 6 h should be allowed. Care should be taken when stirring to ensure that the magnetic pellet or stirring paddle does not tear the dialysis bag.

The volume of dialysis solution to be used depends, of course, on the sample volume and the final concentration of ammonium sulfate desired. If a ratio of 1:100 (sample:dialysis fluid) is used, then at equilibrium the ammonium sulfate concentration will have been reduced 100-fold (this is not strictly true because it ignores the Donnan effect, but will do as an approximation). A second dialysis will then result in a total decrease of 10,000-fold and so on.

The buffer to be used for dialysis is usually dictated by the requirements of the next step in the purification schedule. For example, if this is to be ion-exchange chromatography, then column equilibration buffer is the logical choice.

5. The question of NMWC is also important here, and if the protein of interest has an $M_r < 20,000$, then one of the Spectropor tubings must be used. This will decrease the rate of concentration, but reduce losses of material.

Passage of low-mol wt material through the dialysis tubing can sometimes be used to advantage if it is required to carry out a crude separation of large and small components of a mixture. For example, an effective separation of the peptide components from the protein components of bee venom was achieved using forced dialysis through Visking tubing (2).

6. The optimum size of flask and length of tubing to be used depend on the scale on the experiment to be carried out. For volumes of the order of 100–200 mL, a 500-mL flask is ideal and the total length of tubing in this case would be about 25 cm. A final volume of concentrate of 2–3 mL is easy to achieve. For very large volumes a 5-L flask should be used with correspondingly long lengths of dialysis tubing (otherwise, concentration will take a very long time). With large flasks, it is perfectly feasible to mount up to three funnels in the bung, and thus increase the capacity and speed of the system.
7. If desired, the tubing can be pretreated as described in Note 4 above to remove metal ions.
8. Funnels of the type shown in Fig. 1 can be made from heavy glass tubing of about 25-mm diameter drawn down to a stem of about 7-mm diameter so that the dialysis tubing is a tight push fit; to do this requires an experienced glass-blower. An alternative is to purchase cylindrical dropping funnels of the appropriate size from a laboratory supplier. These will have a tap between the stem and the cylindrical part, but this is not a problem. Whatever the type of funnel used, it is crucial that the end of the stem be fire-glazed; otherwise, the sharp glass edge will inevitably cut the dialysis tubing.

9. The buffer chosen will usually be that for the next step in the purification schedule. It should be noted, however, that the composition of the buffer will change during ultrafiltration owing to efflux of buffer ions originally in the sample so that it cannot be assumed that the concentrated sample is properly equilibrated for further purification.
10. Do not use an oil pump; otherwise, it is likely that the pressure will be reduced too much and the dialysis bag may break (or the flask may implode, but this is not likely if it is made of heavy glass). As a useful guide, evacuate the flask until bubbles start to form in the buffer in the flask; this will occur at about 15 mmHg depending on the ambient temperature.

Care should always be taken when using glass vessels under low pressure. Ideally, a cage should be used, but as a minimum safety glasses should be worn. Do not drop evacuated flasks!

11. Using a single flask of 2-L capacity with a single funnel and about 40 cm of tubing, it should be possible to reduce 500 mL of protein solution to about 5 mL overnight. If the capacity of the funnels available is too small to take the amount of protein solution to be concentrated, then it is simple to set up a syphon between solution in the funnel and extra solution in an external reservoir. Using a single 5-L flask with three funnels each one being replenished from a reservoir by means of a syphon, it is possible to concentrate 2 L of dilute protein solution down to about 20 mL overnight.
12. If left too long, the retentate may reduce in volume to such an extent that it is necessary to wash the protein out of the tubing with a small amount of buffer. On occasion, pure proteins may even crystalize in the tubing (*see* ref. 3 for an example); in this case, care should be exercised in handling the tubing so that the crystals do not puncture it.

References

1. Englard, S. and Seifter, S. (1990) Precipitation techniques. *Methods Enzymol* **182**, 285–300
2. Shipolini, R. A., Callewaert, G. L., Cottrell, R. C., Doonan, S., Vernon, C. A., and Banks, B. E. C. (1971) Phospholipase A from bee venom *Eur J Biochem* **20**, 459–468
3. Barra, D., Bossa, F., Doonan, S., Fahmy, H. M. A., Martini, F., and Hughes, G. J. (1976) Large-scale purification and some properties of the mitochondrial aspartate aminotransferase from pig heart *Eur J Biochem* **64**, 519–526

CHAPTER 11

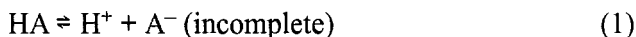
Making and Changing Buffers

Shawn Doonan

1. Introduction

Control of pH is a central consideration in handling proteins, and this requires the use of buffer solutions. Furthermore, a typical protein purification schedule will contain several steps generally needed to be carried out in buffers of different pH values and ionic strengths (*see* Note 1). Hence, it is necessary to have available methods for changing buffers. The present chapter reviews the properties of buffers and preparation of buffer solutions, and then deals with ways in which a protein solution can be changed from one buffer to another.

Buffer solutions consist of a weak acid and a salt of that acid, or of a weak base and a salt of that base. For example, in a solution of a weak acid HA and its sodium salt, the following occur:



The species A^- is referred to as the conjugate base of the acid HA.

Addition of H^+ to the buffer moves equilibrium (1) to the left using A^- supplied by (2), whereas added OH^- (or other base) combines with H^+ provided by equilibrium (1) moving to the right; in either case, change of H^+ concentration, and hence, of pH, is resisted. The Henderson-Hasselbalch equation describes the relationship between the pH, the pK_a of the buffer and the relative concentrations of the free acid and of the salt as follows (*see* Note 2):

$$\text{pH} = \text{pK}_a - \log [(\text{acid})/(\text{salt})] \quad (3)$$

From *Methods in Molecular Biology*, Vol 59 *Protein Purification Protocols*
Edited by S Doonan Humana Press Inc , Totowa, NJ

Hence, if, for example, a solution of a weak acid of concentration (a) *M* is partially neutralized by addition of a strong base to a concentration of (b) *M*, then the result is a buffer solution where the pH is given by:

$$\text{pH} = \text{p}K_a - \log \{[(a) - (b)]/(b)\} \quad (4)$$

It is immediately apparent from Eq. (4) that when (b) = 0.5(a), then pH = p*K_a*, i.e., the pH is numerically equal to the p*K_a* value. The equation also suggests that the pH of a buffer is unaffected by dilution; this is true to a first approximation for some buffers, but with others, the pH changes quite markedly with dilution (*see* Note 3).

What is not so obvious is that the ability of the buffer to resist changes of pH is maximum at the p*K_a* and falls off on either side such that the buffering power is small to negligible outside the range p*K_a* ± 1. This can be seen by considering the parameter β (the buffer capacity or buffer value), which is defined as:

$$\beta = db/d\text{pH} \quad (5)$$

and is given by:

$$\beta = 2.303 \{(a)K_a(\text{H}^+)/[K_a + (\text{H}^+)]^2\} + 2.303 [(\text{H}^+) + (\text{OH}^-)] \quad (6)$$

where (a) is the total concentration of the free acid plus the salt. In the pH range of approx 3–11, the value of β is determined entirely by the first term in the equation, and under those circumstances a more convenient form is:

$$\beta = 2.303 \{(b)[(a) - (b)]/(a)\} \quad (7)$$

This is obtained by taking the first derivative of the Henderson-Hasselbalch equation and inverting it. The buffer value is a measure of the amount of base needed to produce a unit change in the pH of the buffer, and the larger its value, the greater the resistance to pH change. Table 1 shows the buffer values for a solution of a weak acid to which various quantities of base have been added and emphasizes the restricted range of pH values over which the solution has good buffering properties. Obviously, the stronger the buffer, the greater the buffer value, but for practical reasons, buffers of strength >0.1*M* are not usually used.

There are, then, a variety of considerations in selecting a buffer for a particular application. These include:

- 1 The desired pH: The p*K_a* of the buffer must be as close to this as possible and certainly not outside the range pH ± 1.

Table 1
pH and Buffer Values of a 0.1 M Solution
of a Weak Acid ($pK_a = 7$) as a Function
of Concentration of an Added Base b

(b), M	pH	Buffer value
0.005	5.72	0.0109
0.010	6.05	0.0207
0.020	6.40	0.0368
0.030	6.63	0.0484
0.040	6.82	0.0553
0.050	7.00	0.0576
0.060	7.18	0.0553
0.070	7.37	0.0484
0.080	7.60	0.0368
0.090	7.95	0.0207
0.095	8.28	0.0109

2. The pH of the buffer should change as little as possible with temperature, with dilution, and with added neutral salt (*see* Note 4).
3. The buffer should be chemically unreactive.
4. The buffer should not absorb light at 280 nm, particularly if it is to be used for chromatographic procedures where column monitoring will usually be carried out by absorbance measurements.
5. For cation ion-exchange chromatography, an anionic buffer should be used and vice versa.
6. Buffers that might interact with components of the protein mixture (e.g., borate with glycoproteins) should be avoided.

Table 2 gives a list of commonly used buffer compounds with their pK_a values; values for other substances are given in ref. 1. Included in the table are a variety of "Good" buffers (MES, ADA, PIPES, ACES, BES, MOPS, HEPES, TAPS, CHES, CAPS). These zwitterionic buffers are chemically unreactive, do not absorb at 280 nm, and their pH values vary only slightly with temperature and dilution (2). They are, therefore, ideal buffers in many respects, but are very expensive (*see* Note 5) and their use is often not feasible when large volumes of buffer are required for procedures such as ion-exchange chromatography and dialysis.

As mentioned, there will frequently be a need during protein purification for changing the buffer in which a protein mixture is dissolved; this may be to change the pH, the buffer ionic strength, or the concentration of a neutral salt. There are two major ways in which this can be done.

Table 2
Buffer Compounds and Their pK_a Values at 25°C

Compound	Trivial name	pK_a
Phosphoric acid (pK_1)	—	2.15
Glycine (pK_1)	—	2.35
Fumaric acid (pK_1)	—	3.02
Citric acid (pK_1)	—	3.13
Formic acid	—	3.75
Succinic acid (pK_1)	—	4.21
Fumaric acid (pK_2)	—	4.38
Citric acid (pK_2)	—	4.76
Acetic acid	—	4.76
Succinic acid (pK_2)	—	5.64
2-(<i>N</i> -morpholino)ethanesulfonic acid	MES	6.10
Carbonic acid (pK_1)	—	6.35
[Bis(2-hydroxyethyl)imino]- tris(hydroxymethyl)methane	Bis-Tris	6.46
<i>N</i> -2-acetamidoiminodiacetic acid	ADA	6.59
Piperazine- <i>N,N'</i> -bis(2-ethanesulfonic acid)	PIPES	6.76
<i>N</i> -2-acetamido- 2-hydroxyethanesulfonic acid	ACES	6.78
Imidazole	—	6.95
<i>N,N</i> -bis-(2-hydroxyethyl)- 2-aminoethanesulfonic acid	BES	7.09
3-(<i>N</i> -morpholino)propanesulfonic acid	MOPS	7.20
Phosphoric acid (pK_2)	—	7.20
<i>N</i> -2-hydroxyethylpiperazine- <i>N'</i> -2-ethanesulfonic acid	HEPES	7.48
<i>N</i> -ethylmorpholine	—	7.67
Triethanolamine	—	7.76
Tris(hydroxymethyl)aminomethane	Tris	8.06
<i>N</i> -[Tris(hydroxymethyl)methyl]glycine	Tricine	8.05
<i>N,N</i> -bis(2-hydroxyethyl)glycine	Bicine	8.26
3-[[Tris(hydroxymethyl)methyl]- amino]propanesulfonic acid	TAPS	8.40
2-Amino-2-methylpropan-1,3-diol	—	8.79
2-Aminoethylsulfonic acid	Taurine	9.06
Boric acid	—	9.23
Ammonia	—	9.25
Ethanolamine	—	9.50
Cyclohexylaminoethanesulfonic acid	CHES	9.55
3-Aminopropanesulfonic acid	—	9.89
β -alanine	—	10.24
Carbonic acid (pK_2)	—	10.33
3-(Cyclohexylamino)propanesulfonic acid	CAPS	10.40
γ -Aminobutyric acid	—	10.56
Piperidine	—	11.12
Phosphoric acid (pK_3)	—	12.33

Table 3
Standard Buffer Solutions

Buffer	Composition, g/L	Concentration, <i>M</i>	pH		
			5°C	15°C	25°C
Phthalate	10.12 g of $\text{KHC}_8\text{H}_4\text{O}_4$	0.05	4.00	4.00	4.01
Phosphate	3.39 g of KH_2PO_4 + 3.53 g of Na_2HPO_4	0.025	6.95	6.90	6.87
Borate	3.80 g of $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$	0.01	9.40	9.28	9.18

The first of these is dialysis in which the protein solution is enclosed in a bag of a semipermeable membrane, i.e., one that allows the passage of small molecules but not of large ones, and is then equilibrated in two or more changes of a large excess of the target buffer solution. Ultimately, equilibrium will be reached where the internal and external buffers may approximate to the same pH and concentration (*see* Note 6). A related technique uses membrane UF; this is described in Chapter 12.

The second method is gel filtration. This uses a column of porous beads designed such that water and low-mol wt solutes have access to the interior of the beads whereas larger molecules do not. In gel filtration the solution of protein is applied to such a column equilibrated in the target buffer and the proteins are then eluted with target buffer. The protein molecules will move ahead of the buffer in which they were originally dissolved and will emerge in the target buffer. This technique can also be used as a method for fractionation of protein mixtures on the basis of size as described in Chapter 25.

2. Materials

2.1. Preparation of Buffers

1. pH meter capable of accuracy at two decimal places and with temperature compensation (*see* Note 7).
2. Magnetic stirrer and pellet.
3. Standard buffer solutions with pH values bracketing that of the buffer to be made. These can either be prepared using the information in Table 3 or purchased from most suppliers of chemicals.
4. Analar HCl or NaOH pellets depending on the buffer to be made.

2.2. Dialysis

1. Visking dialysis tubing (either 14- or 19-mm inflated diameter; 26- or 31-mm flat width). This is available from most suppliers of laboratory materials (e.g., Fisons, Crawley, UK)

2. Target buffer (at least 200 times the volume of protein solution is required).

2.3. Gel Filtration

1. Sephadex G-25 (medium grade) (*see* Note 8) About 1 g will be required for every 5 mL column volume. This material is produced by Pharmacia Biotech (Milton Keynes, UK), but is available from general suppliers.
2. Appropriate size chromatography column. The packed volume will need to be about five times greater than the volume of protein solution to be treated (*see* Note 9).
3. Peristaltic pump, fraction collector and absorbance detector (optional—*see* Chapter 36).
4. Target buffer (about 10 times the volume of the column bed).

3. Methods

3.1. Preparation of Buffers

It is usually convenient to make stock solutions of buffers that can then be diluted for use (*see* Note 3). Recipes are available for many of the more common buffers (*see* ref. 1), or compositions can be calculated using the Henderson-Hasselbalch equation. More usually, however, buffers are made by weighing out the required amount of the buffering substance, dissolving in water, and then adjusting the pH to the desired value by adding HCl or NaOH as appropriate; other acids or bases may, of course, be used. For example, to make 1 L of 2M sodium acetate buffer, pH 5.0, proceed as follows.

1. Standardize the pH meter using standard buffers of pH 4.01 and 6.87, assuming that the buffer is to be made at room temperature.
2. Weigh out 2 mol (120 g) of glacial acetic acid, and transfer to a 1-L beaker.
3. Add about 800 mL of distilled water, place on a magnetic stirrer, and insert the pH meter electrodes.
4. Slowly add solid NaOH pellets (preferably from a freshly opened bottle to ensure that they are not contaminated with sodium carbonate), making sure that one lot dissolves before adding the next. When the pH has reached about 4.8, allow the buffer to cool to room temperature (heat will have been generated by reaction of acetic acid with the NaOH), and then finally adjust the pH to 5.0 using a solution (about 4M) of NaOH.
5. Transfer the solution quantitatively to a graduated flask or measuring cylinder (the latter is sufficiently accurate), and make up the volume to 1 L.
6. Store in a glass container at 4°C Preservatives are not necessary because the high concentration inhibits bacterial growth.

3.2. Changing Buffers by Dialysis

It should be borne in mind that proteins are polyelectrolytes whose state of ionization varies with pH because of their content of ionizable side chains. Hence, particularly if the protein concentration is high and the solution is at a pH far removed from that of the target buffer, it is not possible to equalize the pH and ionic strength values of the internal and external buffers by dialysis because of the Donnan effect (*see* Note 10). In the following, it is assumed that dialysis will be carried out in a cold room or refrigerator at about 5°C using a diluted stock buffer.

1. Calibrate the pH meter with standard buffers (Table 3) cooled to 5°C.
2. Dilute stock buffer solution to the desired concentration using cold distilled water, and check that the pH has not been altered by dilution or the decrease in temperature. If it has, adjust it with the acidic or basic component of the buffer as appropriate. The volume of buffer should be about 100 times that of the sample to be dialyzed.
3. If the target buffer is to contain a neutral salt, then add this before checking and adjusting the pH.
4. Take a length of Visking dialysis tubing able to hold about 1.5 times the volume of the protein solution, and soak it in the buffer for a few min. For some applications, it may be desirable to pretreat the dialysis tubing and/or use tubing with a lower nominal mol wt cutoff (*see* Note 4, Chapter 10).
5. Tie two tight knots in one end of the tubing, and then pour in the protein solution with the aid of a funnel. Squeeze out air above the protein solution to allow room for expansion, and seal the bag with two more knots.
6. Place the dialysis bag in the buffer solution, and leave for at least 4 h in the cold to reach equilibrium. If a magnetic stirrer is used, be sure that the pellet does not tear the dialysis bag.
7. Replace the buffer with a fresh lot, and repeat the dialysis for another 4 h.
8. Check the pH of the dialyzed solution and, if necessary, adjust it to the target value by very careful addition of dilute acid or base (*see* Note 10). Use constant stirring to ensure that local high concentrations of acid or base do not occur.
9. Remove any insoluble material from the dialyzed sample by centrifugation at about 5000g for 15 min.

3.3. Changing Buffers by Gel Filtration

This is a more rapid procedure (if a column is already available) and, hence, to be preferred particularly if time is important, e.g., if the protein of interest is unstable. It should not be used, however, with crude protein

mixtures, since there is a strong possibility that protein will precipitate during the procedure and ruin the column.

1. Take 1 g of Sephadex G-25 (medium grade) for every 5 mL of desired column volume, and stir carefully into 10 vol of the target buffer. Leave at 5°C overnight for the gel to swell to its maximum extent. Resuspend by stirring, allow to settle, and remove any fines by aspiration.
2. Resuspend the gel in about 2 vol of buffer and pour into the chromatography column (*see* Note 11) with the outflow blocked off. Allow a few centimeters of settled bed to form, and then start flow through the column at the rate to be used for gel filtration (*see* Note 12). As clear liquid forms above the gel suspension, remove it by aspiration and replace it with fresh suspension. After the gel bed has reached the desired height, attach a buffer reservoir and pass two column volumes of buffer through the column to ensure equilibration of the bed.
3. Apply the sample to the column, and then continue elution with about one column volume of the target buffer. Collect appropriate sized fractions (about 1/20th of the column volume).
4. If using an automatic absorbance monitor, the fractions of interest will be obvious. Otherwise, measure the absorbance of individual fractions at 280 nm, and combine fractions containing protein (*see* Note 13).
5. Check the pH of the combined fractions, and if necessary, adjust as described in Section 3.2., step 8 (*see* Note 14).
6. Re-equilibrate the column by passage of two column volumes of buffer.

4. Notes

1. Sometimes, by judicious choice of the sequence of steps, it is possible to avoid changing the buffer between one step and the next. This is desirable, because it saves time and can improve the overall yield of the purification. An example of where this approach has been used is given in ref. 3.
2. This equation is strictly only valid in the pH range 3–11. Outside this range the self-ionization of water becomes significant.
3. The reason for this is that the concentration terms in the Henderson-Hasselbalch equation should properly be thermodynamic activities; these are concentration-dependent to varying extents.
4. K_a values are temperature-dependent, and hence, so are the pH values of buffer solutions. The magnitude of the effect depends on the particular buffer substance chosen. Similarly, added salt can affect the pH of a buffer because of differential effects on the thermodynamic activities of the component buffer ions. If possible, buffers should be selected where these effects are minimal (*see* ref. 1).
5. Prices of “Good” buffers are in the range \$300–750/kg. Common organic acids and bases used as buffers cost about one-tenth of this, and inorganic buffers are even cheaper.

6. This is not strictly true because of the so-called Donnan equilibrium. The protein, which is a nonpermeant polyelectrolyte, will affect the distribution of the permeant ionic species. Consider a protein solution volume V_1 at a pH such that it is negatively charged, and suppose that the concentration of charges on the protein is $C_1 M$; these negative charges will be neutralized by positive ions (e.g., Na^+) also at a concentration of $C_1 M$. If this solution is enclosed in a dialysis bag and placed in a volume V_2 of NaCl solution concentration $C_2 M$, then NaCl will cross the membrane until equilibrium is attained, i.e., until the activities of the NaCl in the two compartments are equal. It can be shown that, to a first approximation, the equilibrium condition is:

$$(\text{Na}^+)_{\text{in}} \times (\text{Cl}^-)_{\text{in}} = (\text{Na}^+)_{\text{out}} \times (\text{Cl}^-)_{\text{out}} \quad (8)$$

where in and out refer to inside and outside of the dialysis bag. If the change of concentration of NaCl inside the dialysis bag is $+x M$, then it follows that the change outside will be $-x (V_1/V_2) M$ and the equilibrium condition is:

$$x(C_1 + x) = \{C_2 - [x (V_1/V_2)]\}^2 \quad (9)$$

Hence if, for example, $C_1 = 20 \text{ mM}$, $V_1 = 10 \text{ mL}$, $C_2 = 100 \text{ mM}$, and $V_2 = 1000 \text{ mL}$, then $x = 89.6 \text{ mM}$, i.e., the NaCl concentration inside the dialysis bag at equilibrium will be 89.6 mM and that outside will be 99.1 mM . Repeating dialysis with a fresh NaCl solution will result in only a marginal increase of internal NaCl concentration to 90.5 mM . This example is, perhaps, a little extreme because it would require a 50 mg/mL solution of protein, average $M_r = 50,000$, each molecule of which carried a net 20 negative charges. It does, however, serve to illustrate the point that the distribution of salts, including buffer ions, in dialysis will be affected by the presence of protein and that even very extensive dialysis will not achieve equality of internal and external concentrations. One consequence of this is that dialyzed solutions of high and low concentrations of the same protein mixture will not behave identically on, for example, ion-exchange chromatography because their ionic strengths and pH values will differ.

7. Note that the temperature compensator corrects for the conversion of measured emf to pH and not for the variation of buffer pH with temperature. If Tris buffers are to be used and very accurate pH values are required, then special glass electrodes are needed that do not respond to Tris itself; these are available from several suppliers. Care must be taken of glass electrodes if good pH measurements are to be made. They must never be allowed to dry out (store in saturated KCl solution), and protein must not be allowed to accumulate on them (wash with detergent solution if this occurs).
8. This exclusion limit is appropriate for proteins of $M_r > 20,000$. For smaller proteins, Sephadex G-10 should be used. This latter material produces a smaller bed volume/g dry wt and hence is more expensive to use.

9. For desalting or changing buffers with small volumes of protein solution (up to about 2 mL), Pharmacia Biotech markets prepacked Sephadex G-25 columns in either disposable or reuseable form. These are very convenient, but moderately expensive. Information can be obtained from the company's technical publications.
10. The situation is complex and it is very difficult to analyze precisely because of the change in net charge of a protein with changing pH. However, it should be clear from the discussion in Note 6 that, since proteins will carry a net charge at all pH values except the isoelectric point, the presence of protein inside the dialysis bag will affect the distribution of ions, including those of the buffer. Therefore, for example, dialysis of a negatively charged protein mixture with a buffer formed from a weak acid will lead to a situation where the concentration of conjugate base is lower inside than outside, and hence, the pH inside will be lower than in the bulk buffer. The higher the protein concentration and the greater the difference between the starting pH of the protein solution compared with the target pH, the greater the effect will be. Hence, it is not possible to obtain complete equality of pH and buffer concentration by dialysis even for relatively dilute protein solutions. Given that for most applications the pH rather than the ionic strength is the key factor, it is best to dialyze to equilibrium, and then to adjust the pH to the target value if necessary by addition of acid or base.
11. The ratio of the length to the diameter of the column should be in the range 10:1–20:1. Longer but narrower columns may give slow flow rates, whereas short, fat columns may give poor separations and greater dilutions of the emergent protein solution because of imperfections in the packing. Pharmacia Biotech markets columns of these dimensions, as do other companies, but homemade columns provide a cheaper satisfactory alternative (see Chapter 36).
12. Flow rates should be approx 0.1–0.2 mL/min/cm² of cross-sectional area. Faster flow rates will lead to incomplete exchange between the bulk solvent and the beads, whereas slower values will result in more band spreading and dilution.
13. If a column is to be used several times for desalting or for exchange of buffers, then it may be convenient to calibrate it by passing through a mixture of a standard protein (e.g., bovine serum albumin) and a salt (e.g., NaCl) and determining the volume ranges over which the protein and the salt emerge. For detecting the protein absorbance values at 280 nm are used, whereas detection of the presence of NaCl could be done by addition of silver nitrate to aliquots of the fractions and observing the precipitation of silver chloride; if the buffer contains chloride or phosphate ions (silver phosphate is insoluble also), then some alternative low-mol wt substance

such as acetone (which can be determined spectroscopically) would have to be used. Thereafter, provided the same equilibration buffer is used, the protein fraction can be collected simply on the basis of elution volume.

14. The point here is that protein solutions are themselves buffers by virtue of the ionizable side chains that they contain. In gel filtration, the protein may be transferred from a buffer at one pH to an only slightly larger (about 1.5 times) volume of a buffer at a different pH. Unless the second buffer is very strong it is unlikely to have sufficient buffering capacity to effect the required change in pH particularly if the starting and target pH values are widely different. For example, a change from pH 8.0 to 6.0 will require protonation of all or most of the histidine residues in a protein, thus withdrawing protons from the target buffer with a consequent rise in pH. Hence, again the pH should be checked after gel filtration and adjusted if necessary.

References

1. Dawson, R. M. C., Elliot, D. C., Elliot, W. H., and Jones, K. M. (1986) *Data for Biochemical Research*, 3rd ed, Oxford University Press, Oxford, UK, pp 417–448
2. Good, N. E, Winget, G. D., Winter, W., Connolly, T. N., Izawa, S, and Singh R. M. M. (1966) Hydrogen ion buffers for biological research. *Biochemistry* **5**, 467–477.
3. Cronin, V B, Maras, B., Barra, D, and Doonan, S. (1991) The amino acid sequence of the aspartate aminotransferase from baker's yeast (*Saccharomyces cerevisiae*) *Biochem J* **277**, 335–340

CHAPTER 12

Purification and Concentration by Ultrafiltration

Paul Schratter

1. Introduction

Membrane ultrafiltration (UF) is a pressure-modified, convective process that uses semipermeable membranes to separate species in aqueous solutions by molecular size, shape, and/or charge. It separates solvents from solutes (i.e., the dissolved species) of various sizes. The result of removing solvent from a solution is solute concentration or enrichment. Repeated or continuous dilution and reconcentration are used to remove salts or exchange solvent (in such applications as buffer exchange). Definitions of some terms used in UF are given in Table 1.

UF is a low-pressure procedure, generally more gentle for the solutes than nonmembrane processes. It is more efficient than such processes, and can simultaneously concentrate and desalt solutions. It does not require a phase change, which often denatures labile species and can be performed at cold-room temperatures.

UF should be viewed as an excellent tool for efficient separation of biological substances into groups, according to molecular weight and size. For finer separation, it must be followed by a more selective process, such as chromatography or electrophoresis.

1.1. Membrane Processes

UF is one of a spectrum of membrane separation techniques that include reverse osmosis, dialysis, and microfiltration. Reverse osmosis separates solvents from low-mol-wt solutes (typically <100

From *Methods in Molecular Biology*, Vol 59 *Protein Purification Protocols*
Edited by S. Doonan Humana Press Inc., Totowa, NJ

Table 1
Some Definitions of Terms Used in UF^a

Concentration	Enrichment of a solution by solvent removal
	The relative amount of a molecular species in a solution, expressed in percent
Concentration polarization	Accumulation of rejected solute on the membrane surface, depends on interactions of pressure, viscosity, crossflow (tangential) velocity, fluid flow conditions, and temperature.
Cutoff (MWCO)	The molecular weight at which at least 90% of a globular protein is retained by the membrane
Fouling	Irreversible decline in membrane flux owing to deposition and accumulation of sub-micron particles and solutes on the membrane surface, also, crystallization and precipitation of small solutes on the surface and in the pores of the membrane, not to be confused with concentration polarization
Permeate	The solution passing through the membrane, containing solvent and solutes not retained by the membrane
Rejection	The fraction of solute held back by the membrane.
Retentate	The solution containing the retained (rejected) species.
Yield	Amount of species recovered at the end of the process as a percentage of the amount present in the sample

^aTable courtesy of Amicon, Inc

Dalton) at relatively high pressures. Reverse osmosis membranes are normally rated by their retention of sodium chloride, whereas UF membranes are characterized according to the molecular weight of retained solutes.

Dialysis is a diffusive process employing a second liquid (dialysate) on the opposite side of the membrane from the sample. The permeation rate of molecules from sample to dialysate is in direct ratio to their concentration and inversely proportional to their molecular weights. Therefore, the rate of transport of a salt through the dialysis membrane diminishes as salt concentration in the sample declines during the process, so that desalting by dialysis tends to be quite slow. In UF, all completely

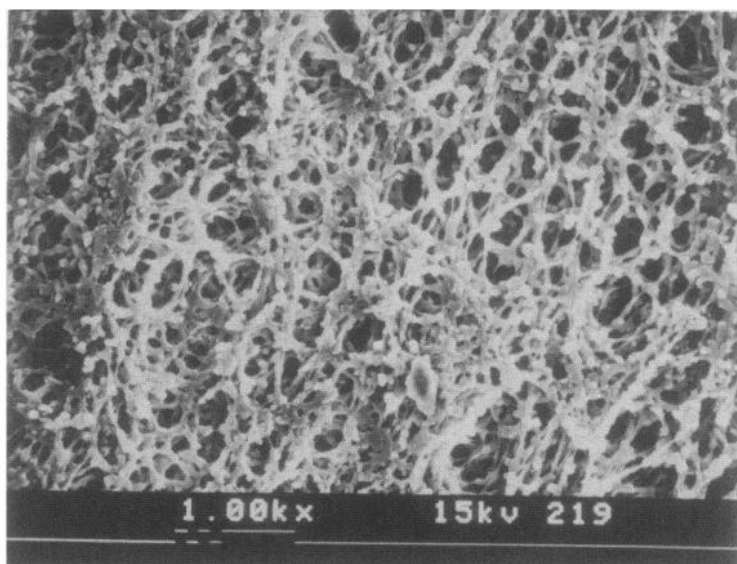


Fig. 1. Cross-section of microporous membrane. Particles are trapped on its surface or within pores. Electron micrograph.

membrane-permeating species pass equally with the solvent, independent of their concentration.

Microporous membranes (microfilters) are generally rigid, continuous meshes of polymeric material with pore diameters that are two or three orders of magnitude larger than those of ultrafilters. Species are either retained on the membrane surface or trapped in its substructure (Fig. 1). Microfilters retain bacteria, colloids, and particles upward of $0.025\text{ }\mu\text{m}$ in diameter, depending on rated pore size. Pore sizes typically used for microfiltration are 0.22 and $0.45\text{ }\mu\text{m}$.

1.2. UF Materials and Devices

UF is fundamentally a very simple process. It marries the selective permeability of a membrane structure to a device or system that applies the required pressure, minimizes buildup of retained material on the filter, and provides for access and egress of the fluid. The surface of the UF membrane contains pores with diameters small enough to distinguish between the sizes and shapes of dissolved molecules. Those above a predetermined size range are rejected, whereas those below that range pass through the membrane with the solvent flow.

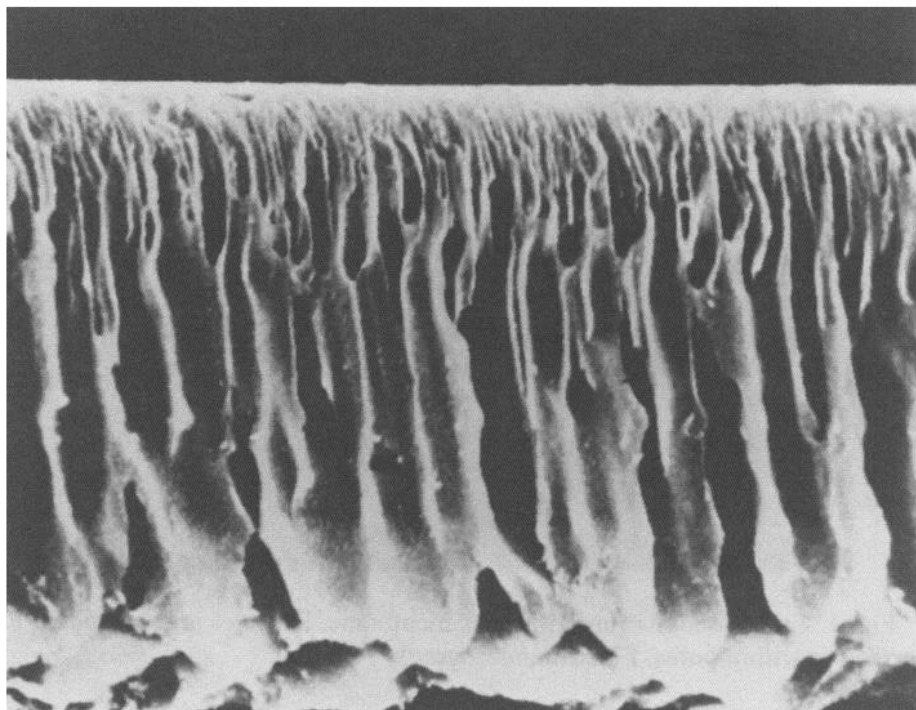


Fig. 2. Cross-section of anisotropic UF membrane. UF takes place in the top layer. Open-celled structure is highly permeable. Electron micrograph.

1.2.1. Membrane Ultrafilters

UF membranes are made of various polymers. They generally have two distinct layers. On the side in contact with the sample or fluid stream is a very thin ($0.1\text{--}1.5\text{ }\mu\text{m}$) dense “skin” with extremely fine pores whose diameters are in the range of $10\text{--}400\text{ }\text{\AA}$ (1×10^{-6} to $4 \times 10^{-5}\text{ mm}$). Below this is a much thicker ($50\text{--}250\text{ }\mu\text{m}$) open-celled substructure of progressively larger voids, largely open to the filtrate side of the ultrafilter (Fig. 2). Any species capable of passing through the pores of the skin (whose sizes are precisely controlled in manufacture) can therefore freely pass the membrane. That arrangement provides a unique combination of selectivity and exceptional throughput at modest pressure. It resists clogging because retained substances are rejected at the smooth membrane surface.

Most membranes are cast on tough, porous substrates for improved handling qualities and repeated use. They offer dependably controlled

retention, water permeability, and solute transport. The best membranes are inert, noncytotoxic, and do not denature biological materials.

Some membranes with high flow characteristics are made of inert, non-ionic polymer. They do not adsorb ionic or inorganic solutes, but may adsorb steroids and hydrophobic macromolecules. Advanced hydrophilic membranes have exceptionally low nonspecific protein binding properties. They should be used where maximum solute recovery is of special importance.

1.2.2. Devices for UF

1.2.2.1. STIRRED CELLS

Pressurized cells are a convenient means of UF for volumes in the range of 3 mL to 2 L. They are capable of final concentrate volumes of 50 μ L to 60 mL, with concentration factors typically in the range of 60–80-fold.

A stirred cell (Fig. 3) is generally a vessel containing the solution to be filtered with a means of installing the membrane at the bottom, supported by a polymer grid. An access port permits pressurization, normally with nitrogen, in the range of 2.7–3.4 atm (40–50 psi). The pressure on the surface of the liquid forces the sample through the ultrafilter where separation between retained and passing solutes takes place. Formation of a layer of retained material on the membrane surface is minimized by means of a magnetic stirrer that is propelled by mounting the cell on a magnetic stirring table. The gentle stirring action assures minimal exposure of labile solutes to degradation by shear effects. The cell can also be connected to a pressurized reservoir that continuously refills the cell as filtrate flows from it, for desalting or buffer exchange.

Stirred cells accommodate membranes of various diameters, normally from 25–90 mm. Choice of membrane diameter involves two conflicting aspects. The larger the membrane (and therefore the cell), the faster the run will be accomplished. However, if the molecules to be concentrated are dilute and their maximum recovery is very important, a large membrane and cell-wall surface area may expose them to nonspecific adsorption and possibly significant loss. Therefore, if speed is important, a large cell should be used; if high recovery from a dilute solution is vital, the smallest possible cell (and membrane area) should be employed. Since large membranes and cells are more costly, that may also be a factor in size selection. Normally, the choice is a trade-off between the two extremes.

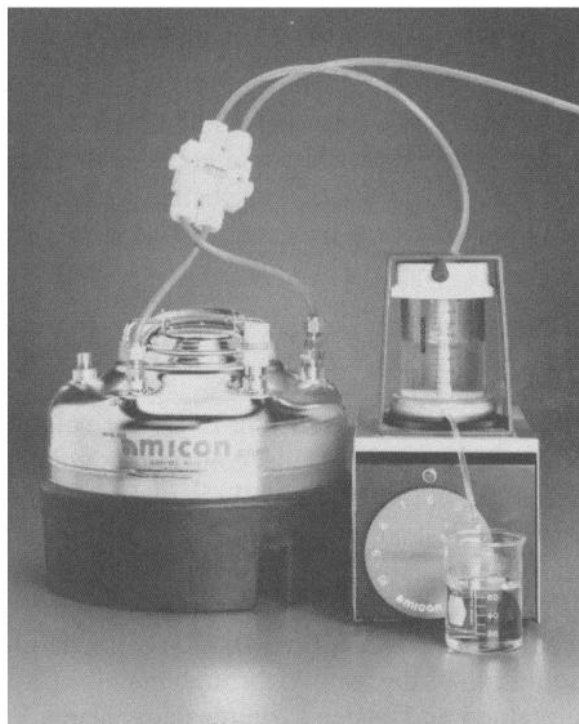


Fig. 3. Stirred UF cell on magnetic stirrer. The reservoir and selector valve are used for desalting or buffer exchange.

Stirred cells are excellent for solutions with up to 10% solute concentration. They are used widely for concentration or desalting of dilute proteins, enzymes, polypeptides, viruses, yeasts, bacteria, and so forth.

1.2.2.2. CENTRIFUGAL DEVICES

A selection of devices is offered by UF equipment manufacturers that employs centrifugal force to exert the needed pressure on the sample to obtain UF. The smallest of these devices—for initial volumes in the micro-liter range and up to 2 mL—consist of small, capped sample reservoir tubes with the UF membrane sealed across the bottom, supported by a polymer grid. The tube fits into a vial to capture the filtrate (Fig. 4). These units only require ordinary laboratory centrifuges, with centrifugal force of up to 14,000g. Use of a fixed-angle centrifuge rotor, rather than a swinging-bucket type, can reduce the time required for a separation run. Since

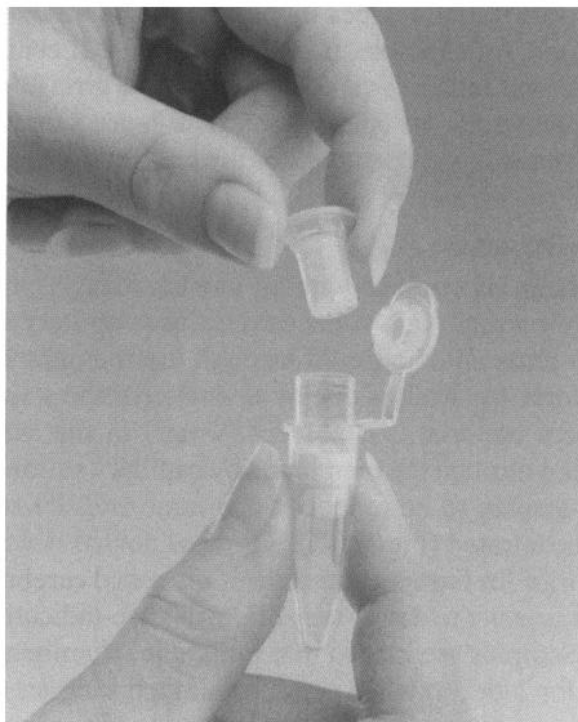


Fig. 4. Centrifugal microconcentrator with microporous insert (Micropure, top) for simultaneous separation of protein from electrophoresis gel and protein concentration or desalting.

centrifuge rotors can hold multiple units, multiple samples can be processed quickly at the same time. Some devices include a brief extra centrifugation step with the UF element reversed so that the side containing the retained material faces the filtrate vial. This drives every bit of the retentate into the vial for maximal recovery.

Centrifugal ultrafilters presently range in volume capacity from 0.5–15 mL. The smallest allow concentration from 0.5 mL initial volume to as little as 5 μ L (100-fold concentration). Normal spin time for concentration at room temperature is 6–60 min, depending on the selected molecular weight cutoff (MWCO) as well as solute viscosity. Use of low-adsorption membranes and plastics makes these devices very efficient, typically delivering solute recoveries of over 90%. They concentrate the product without change in ionic environment or denaturation.

Some larger centrifugal devices include more elaborate mechanisms for separation. One, for example, is so constructed that centrifugal force drives dense material in the sample away from the ultrafilter surface. It is, therefore, effective in concentrating particle-laden samples with high recovery. This is useful, for example, for the concentration of cell supernatants, lysates and extracts.

1.2.2.3. STATIC DEVICES

For concentration of samples without any centrifuge, pressurization, or other accessory equipment, some devices use capillary action as the driving force to transport the liquid through the membrane. By letting the membrane form the wall of one or several chambers and backing it with an absorbent material, the solvent (water) in the samples in the chambers is pulled through the membrane by capillary action. This causes the individual samples to be reduced in volume and the retained molecules to be concentrated (Fig. 5). This type of device is widely used in clinical laboratories for the concentration of urine and cerebrospinal fluid in order to make it easier to detect very dilute disease-indicating species in those samples. Samples are loaded into individual chambers. The unit is left unattended for 1 or 2 h when the concentrated samples can be withdrawn with a pipet. A treatment near the bottom of the membrane prevents accidental drying of the sample. Graduation lines indicate the degree of concentration, which may be up to 200-fold. These devices can also be used to desalt the sample by repeated dilution and concentration.

1.3. Operating Parameters

1.3.1. Molecular Weight Cutoff

The MWCO of a membrane is defined as the molecular weight of hypothetical globular solutes (proteins) that will be 90% rejected by the membrane (Table 2). For example, a 10,000 MWCO ultrafilter will nominally reject 90% of molecules with a mol wt of 10,000 Dalton. Since rejection is actually a function of physical size, shape, and electrical characteristics of the molecule, the MWCO is only a convenient indicator, based on model solutes. Linear molecules, such as polysaccharides, will tend to slip through a membrane that would reject globular molecules of the same molecular weight.

Although two membranes can be claimed to have the same cutoff, they can exhibit quite different rejection behavior owing to distribution of pore diameters. Solute retention is not absolute. A "sharp" cutoff mem-

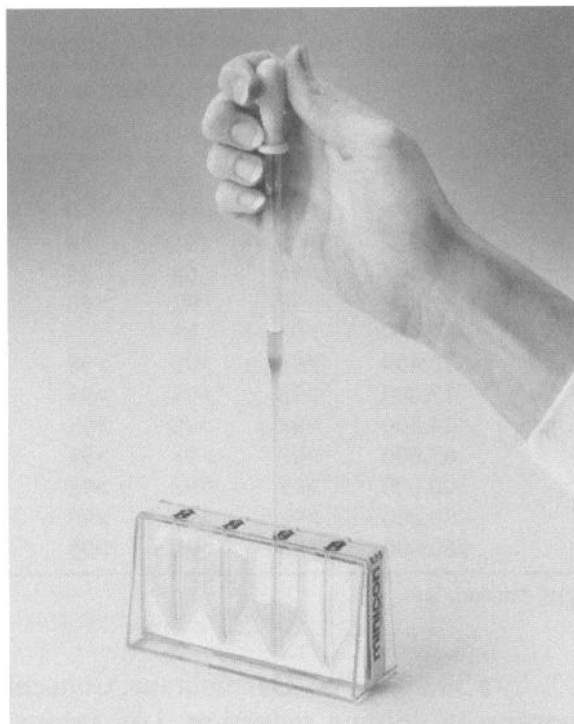


Fig. 5. Static UF device for use in clinical sample concentration.

brane will have minimal retention for species below its nominal MWCO rating. A “diffuse” cutoff membrane can significantly retain species of a size below the nominal MWCO or allow passage of some species above its cutoff. For concentration of retained species, either sharp or diffuse cutoff membranes will generally work equally well, but where the permeate is of interest, the final product may be markedly different. To determine cutoff sharpness, solute rejection tables for different membranes should be compared (*see Note 1*).

1.3.2. Solute Retention

Retention or rejection of a solute by an UF membrane defines the degree to which given molecules will be held back from the passing solution by the membrane and, hence, remain in the “retentate.” For each membrane of a given MWCO, there is a specific degree of rejection of biomolecules. For example, cytochrome *c* ($M_r = 12,400$) may be rejected >98% by a 3000 MWCO membrane, >95% by a 10,000 MWCO mem-

Table 2
Typical Solute Rejection with UF Membranes^a

Solute	M_r	Membrane type				
		YC05	YM1	YM3	YM10	YM30
NaCl	58	<20	0	0	—	—
Dextrose	180	>70	0	0	—	—
Sucrose	342	>85	45	20	—	—
Raffinose	504	95	65	25	—	—
Bacitracin	1400	98	92	>80	20	—
Inulin	5000	>98	95	—	25	—
Cytochrome <i>c</i>	12,400	>98	>98	>98	>95	<15
Myoglobin	17,000	>98	>98	>98	>98	—
α -Chymotrypsinogen	24,500	>98	>98	>98	>98	>80
Albumin	67,000	>98	>98	>98	>98	>98
IgG	160,000	>98	>98	>98	>98	>98
Apo ferritin	480,000	>98	>98	>98	>98	>98
IgM	960,000	>98	>98	>98	>98	>98

^aTable courtesy of Amicon, Inc

brane, and <15% by a 30,000 MWCO membrane. Concentration proceeds in direct proportion to volume reduction, i.e., solute concentration doubles at 50% volume reduction. For guidance, manufacturers provide rejection tables or curves in their literature.

For freely membrane-permeating species, such as sodium chloride, the concentration of solute in the retentate and the permeate will be equal. If, for example, a solution containing 1% protein and 2% salt is processed with a membrane that is totally retentive to the protein, doubling the protein concentration by reducing the starting solution by 50% will result in a retentate containing 2% protein and 2% salt.

Many biological macromolecules tend to aggregate, or change conformation, under varying conditions of pH and ionic strength, so that their effective size may be much larger than that of the "native" molecule. This will cause increased retention at the membrane. The degree of hydration, counterions, and steric effects can also cause molecules with similar molecular weights to exhibit very different retention behavior.

Solute/solvent and solute/solute interactions in the sample can also change effective molecular size. For example, some proteins will polymerize under certain concentration and buffer conditions, whereas others (such as heme proteins) may dissociate into corresponding subunits.

Ionic interactions or p-p stacking can cause small molecules to behave similarly to molecules of greater M_r . When this occurs, as in the case of phosphate ions with a 500 MWCO membrane, the small molecules may not effectively permeate the membrane.

1.3.3. Concentration Polarization

As solute concentration increases during the process of enrichment, solute at the membrane surface forms a gel that is permeable to solvent under pressure. This effect is called concentration polarization. At moderate to high concentration of retained solutes, the flow resistance of the gel layer will reach a level where it significantly exceeds that of the membrane, in effect forming a secondary membrane. As solute continues to accumulate at the membrane/liquid interface, resistance grows. When net transport of solute by convection equals the back diffusion of solute toward the bulk solution, owing to the concentration gradient, further increase in transmembrane pressure will not increase flow through the membrane and may cause it to decrease.

All efficient UF devices or systems must provide the means to minimize the effect of concentration polarization. The most important of these are magnetic stirring, pumped tangential flow across the membrane surface in narrow channels, and positioning the membrane surface at an acute angle with respect to the force vector acting on the fluid. The latter is achieved in centrifugal UF devices by using an angle-head rotor. Centrifugal force causes the gel film to slide across the angled surface, keeping the rest of the membrane surface relatively clear for permeation by solvent and membrane-passing molecules at relatively high rates. Cone-shaped membranes also employ the force vector at an angle during centrifugal separation.

Flow rate decrease owing to concentration polarization should not be confused with the effect of membrane fouling. Fouling is the deposition and accumulation of submicron particles and solute on the membrane surface, or crystallization and precipitation of smaller solutes on or within the pores of the membrane. There may, in addition, be chemical interaction with the membrane.

1.3.4. Maximizing Solute Recovery

Although UF membranes are normally inert, adsorptive losses may occur. Additional losses, caused by formation of concentrated solute gel or cake on the membrane, can be counteracted by polarization control, as

indicated in Section 1.3.3., by operating at modest pressures, and by a final agitation cycle at zero transmembrane pressure.

Effects of adsorption are more noticeable with dilute solutions where adsorption may severely diminish the amount of the desired product. Since adsorption is largely a function of membrane and device surface area, the relation of sample concentration and volume to surface area should be considered before choosing a system. Small, dilute samples should be concentrated by using membranes or devices with minimum surface area, while maintaining reasonable flow characteristics.

Buffer can affect membrane adsorption. Phosphate buffers can cause increased losses. Tris or succinate buffers allow better recovery. This may relate to lyotropic effects on hydrophobic bonding to the membrane or device.

Although rejection is used to characterize membrane performance, it does not always directly correlate with solute recovery from a sample or volume. Actual solute recovery—the amount of original material recovered after UF—is generally based on mass balance calculations (*see* Note 2).

1.3.5. Temperature and pH

Raising the operating temperature normally increases UF rates. Higher temperature increases solute diffusivity (typically 3–3.5%/°C for proteins) and decreases solution viscosity. One normally operates at the highest temperature tolerated by the solutes and the equipment. However, UF equipment is often used in cold rooms.

Changing solution pH often changes molecular structure. This is especially true for proteins. At the isoelectric point, the protein begins to precipitate in some cases, causing a decrease of filtrate flow.

1.4. Applications

1.4.1. Concentration

In macromolecular concentration, the membrane enriches the content of a desired biological species or provides filtrate cleared of retained substances. Microsolute (e.g., salt) is removed convectively with the solvent. Pressure, created by external means, forces liquid through the ultrafilter. Solutes larger than the nominal MWCO of the membrane are retained. The required pressure can be generated by use of compressed gas, pumping, centrifugation, or capillary action. With dilute solutions (1 mg/mL or less), flow rates are directly proportional to applied pressure. At higher concentrations, increased viscosity and polarization con-

Table 3
Desalting by Repetitive Concentration and Redialysis^a

Spin number ^b	Protein recovery	NaCl concentration
Start point	—	500 mM
1	95.1%	140 mM
2	94.2%	25 mM
3	94.0%	5 mM

^aTable courtesy of Amicon, Inc

^bStart: 15 mL of γ -globulin. Spin 1: reduced to 3 mL and rediluted to 15 mL. Spins 2 and 3: repetition of reduction to 3 mL and dilution to 15 mL. After third spin, salt concentration reduced 100-fold.

centration act to reduce the flow, requiring steps to reduce concentration polarization (see Note 3).

1.4.2. Salt Removal or Buffer Exchange (Diafiltration)

Removal of small molecules from a solution by alternating UF and redilution, or by continuous UF and dilution to maintain constant volume, is called diafiltration.

Ultrafilters are ideal for removal or exchange of salts, sugars, nonaqueous solvents, separation of free from protein-bound species, removing materials of low molecular weight, or rapid change of ionic and pH environment. In contrast with dialysis, the rate of microsolite removal or "wash-out" by diafiltration is a function of the UF rate, and independent of microspecies (e.g., salt) concentration. This greatly reduces desalting times by using convective salt removal or exchange at flow rates equal to those of solvent passage. Diafiltration is also used for efficient microsolite exchange, or "wash-in."

Membrane-permeating solutes (i.e., those significantly smaller than the cutoff, especially salts) pass through the membrane pores at the same rate as water. Transport through the membrane is by convection, not by diffusion, so that the rate of permeation is independent of molecular size.

Diafiltration washes microspecies from the solution, purifying the retained species (see Notes 4 and 5). In the discontinuous method, the sample is diluted before concentration or it is diluted after concentration and re-concentrated; this can be repeated one or more times, each time obtaining further desalting or solvent exchange (Table 3). Small volumes may be easily desalted in one step by sample dilution before concentration. The continuous method of diafiltration is to connect the UF device (such as a

stirred cell) to a pressurized reservoir containing solvent, normally buffer or water. As filtration proceeds, solvent automatically flows from the reservoir into the device, at the same rate as the rate of filtration.

UF does not change salt concentration or buffer composition. A solution volume with 100 mM salt still contains 100 mM salt after concentration. Discontinuous diafiltration (rediluting the retentate with water and concentrating again) effectively decreases the salt concentration of the sample by the concentration factor of the UF. To achieve more complete salt removal, multiple concentration and redilution are required. For example, if a 1-mL sample containing 100 mM salt is diluted to 2 mL before concentration in a centrifugal device, the salt concentration in the 2-mL sample will be 50 mM. When reduced to 25 μ L (80 times), the concentrate will still contain 50 mM salt. If more complete salt removal is desired, the sample can be rediluted with water or buffer to 2 mL before reconcentration. At this point, the salt concentration will have been reduced to 0.625 mM (50 mM/80), which will remain after the second concentration to 25 μ L. Each further such dilution and reconcentration step would reduce salt concentration by 1/80, in the present example. For most samples, three concentration/reconstitution/reconcentration cycles will remove about 99% of the initial salt content. With very small sample volumes, dilution of the sample before the initial concentration step can often decrease salt concentration to an acceptable level.

1.4.3. Separation of Free from Protein-Bound Microsolute

In the past, free drug levels in serum or plasma samples were not widely measured, partly for want of a convenient means of separating free from protein-bound drug. The chosen technique was generally equilibrium dialysis, a time-consuming procedure, subject to effects of dilution and buffers. However, this method does not directly indicate the free drug concentration in the sample. Such other techniques as ultracentrifugation and gel filtration are no less time consuming, and there is inadequate standardization of results.

Today's better alternative for free/bound separation is UF with a centrifugal UF device specially designed for filtrate recovery. Spun in a standard laboratory centrifuge (preferably angle-head), free drugs readily pass the membrane for collection and analysis. The sample is not diluted in the process. Multiple samples are conveniently handled, typically in 10-min runs/set.

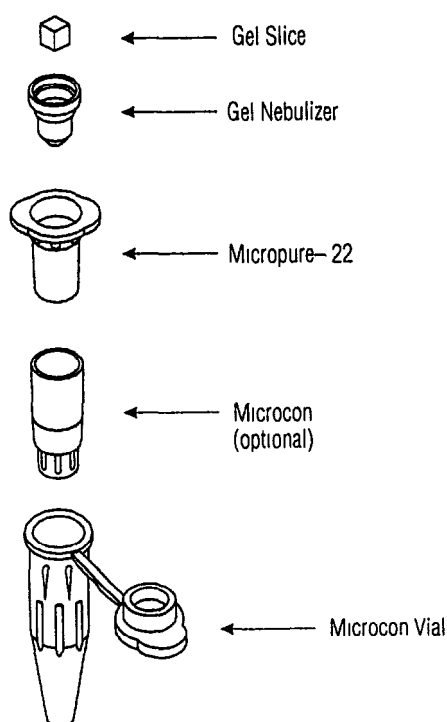


Fig. 6. Stacked elements of device for extraction of DNA, RNA, or proteins from gel.

1.4.4. Recovery from Electrophoresis Gels

The availability of microporous inserts for centrifugal UF devices offers an easy method of recovering proteins (or other macromolecules) from electrophoretic gels. The gel piece containing the protein of interest is crushed or macerated to increase its surface area, then placed into the insert, and mixed with an elution buffer. During subsequent incubation, protein diffuses out of the gel. Centrifugation of the combined devices causes the buffer containing the protein to flow through the microporous membrane of the insert, which retains the gel particles. At the membrane surface of the UF device, the protein is retained for recovery, while the buffer and any salts pass into the filtrate vial. An added new device, placed into the insert described (Gel Nebulizer), makes the process even easier by converting pieces of gel into a spray of fine particles during centrifugation (Fig. 6). This makes extraction of the proteins more efficient.

1.4.5. Purification and Fractionation

Macromolecular mixtures may be separated into size-graded classes by UF, provided the species to be separated have at least a 10-fold difference in molecular weight (*see* Note 6). This can be accomplished either by direct UF, where the permeating solute is obtained in its initial concentration in the ultrafiltrate, or by diafiltration, where the retained solute is obtained in its initial concentration in the retentate and the diluted permeating solute in the ultrafiltrate. Normally, polarization effects require predilution or multiple dilution and reconcentration of the sample.

1.4.6. Detergent Removal

UF membranes efficiently remove detergents from protein solutions. The chemical nature of most detergents causes micelle formation above a critical concentration limit (Critical Micelle Concentration), causing aggregation of the detergent and leading to gross changes in molecular structure. This affects the amount of the detergent that can be removed from solution with membranes of specific cutoff (*see* Note 7).

2. Materials

All UF equipment and membranes may be obtained from Amicon (Beverly, MA). In addition, the following may be required depending on the application:

1. Oxygen-free nitrogen with pressure regulator (for stirred cells) (*see* Note 8).
2. Microcentrifuge (variable speed) and microcentrifuge tubes.
3. Homogenizer (e.g., Eppendorf fitting pestle, VWR cat. no. KT749515-0000 or KT749520-000 and pestle mixer motor, VWR cat. no. KT749540-0000).

The following solutions are required for staining and recovering proteins from polyacrylamide gels:

1. Gel-staining solution: 0.1% Coomassie brilliant blue R250, 50% methanol, 10% acetic acid
2. Destaining solution: 7% acetic acid, 12% methanol
3. Wash solution: 50% methanol.
4. Extraction buffer: 100 mM NaHCO₃, 8M urea, 3% SDS, 0.5% Triton X-100 (reduced), 25 mM DTT.

3. Methods

3.1. Using a Stirred Cell for Concentration

1. Prefilter or centrifuge any solution containing particulate matter, such as cell debris or precipitates (*see* Note 9).
2. Fill a cell of appropriate size (*see* Section 1.2.2.) with the sample, and then secure it on a magnetic stirring table. Connect the inlet line to a regulated

- gas pressure source providing 2.7–3.4 atm (40–50 psi). Nitrogen is recommended (*see* Note 8).
3. Pressurize the cell according to instructions. Keep within cell pressure limits. When operating with either hazardous or specially valuable materials, always pressure-check the cell first to assure that all components are properly assembled.
 4. Turn on the stirring table, and adjust the stirring rate until the vortex created is approx one-third the depth of the liquid volume.
 5. When the run is completed, continue stirring for a few min after depressurization to maximize recovery of retained substances. This will resuspend the polarized layer at the membrane surface.

3.2. Protein and Peptide Recovery from Polyacrylamide Gels

The following protocol is recommended by Amicon for its Micropure™ microporous inserts and Microcon® microconcentrators:*

1. Stain the gel with staining solution. Remove excess dye by soaking the gel in destaining solution with gentle agitation for approx 2 h or until a clear background is obtained.
2. Cut out the gel piece containing the band of interest with a clean scalpel or razor blade.
3. Place the gel piece in a microcentrifuge tube. Remove dye from the protein by adding 1 mL of wash buffer and sonicating for 5–15 min at 50–60°C or until the gel clears.
4. Remove the wash buffer. Add 50–100 µL of extraction buffer. Incubate for 20–30 min at 65°C.
5. Homogenize the gel using a motor-driven pestle homogenizer.
6. Incubate the tube with homogenized gel in a bath at 50–60°C for 2–3 h or overnight.
7. Place the Micropure-0.22 insert into the Microcon sample reservoir (use Microcon-100 or Microcon-50 for large proteins and Microcon-10 or Microcon-3 for polypeptides). Add 100 µL wash solution. Transfer the gel slurry into Micropure with a pipet. Rinse the tube with 100 µL of wash solution to remove remaining slurry. Transfer to Micropure.
8. Spin the assembly until all liquid is removed from the Micropure insert (13,000g for 20–30 min in Microcon-10; for others, *see* the Centrifugation Guidelines in Table 4). Discard Micropure. Proteins or peptides are retained above the Microcon membrane, depending on selected membrane MWCO.

*All Amicon tradenames mentioned are trademarks of which Minicon, Centricon, and Microcon are registered

Table 4
Centrifugation Guidelines^a

Membrane MWCO ^b	Maximum g-force	Spin times ^c	
		4°C	25°C
3000	14,000	185	95
10,000	14,000	50	35
30,000	14,000	20	12
50,000	14,000	10	6
100,000	3000 ^d	35	15

^aTable courtesy of Amicon, Inc

^bMWCO in Daltons

^cSpin time in minutes, 500-μL samples concentrated to 10 μL

^d500g for DNA/RNA samples

9. To remove the extraction buffer from the protein, add 400 μL of wash solution to the Microcon units. Spin as indicated in the Centrifugation Guidelines (Table 4). To complete buffer exchange, add 400 μL of the desired buffer and spin at 13,000g for 30 min (for Microcon-10; for others, see Centrifugation Guidelines in Table 4).
10. To recover concentrated sample, invert Microcon into a new vial. Spin at 1000g for 3 min.

3.3. Concentration of Antibodies from a Hybridoma

By inserting a microporous filter into a centrifugal UF unit, the sample can be run through two stages of separation at the same time. The insert, with a porosity of 0.22 or 0.45 μm, retains all particulate matters (such as bacteria, cell fragments, or electrophoretic gel particles), but lets the solution containing the solutes of interest—such as proteins—flow into the UF element where the protein is concentrated. The small microporous insert can also be used by itself, with a standard microcentrifuge vial, to free samples of particles. The following is a protocol offered by Amicon for its Micropure microporous inserts and Microcon microconcentrators.

1. Place the Micropure-0.45 insert into the Microcon sample reservoir (use Microcon-50 for IgG or Microcon-100 for IgM).
2. Pipet up to 350 μL of sample into the Micropure insert.
3. Spin at 500g for 5 min or until there is no more liquid in Micropure insert.
4. Discard Micropure. Antibody is retained above the membrane in the Microcon.
5. If the antibody is to be desalted or if buffer is to be exchanged, fill the Microcon sample reservoir with 450 μL of desired buffer. Spin Microcon-50 at 12,000g for 8 min. For Microcon-100, spin at 3000g for 15 min.